

(12) **United States Patent**
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(54) **TERMINATION OF A SHIELD WITHIN AN IMPLANTABLE MEDICAL LEAD**

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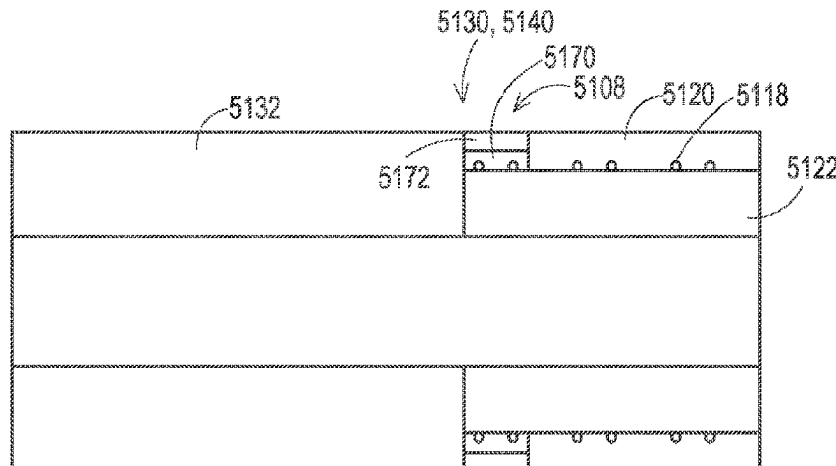
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(57) **ABSTRACT**

A shield located within an implantable medical lead may be terminated in various ways at a metal connector. The shield may be terminated by various joints including butt, scarf, lap, or other joints between insulation layers surrounding the lead and an insulation extension. The shield may terminate with a physical and electrical connection to a single metal connector. The shield may terminate with a physical and electrical connection by passing between an overlapping pair of inner and outer metal connectors. The metal connectors may include features such as teeth or threads that

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penetrate the insulation layers of the lead. The shield may terminate with a physical and electrical connection by exiting a jacket of a lead adjacent to a metal connector and lapping onto the metal connector.

17 Claims, 83 Drawing Sheets

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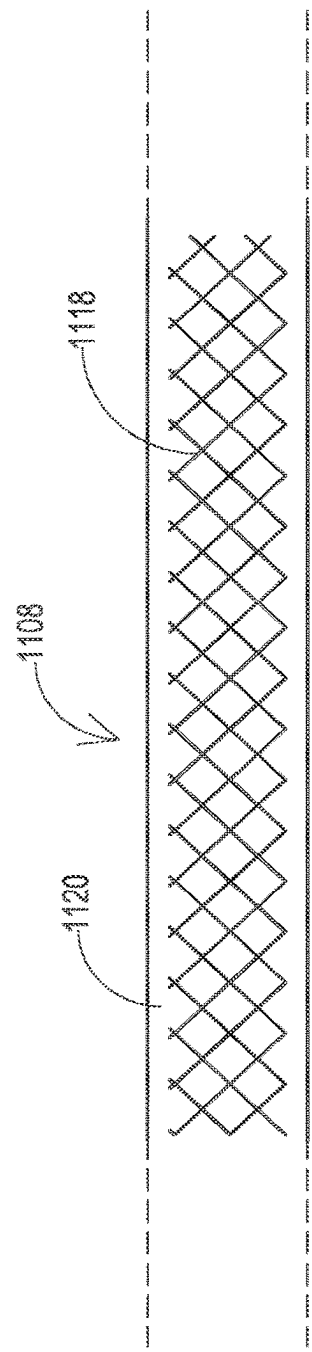
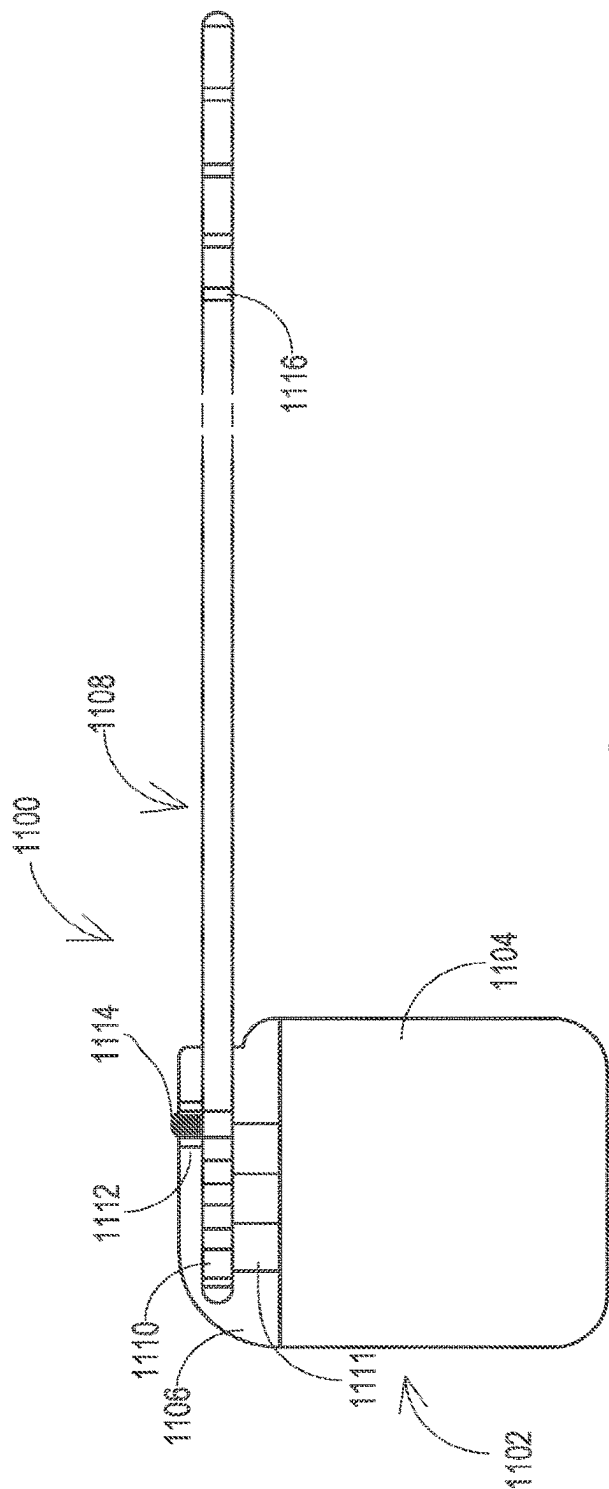
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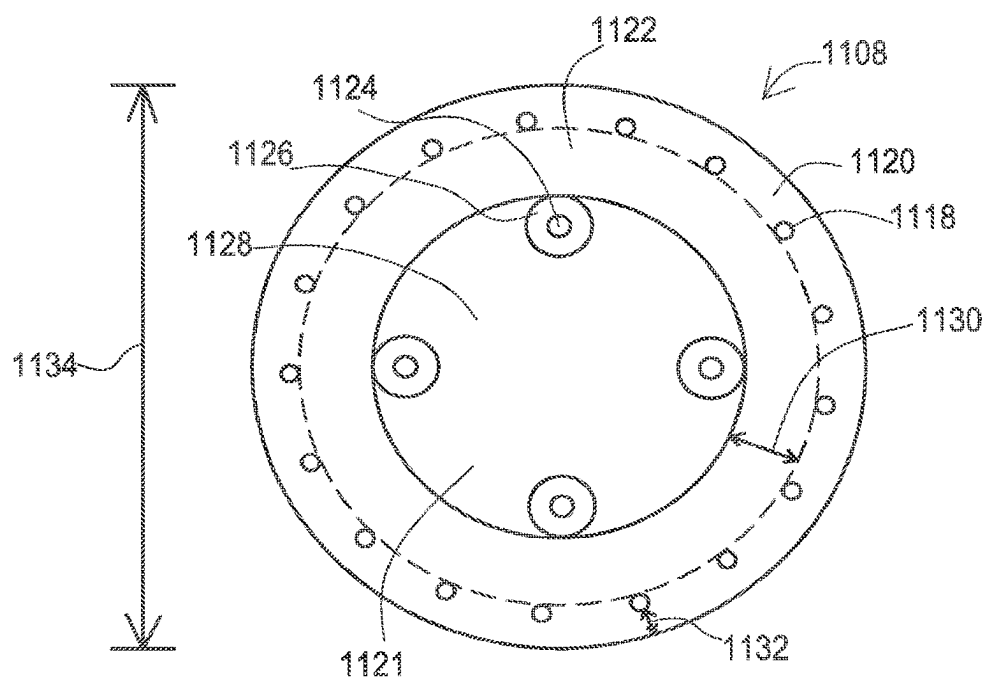


FIG. 2B

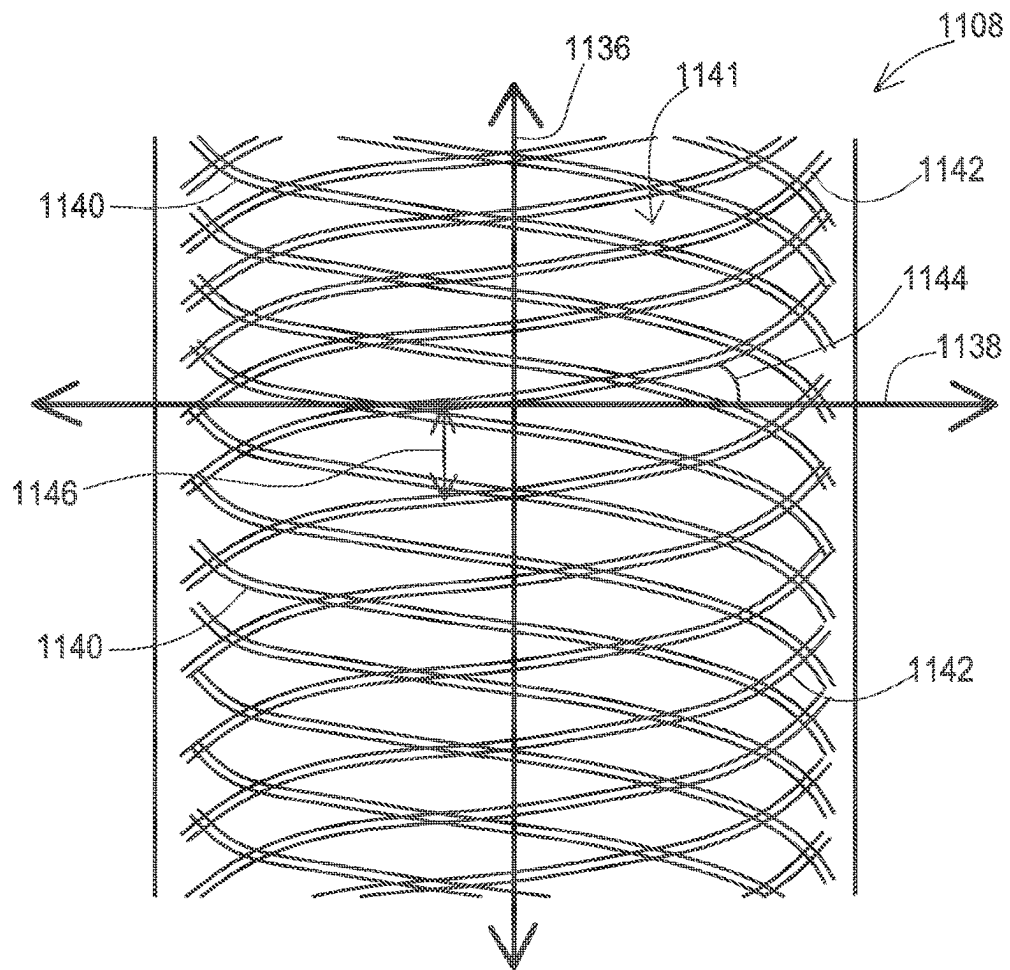


FIG. 2C

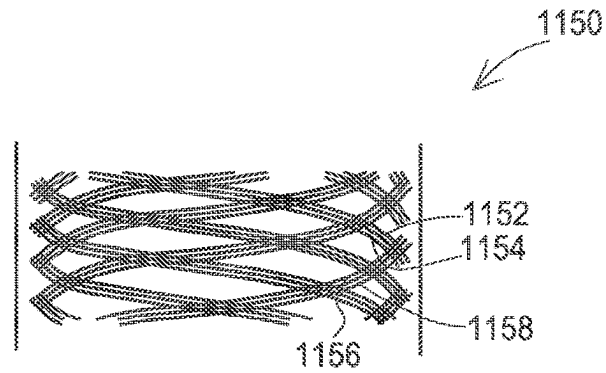


FIG. 2D

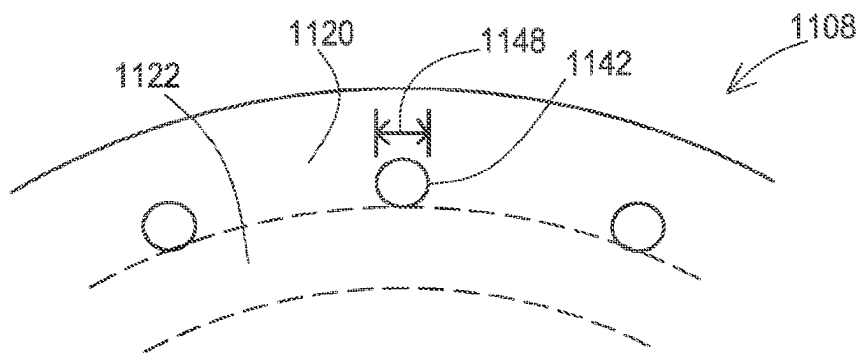


FIG. 2E

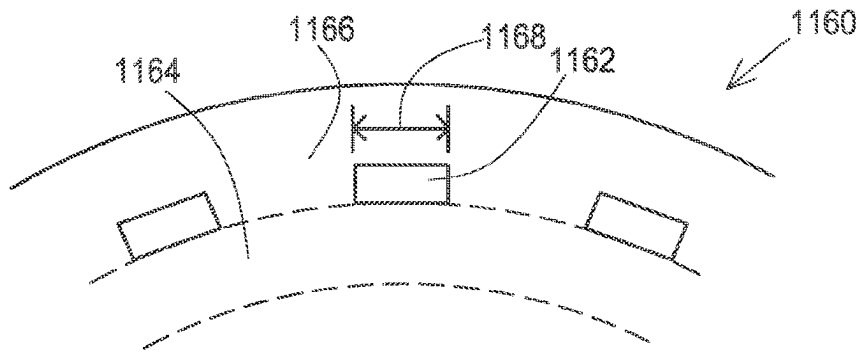


FIG. 2F

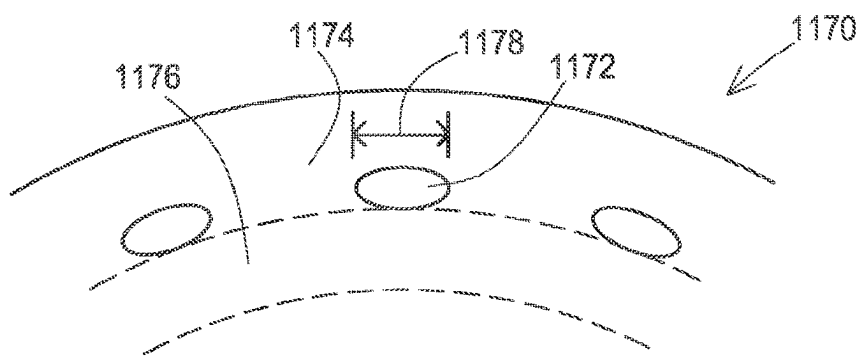


FIG. 2G

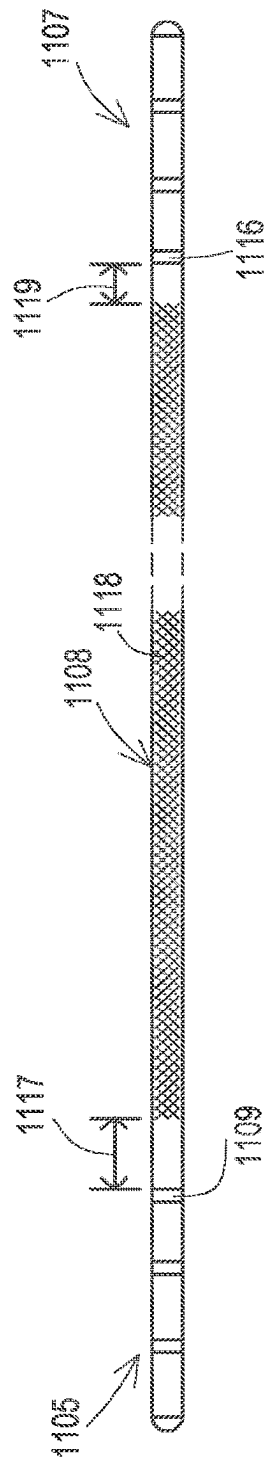
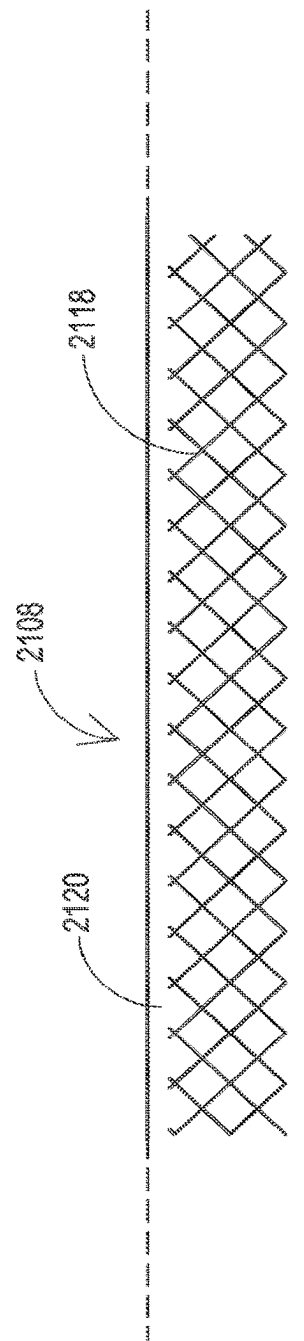
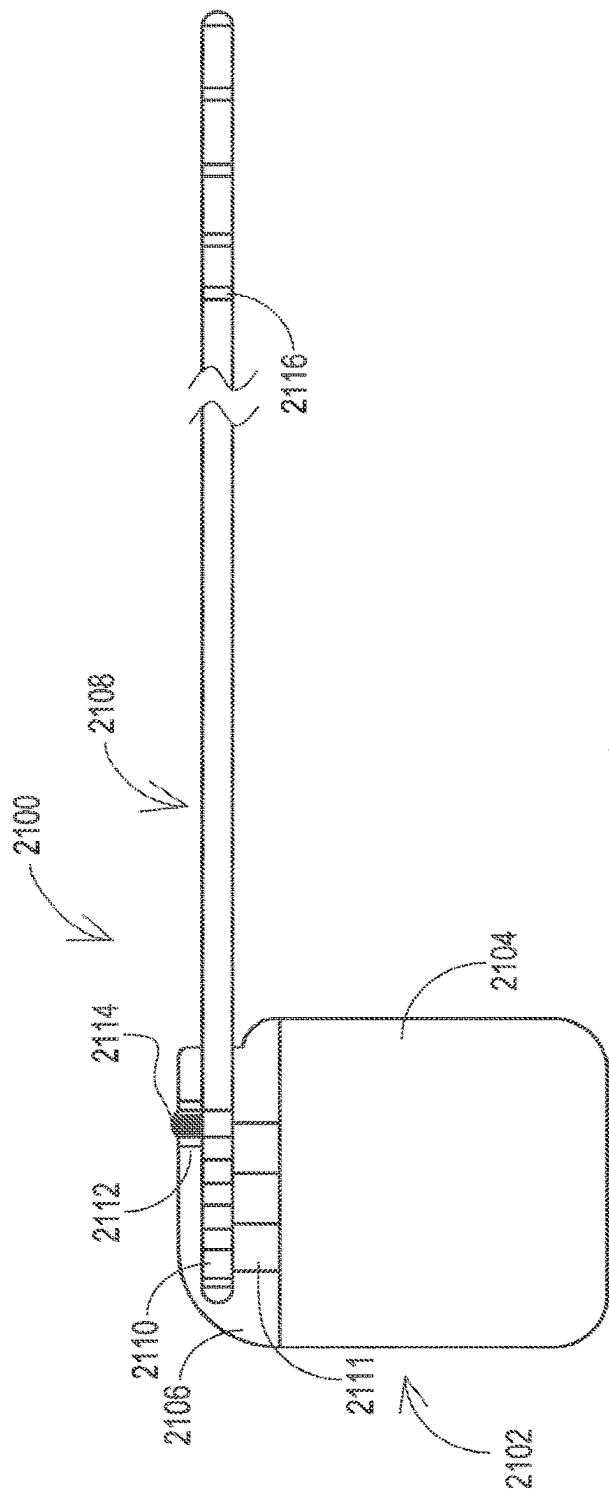


FIG. 2H



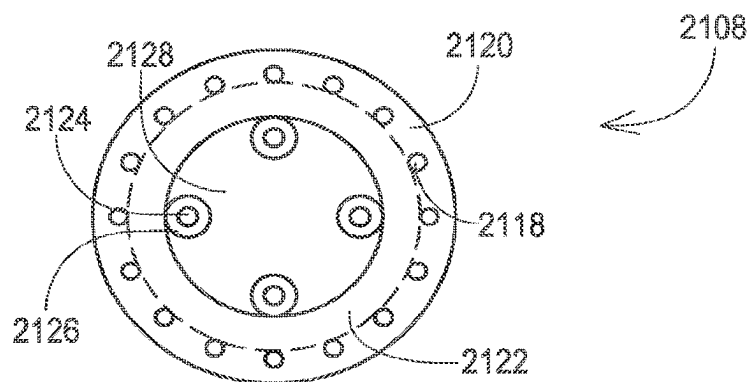


FIG. 4B

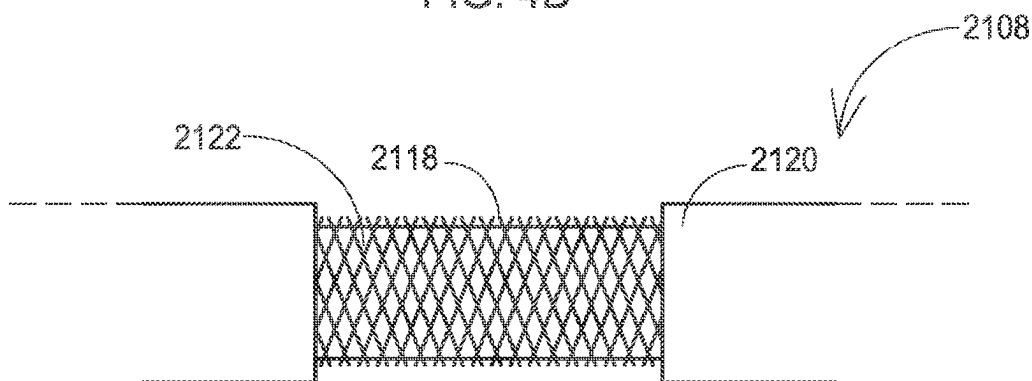


FIG. 4C

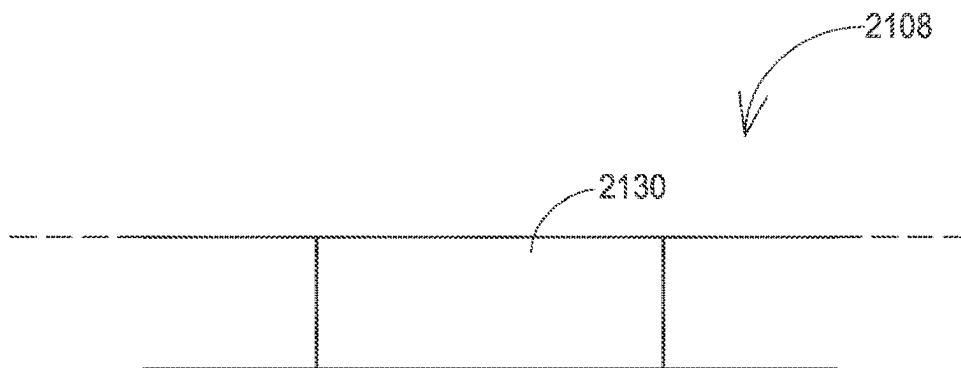


FIG. 4D

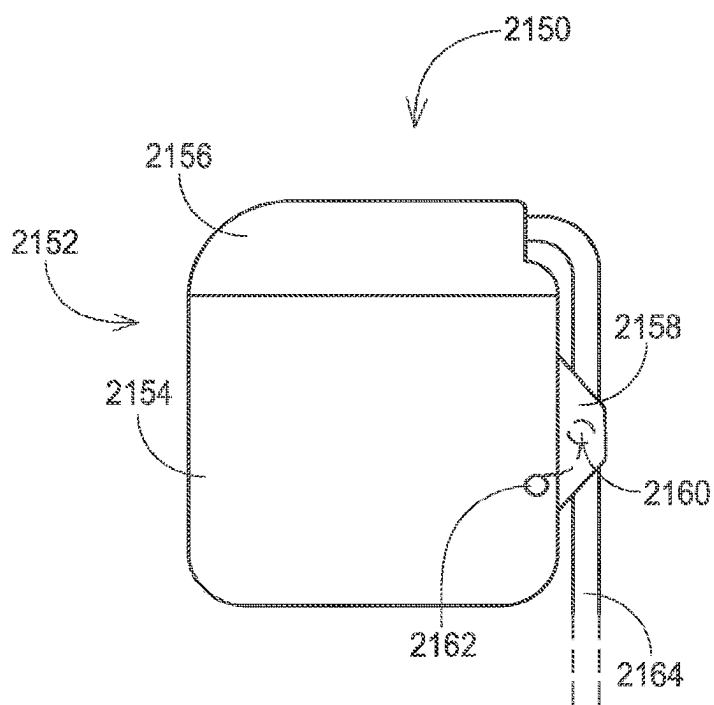


FIG. 5A

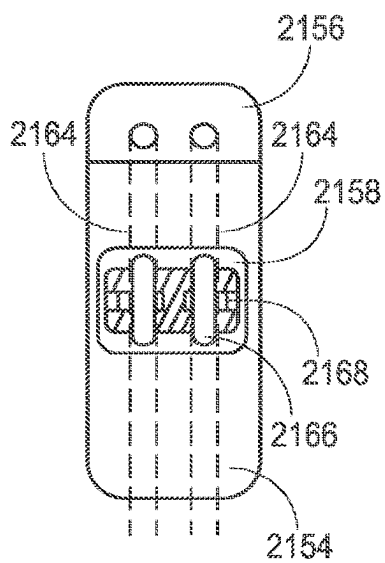


FIG. 5B

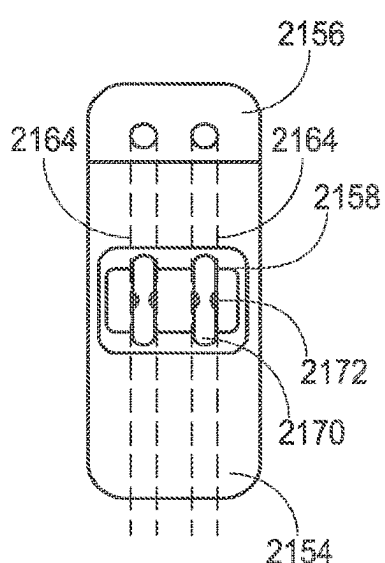


FIG. 5C

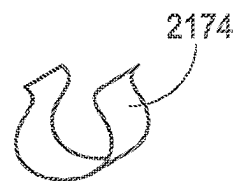


FIG. 6

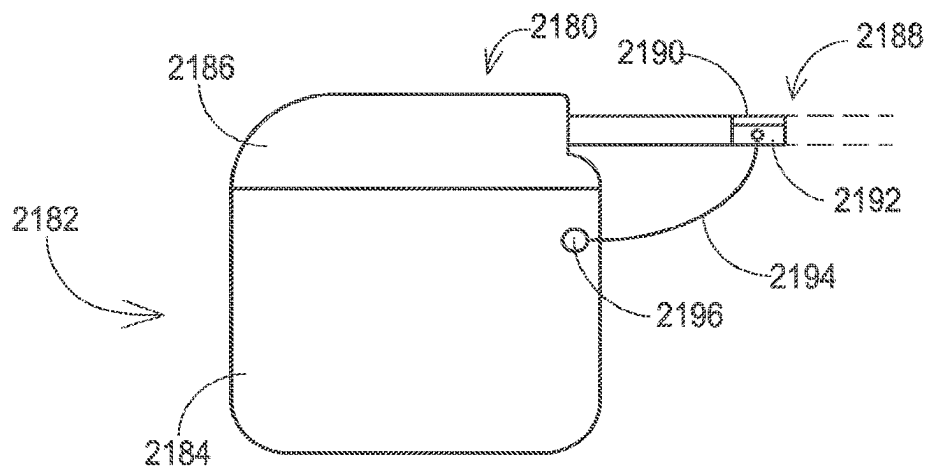


FIG. 7A

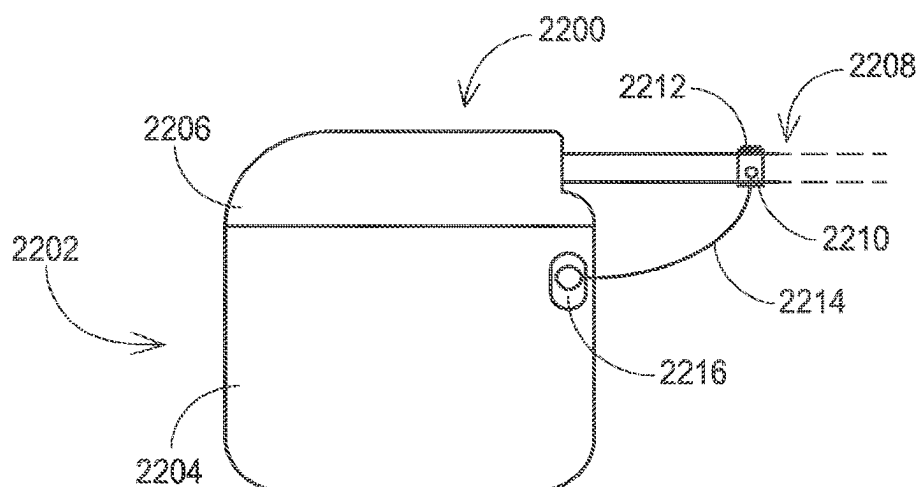


FIG. 7B

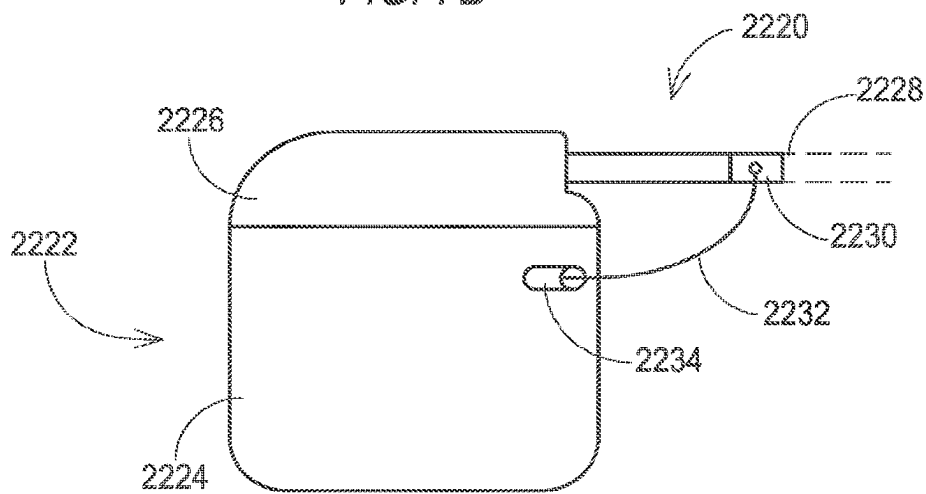


FIG. 7C

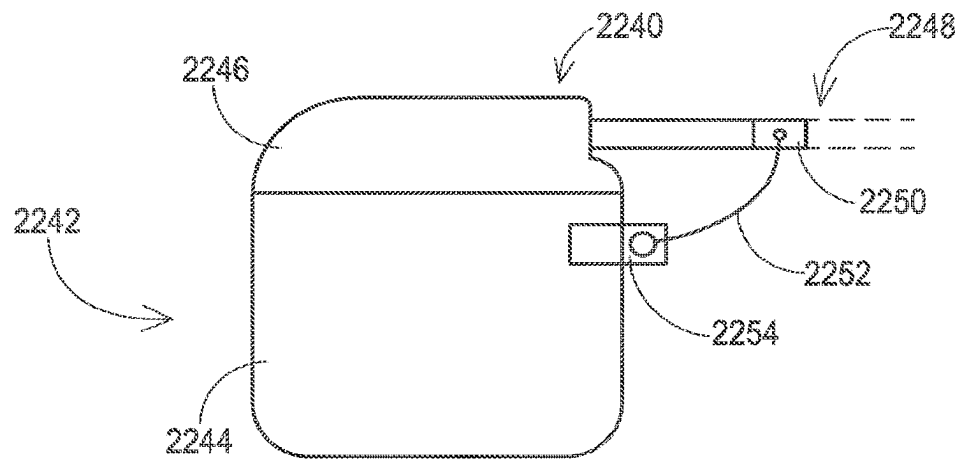


FIG. 8A

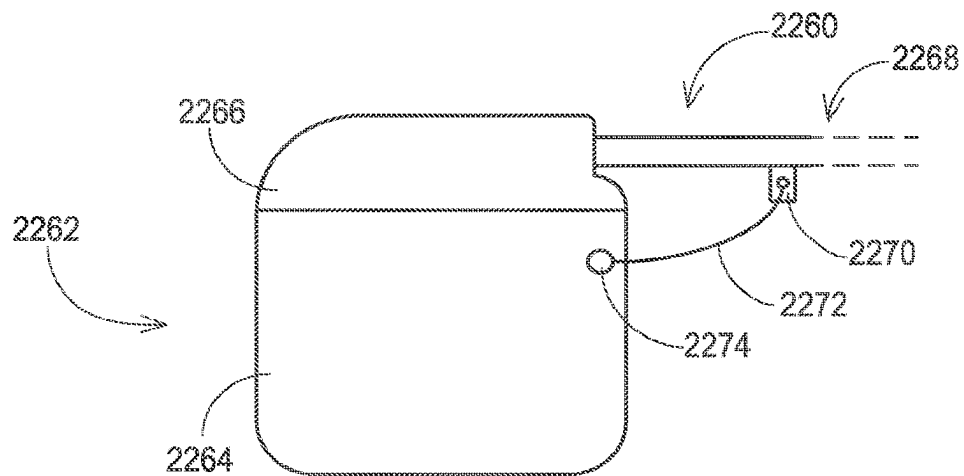


FIG. 8B

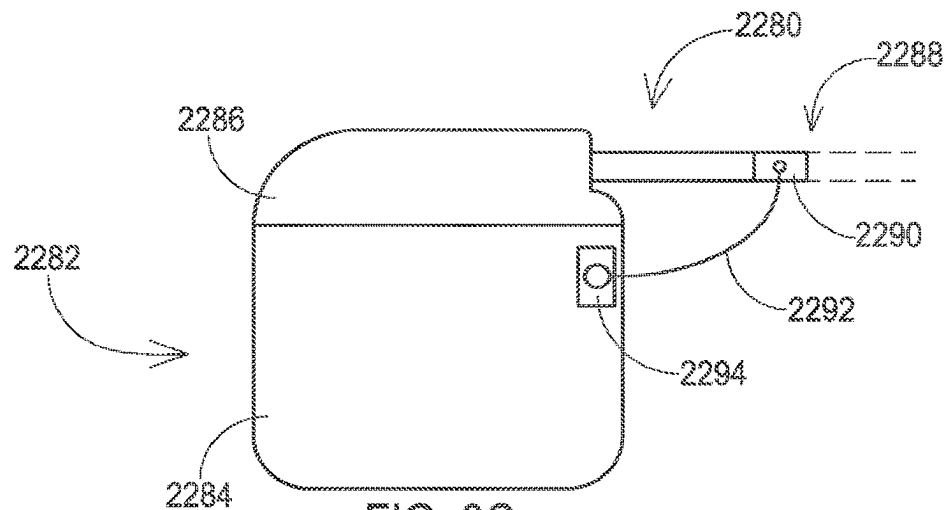
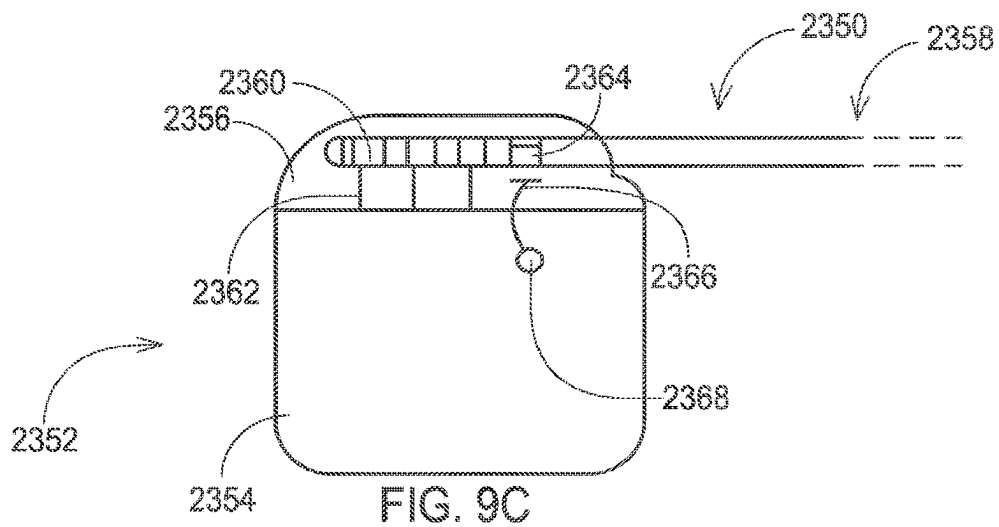
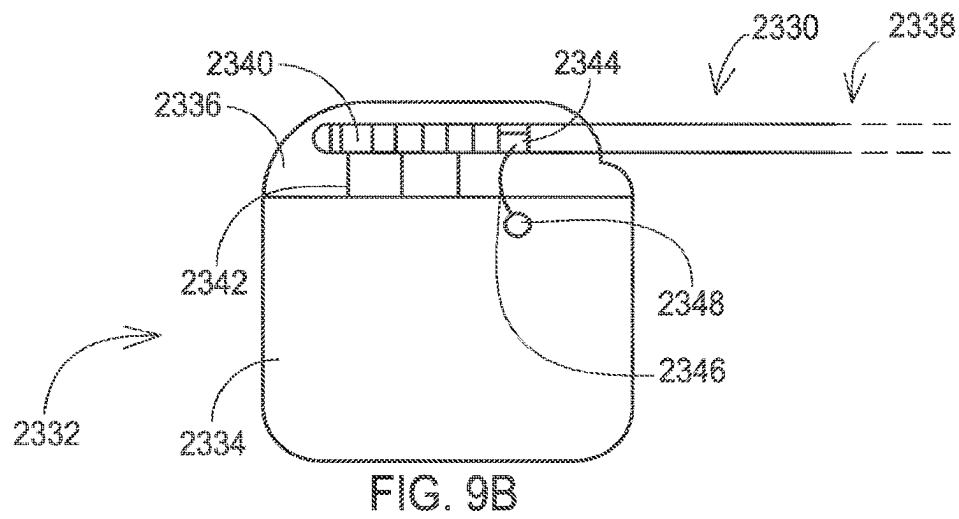
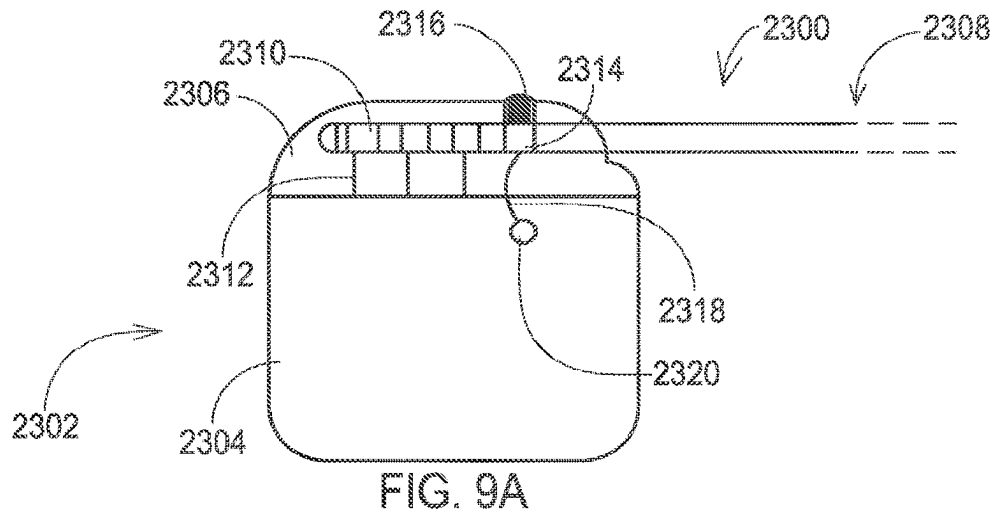


FIG. 8C



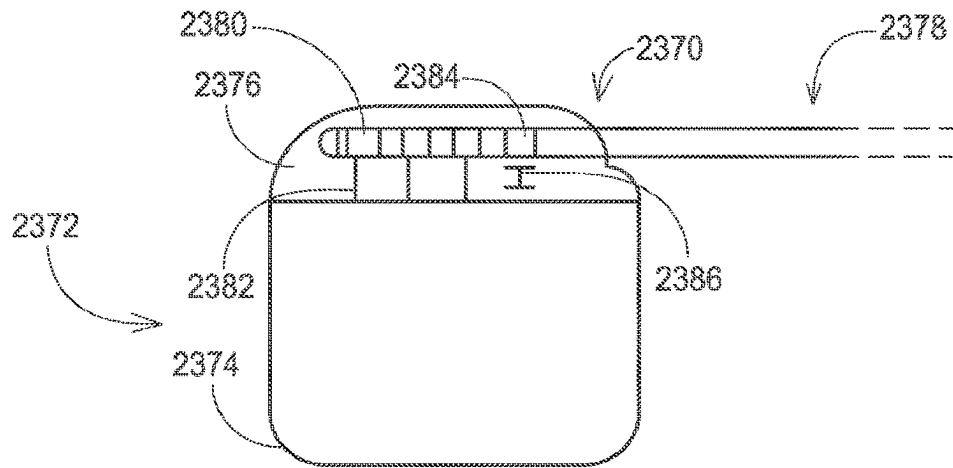


FIG. 9D

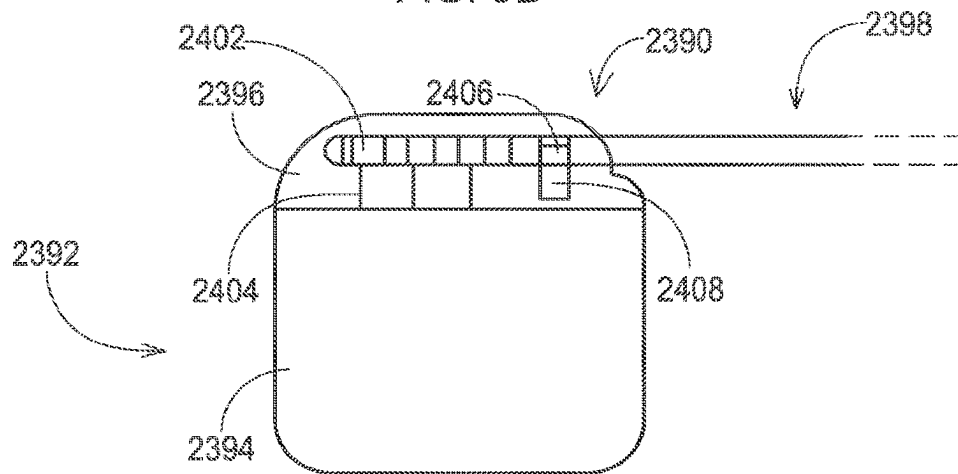


FIG. 9E

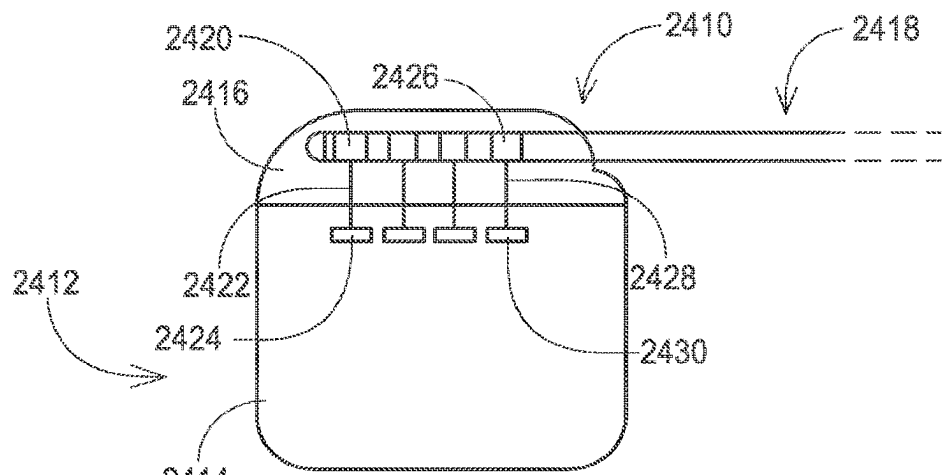
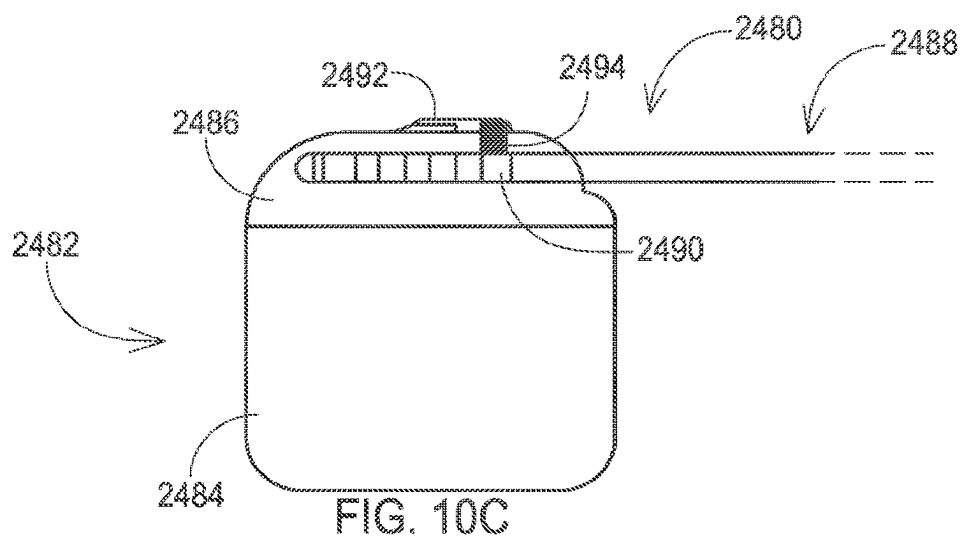
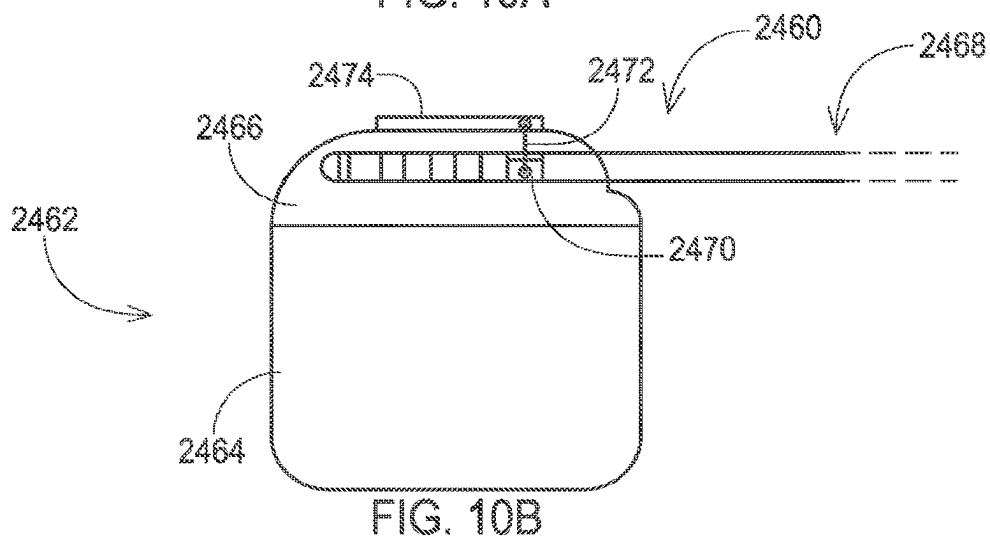
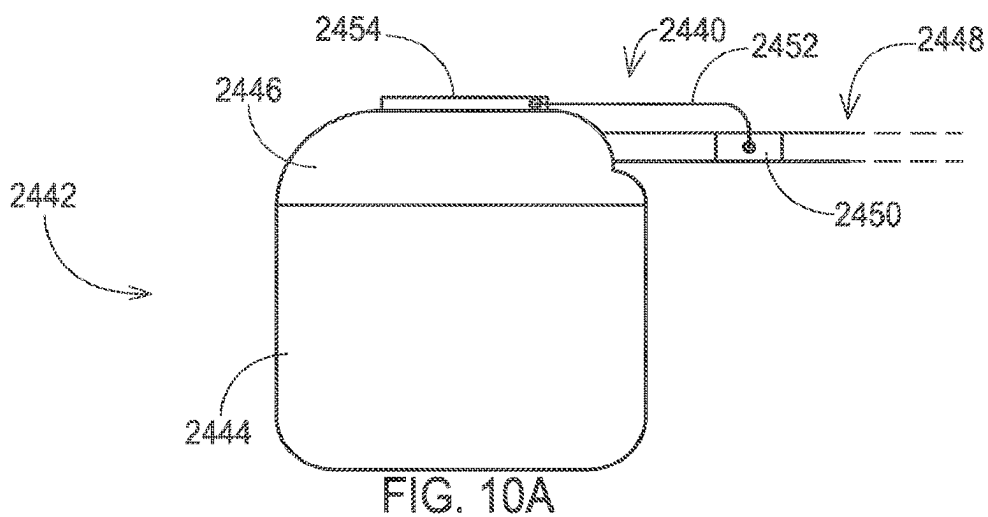
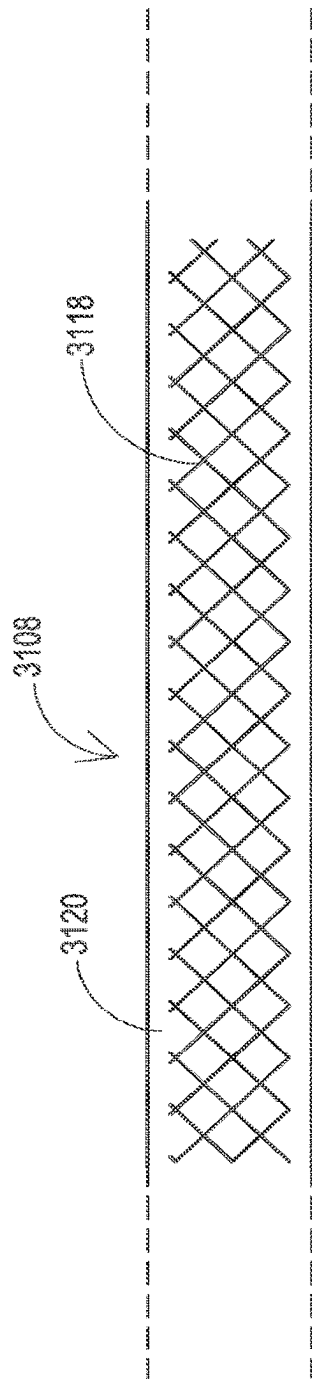
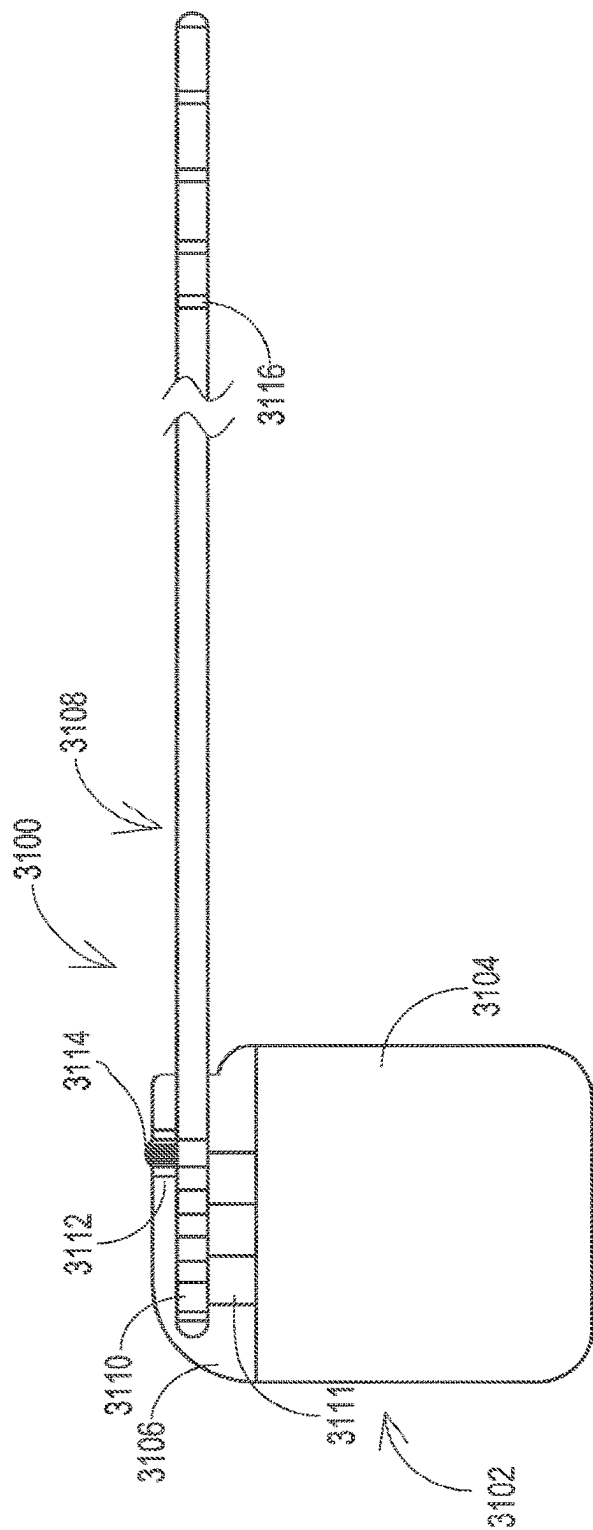


FIG. 9F





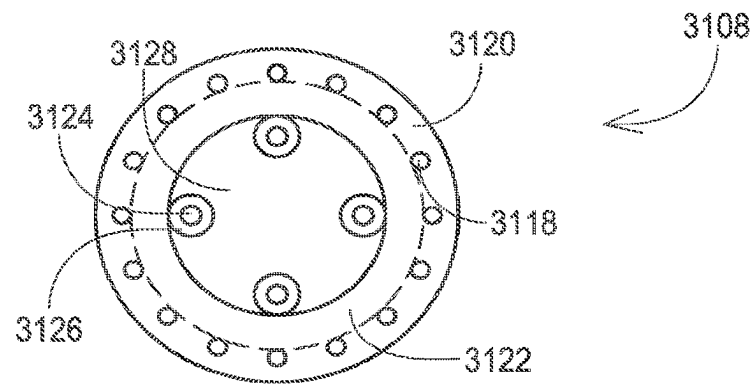


FIG. 12B

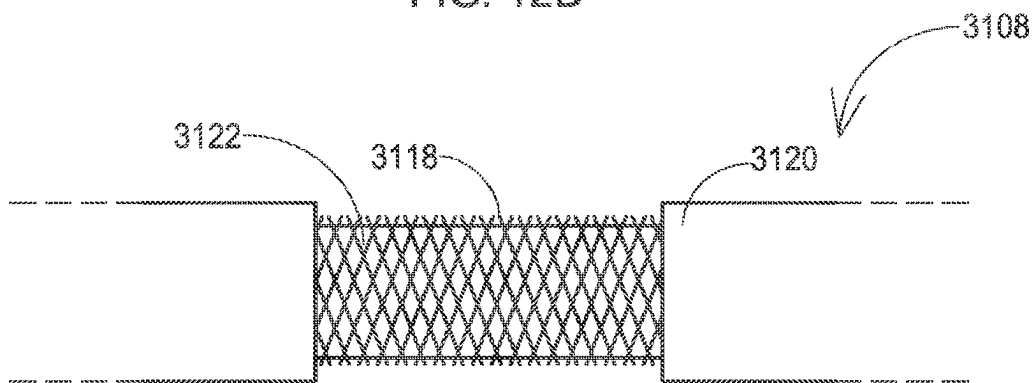


FIG. 12C

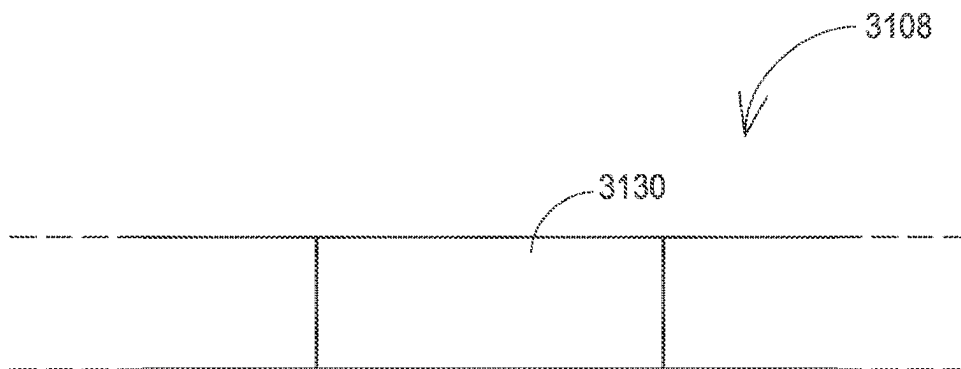


FIG. 12D

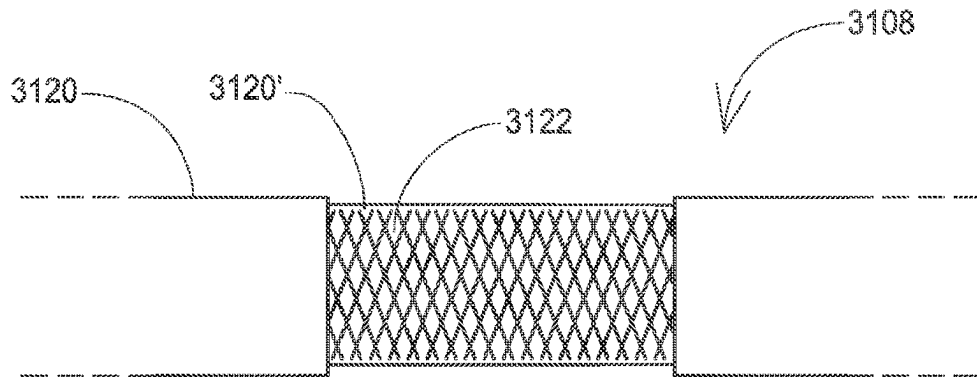


FIG. 12E

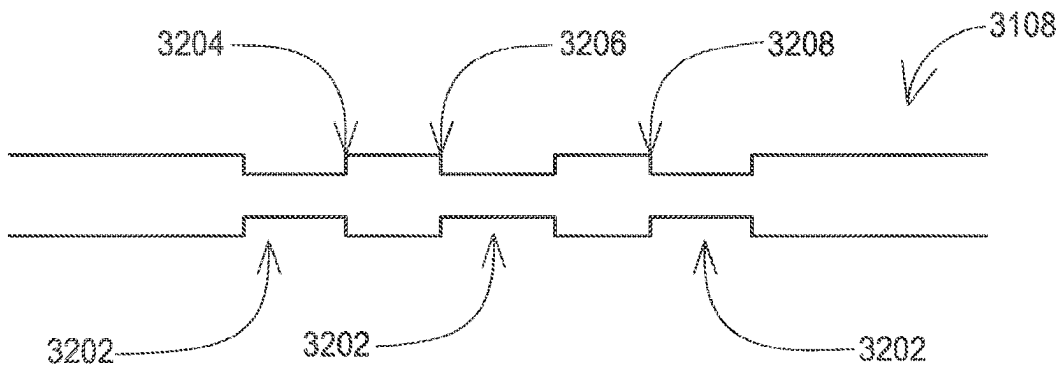


FIG. 12F

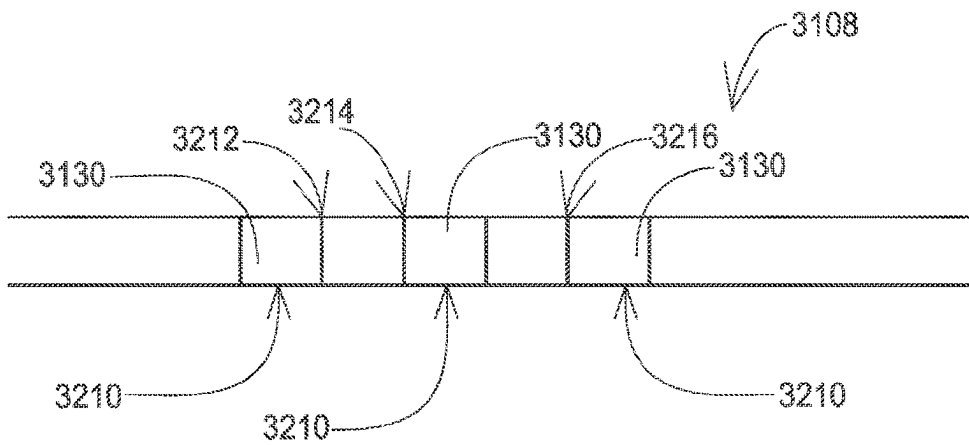


FIG. 12G

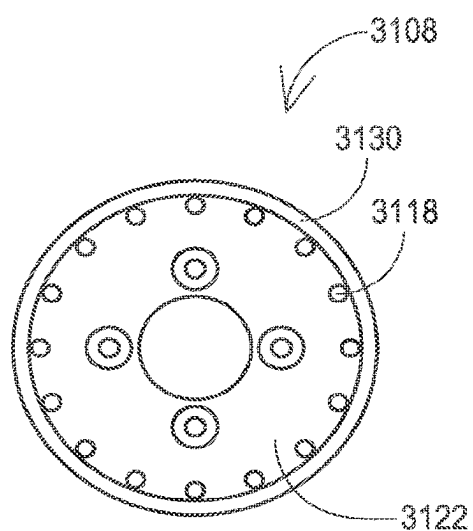


FIG. 12H

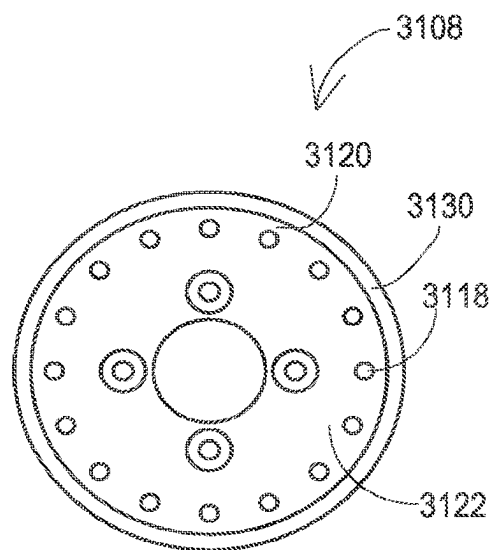


FIG. 12J

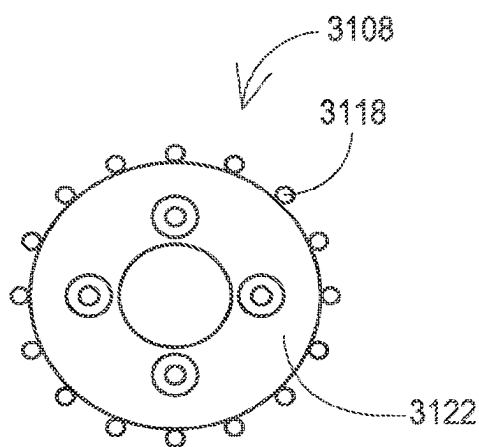


FIG. 12I

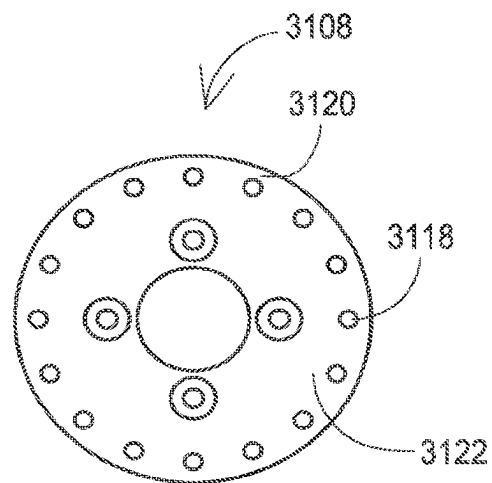


FIG. 12K

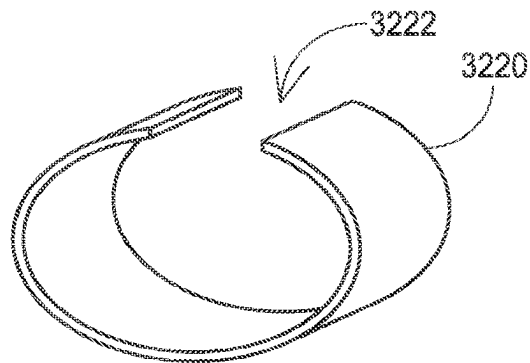


FIG. 13A

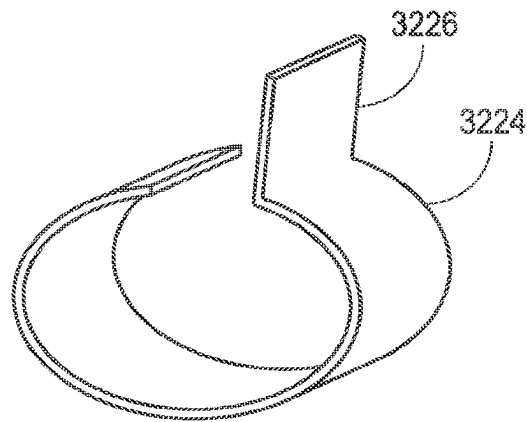


FIG. 13B

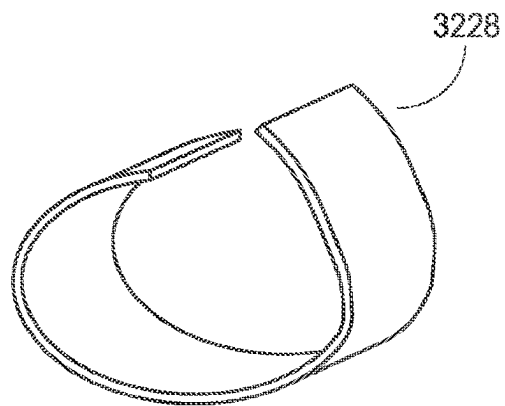
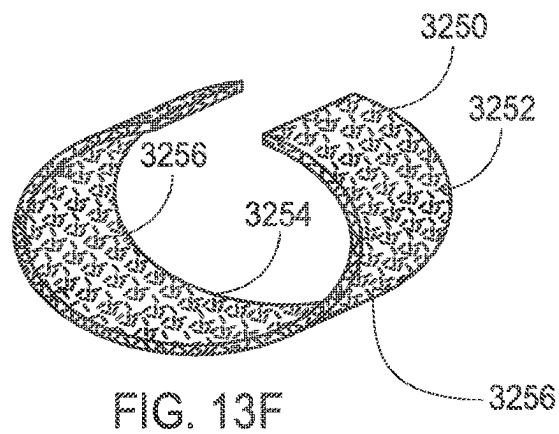
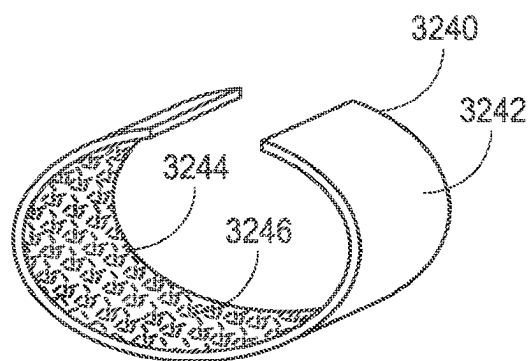
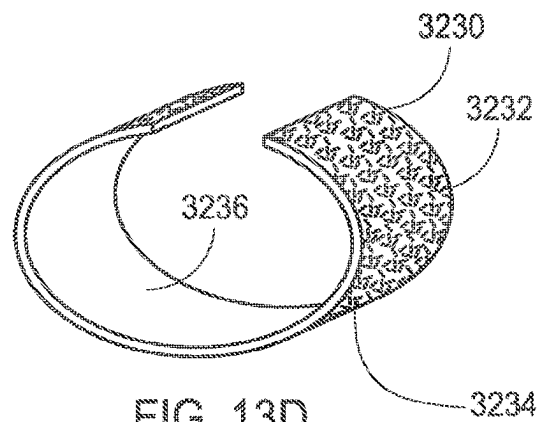
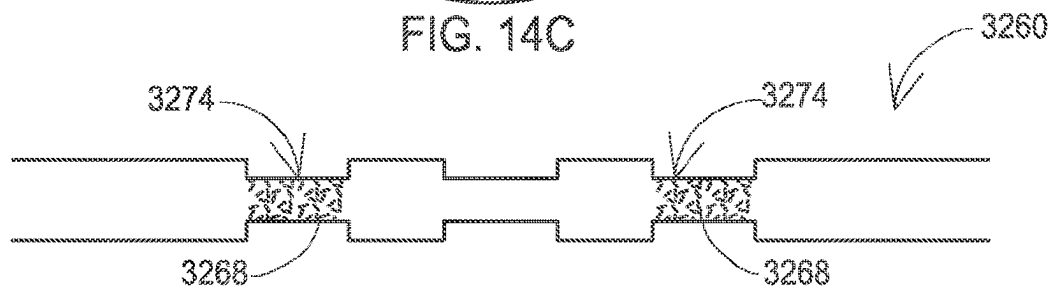
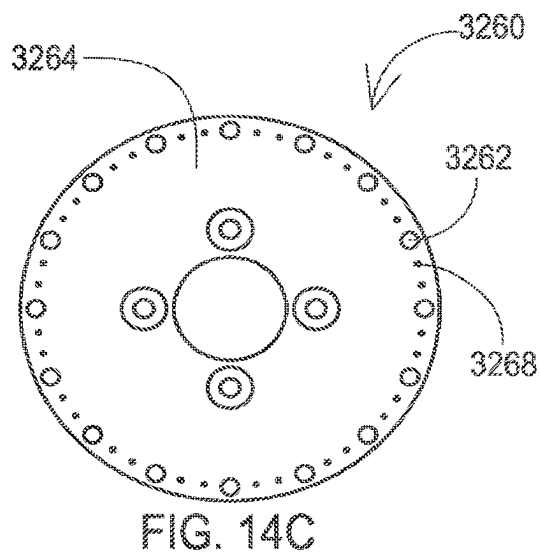
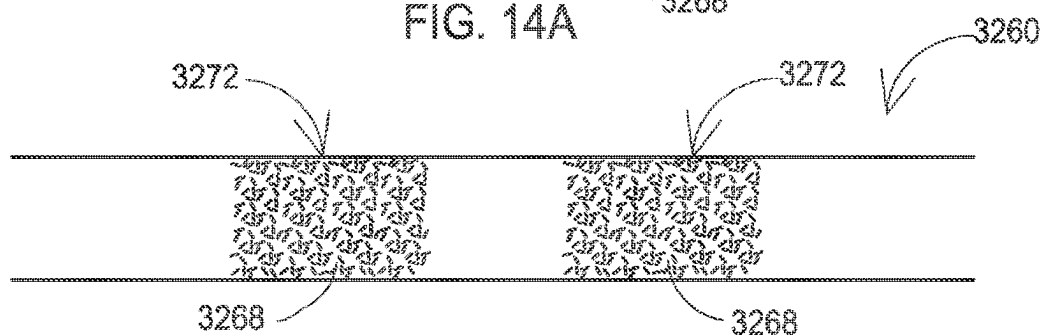
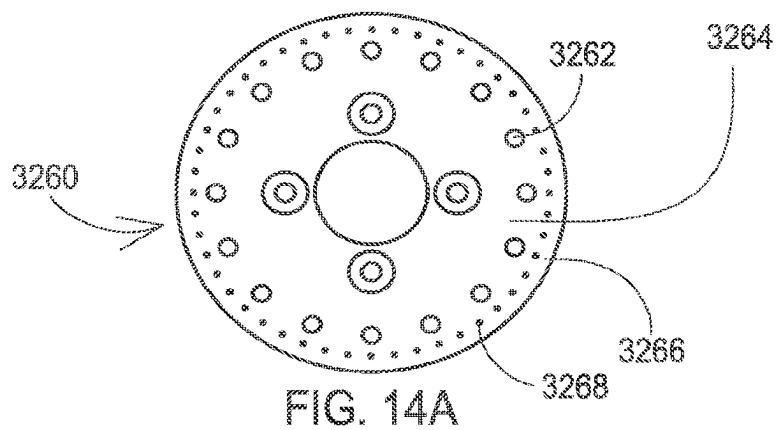


FIG. 13C





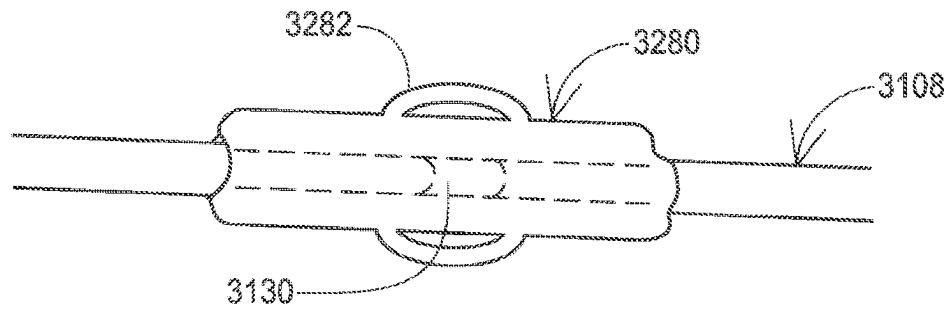


FIG. 15A

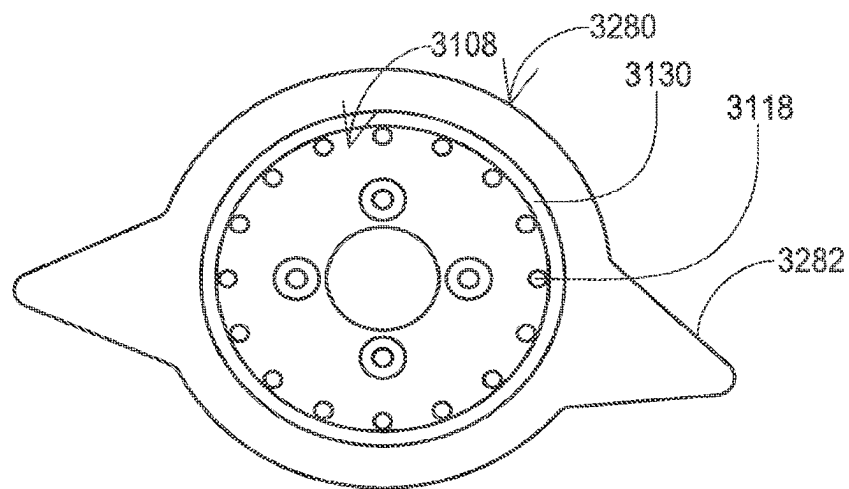


FIG. 15B

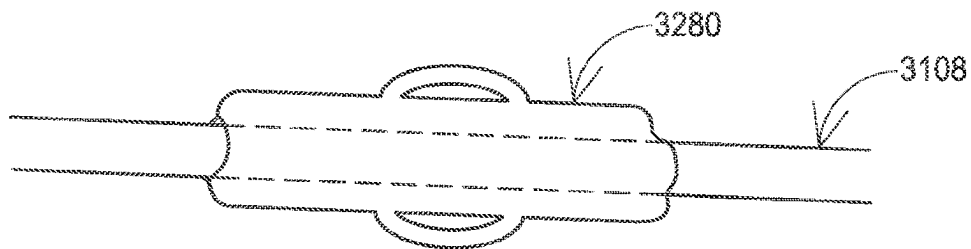


FIG. 15C

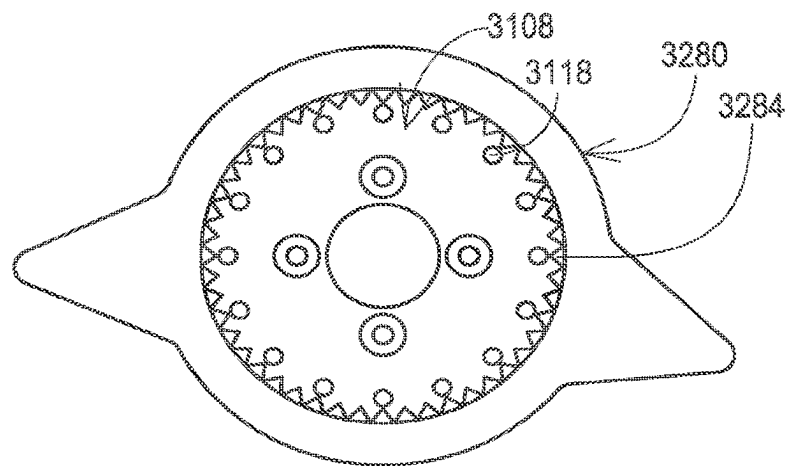


FIG. 15D

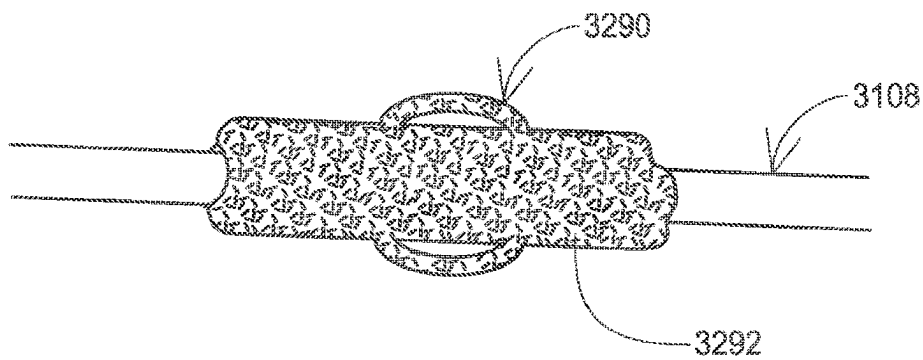


FIG. 15E

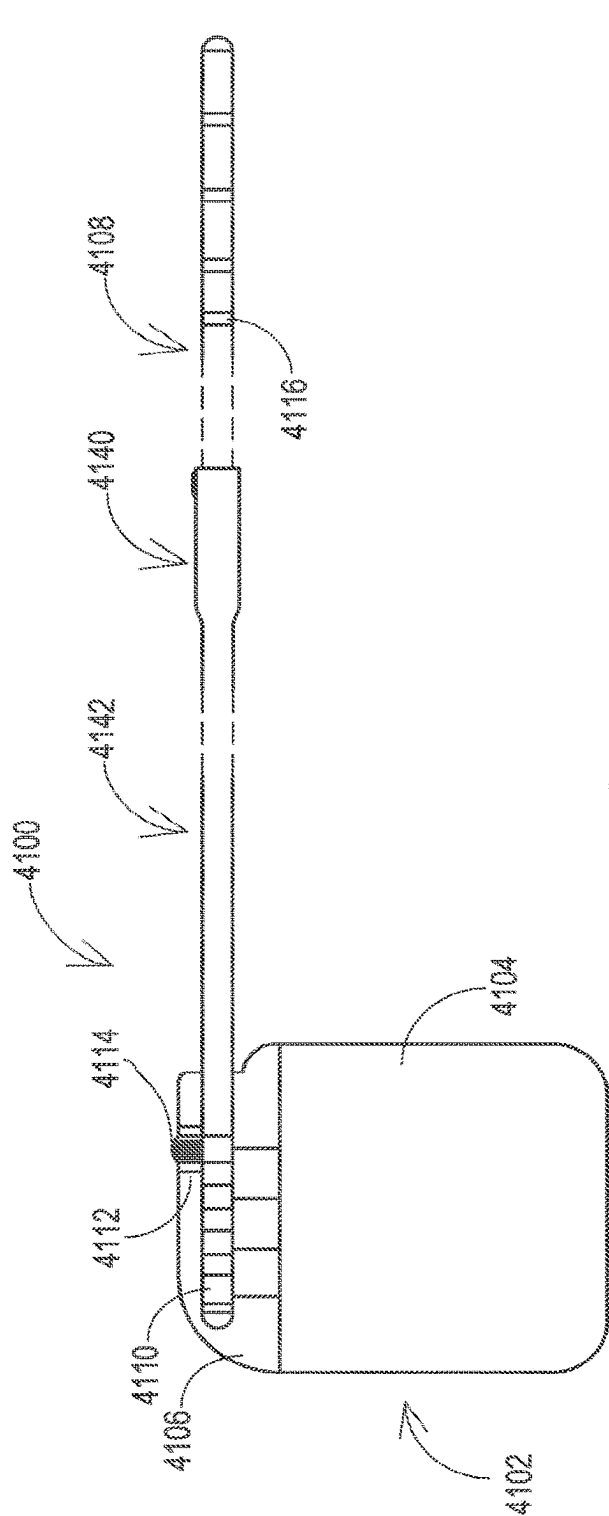


FIG. 16

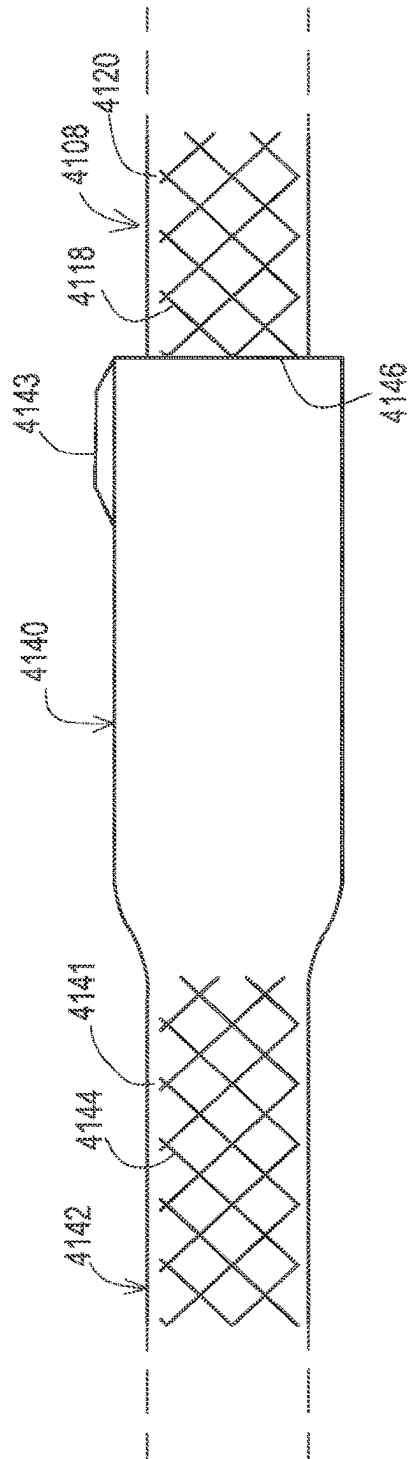


FIG. 17A

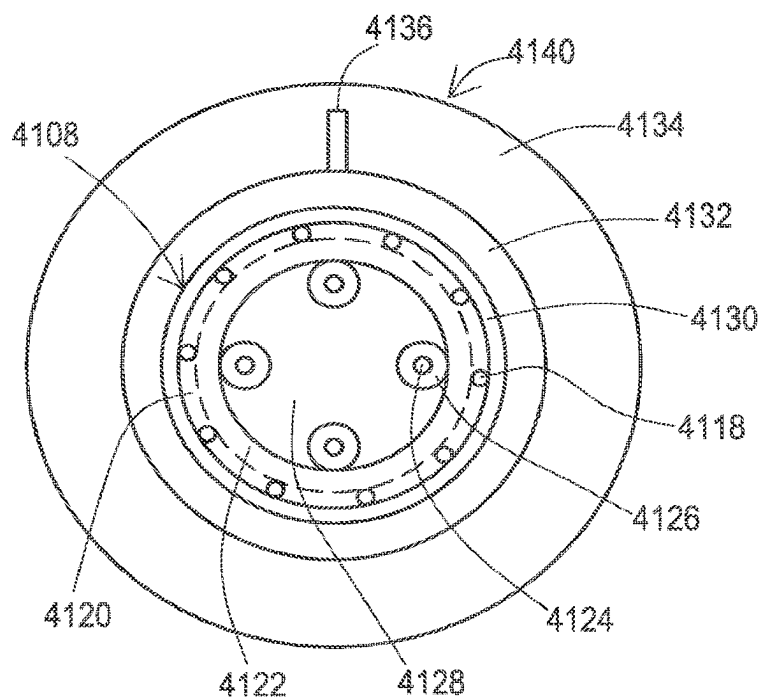


FIG. 17B

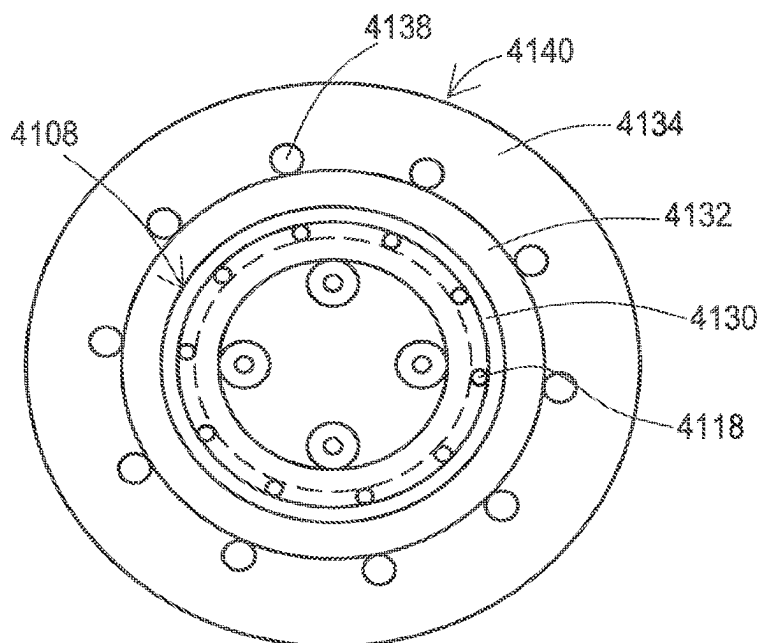


FIG. 17C

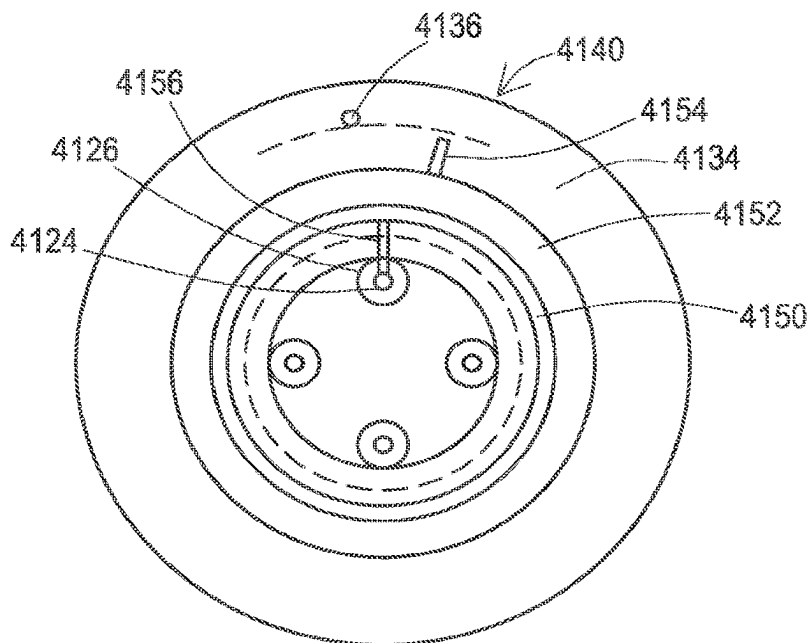


FIG. 17D

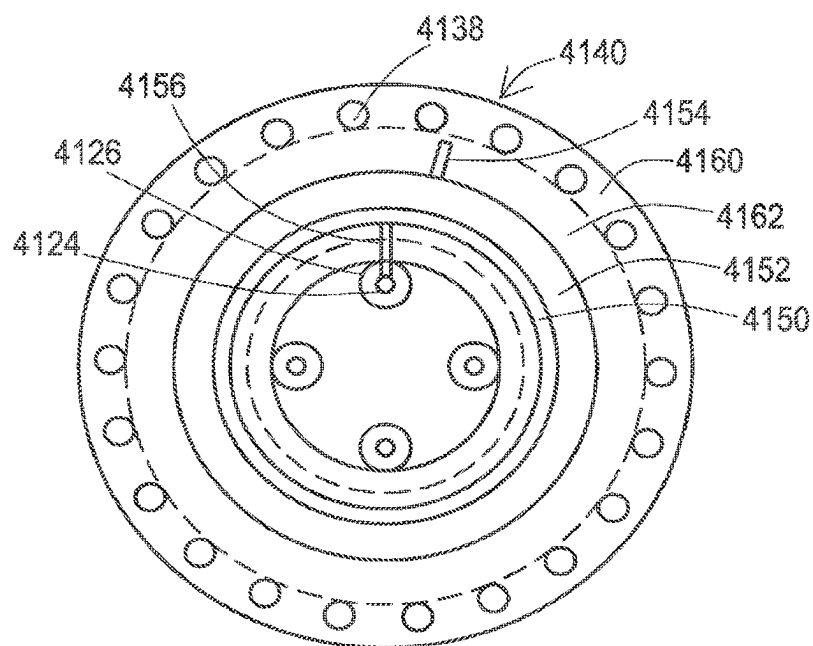


FIG. 17E

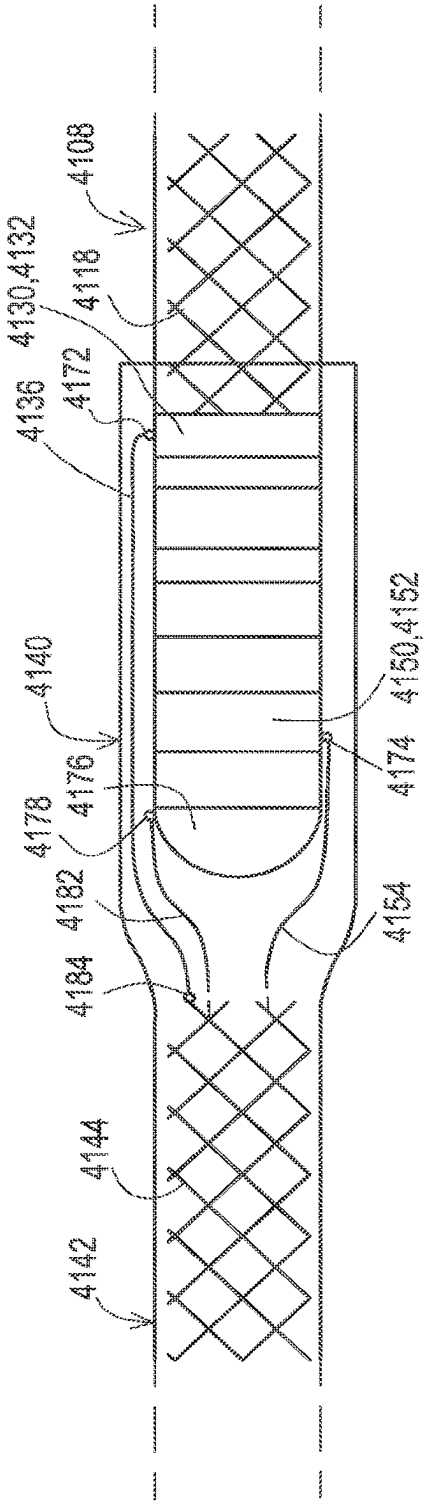


FIG. 17F

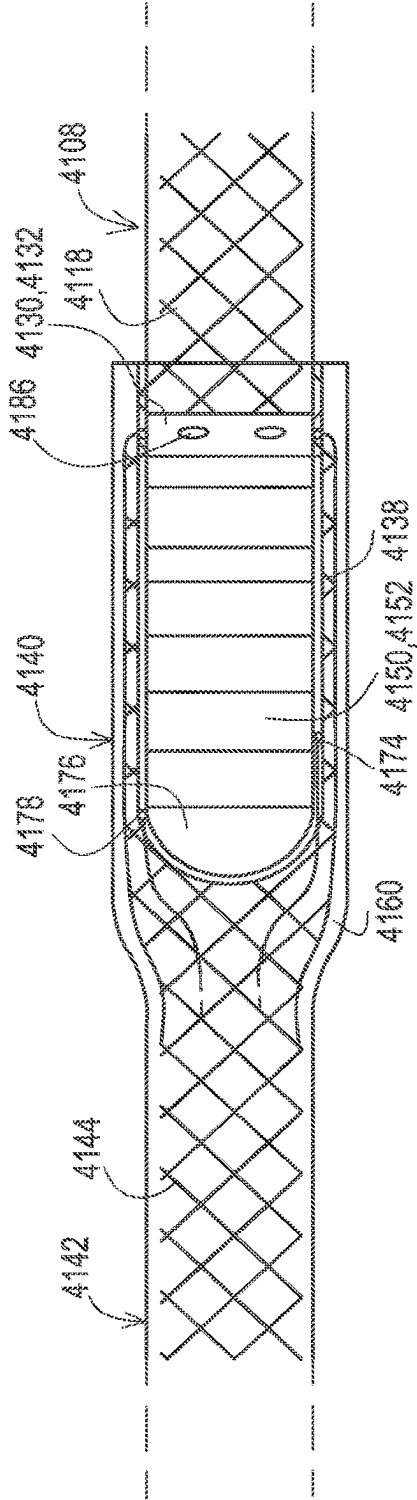


FIG. 17G

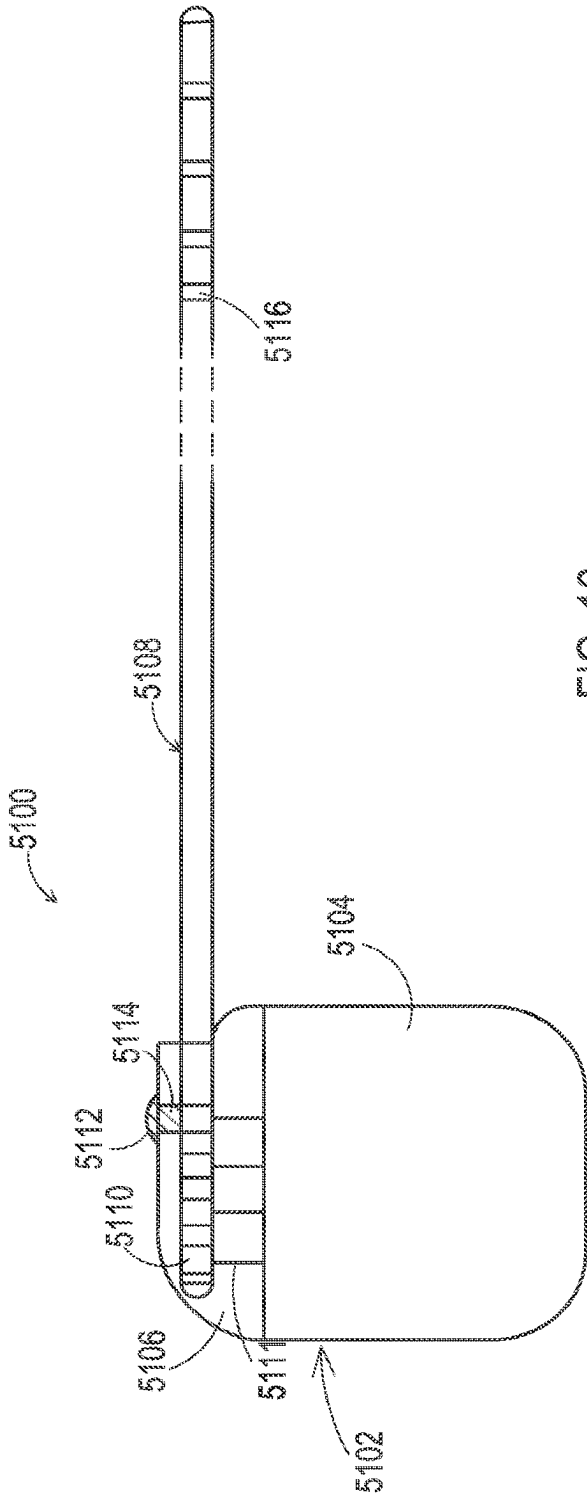


FIG. 18

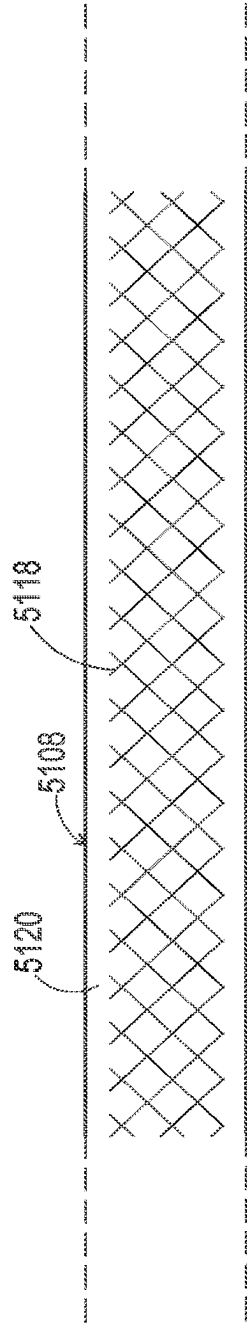


FIG. 19A

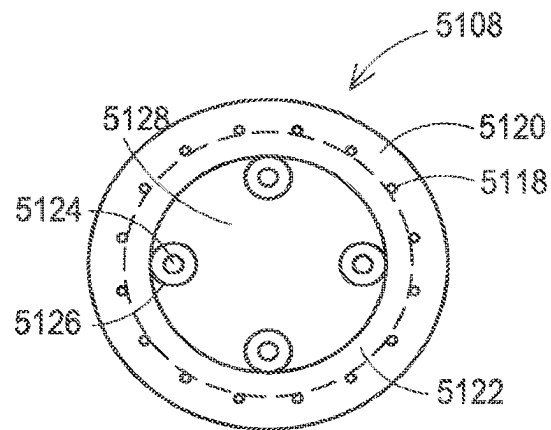
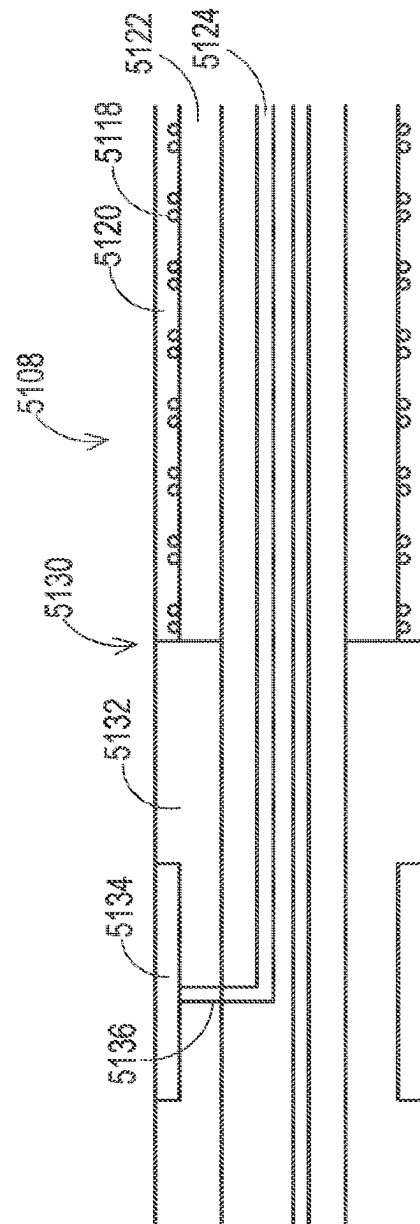


FIG. 19B



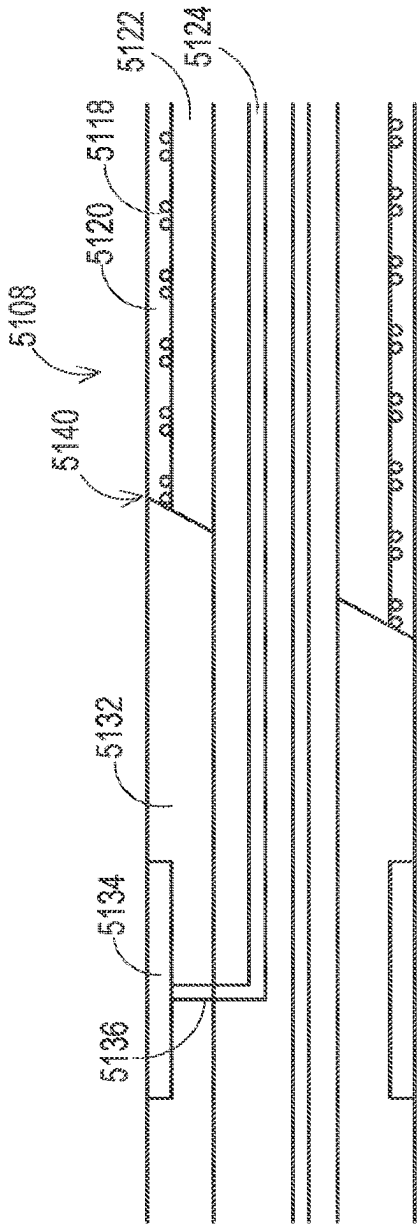


FIG. 21

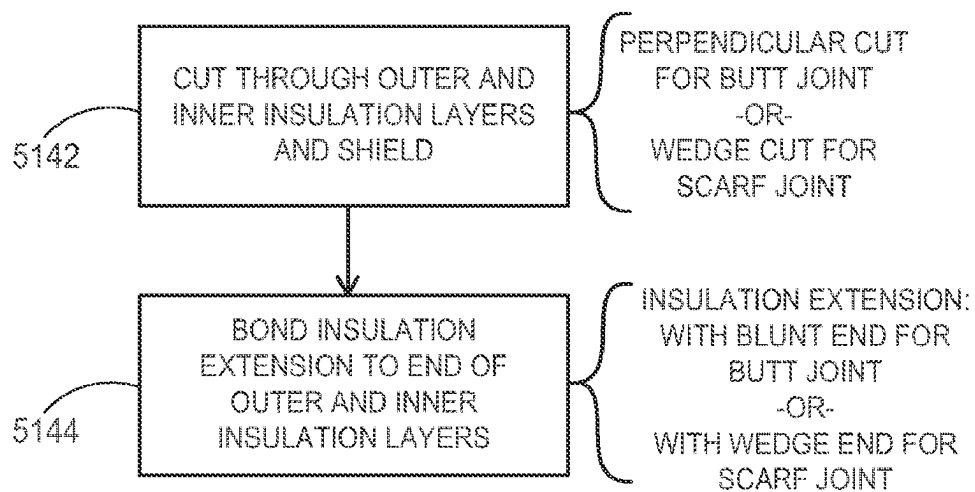
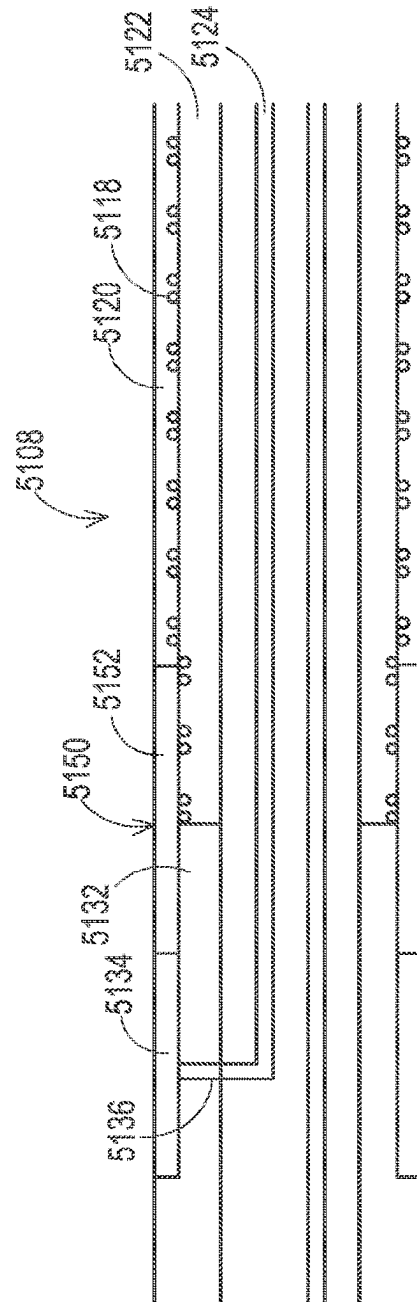


FIG. 22



2
3
4

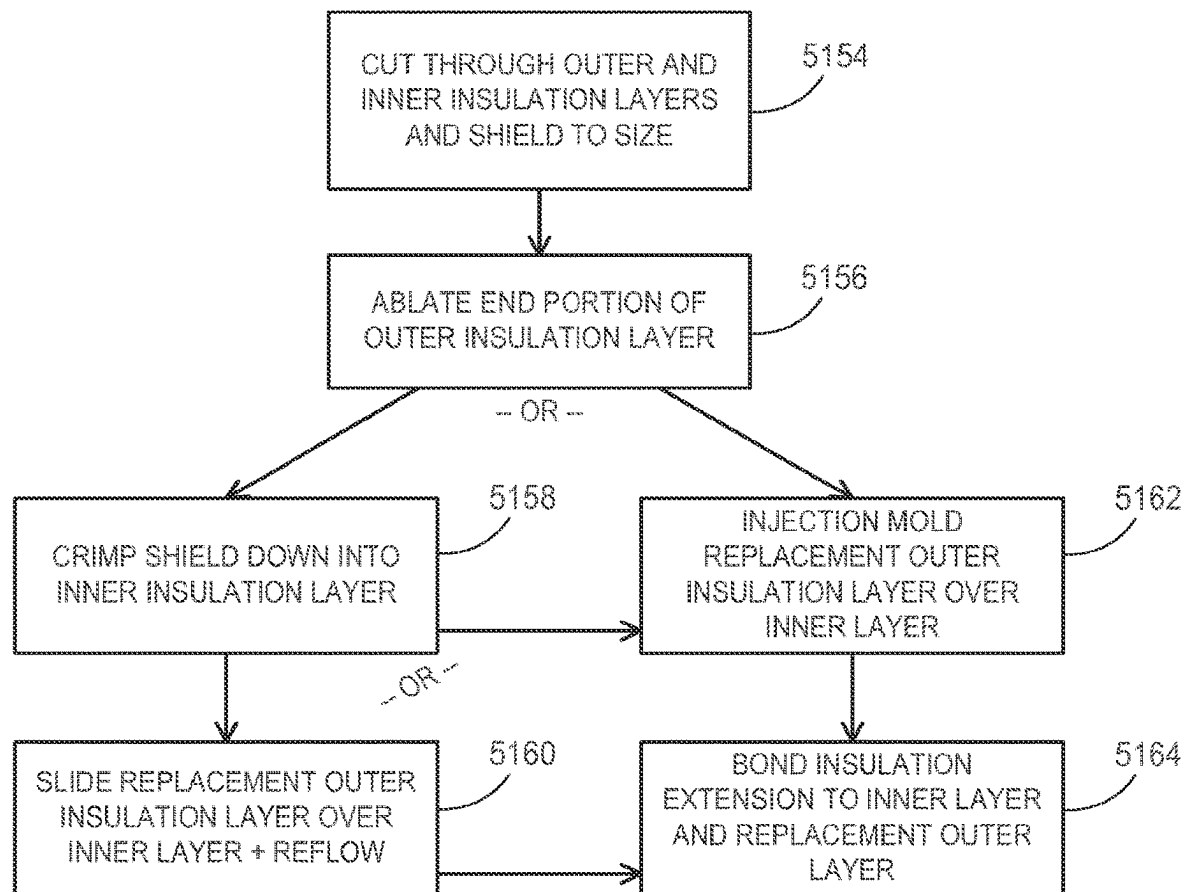


FIG. 24

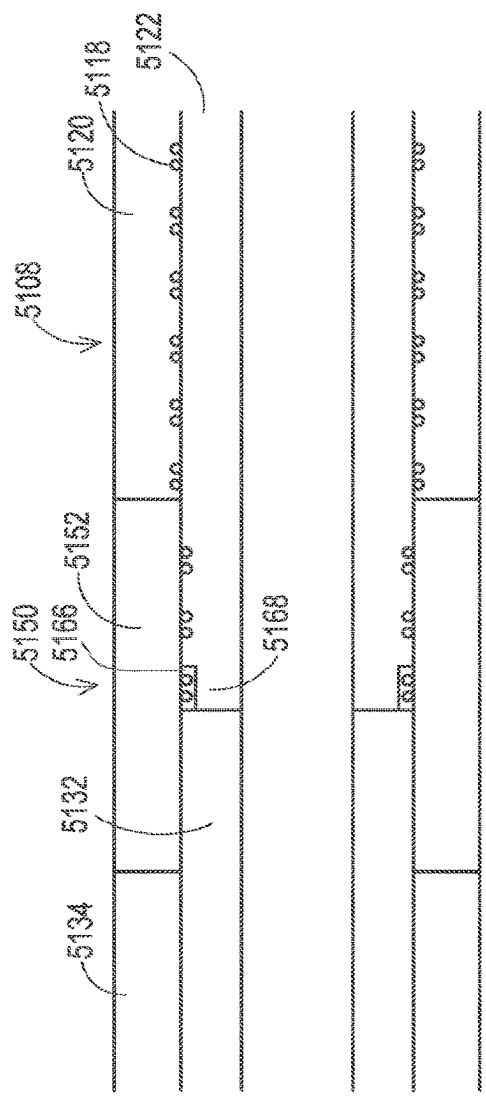


FIG. 25

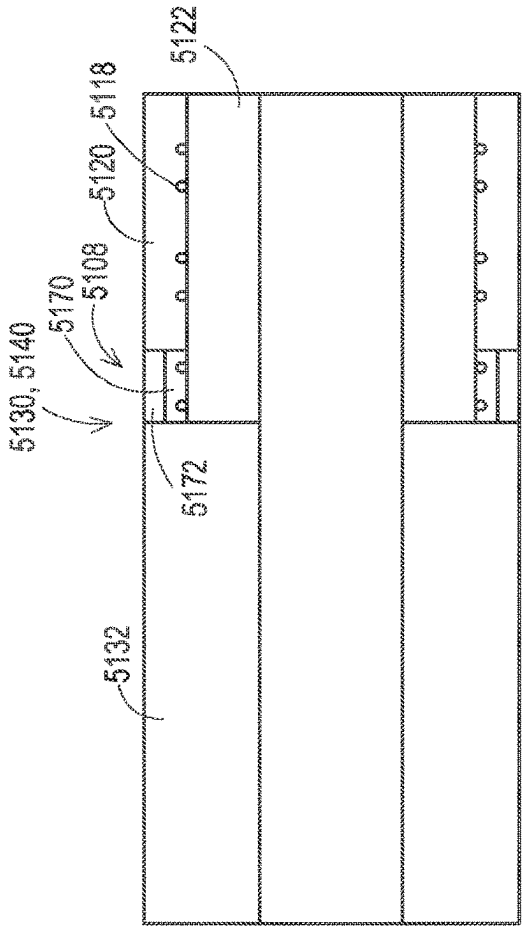


FIG. 26

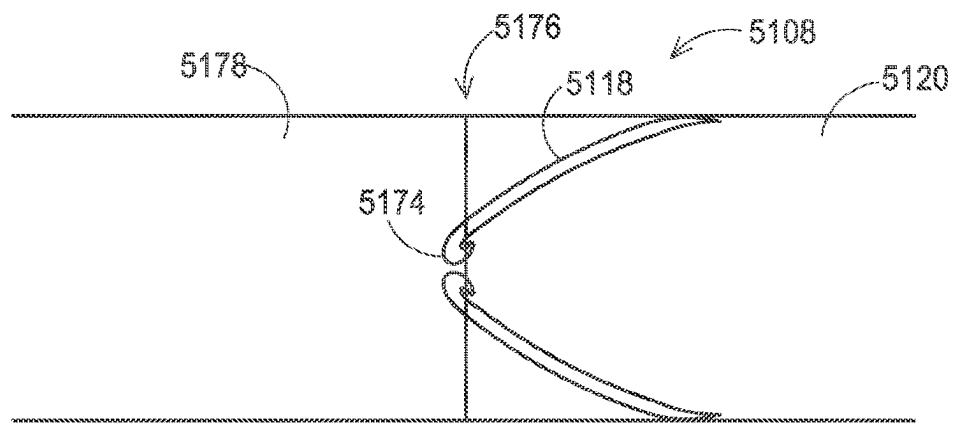


FIG. 27

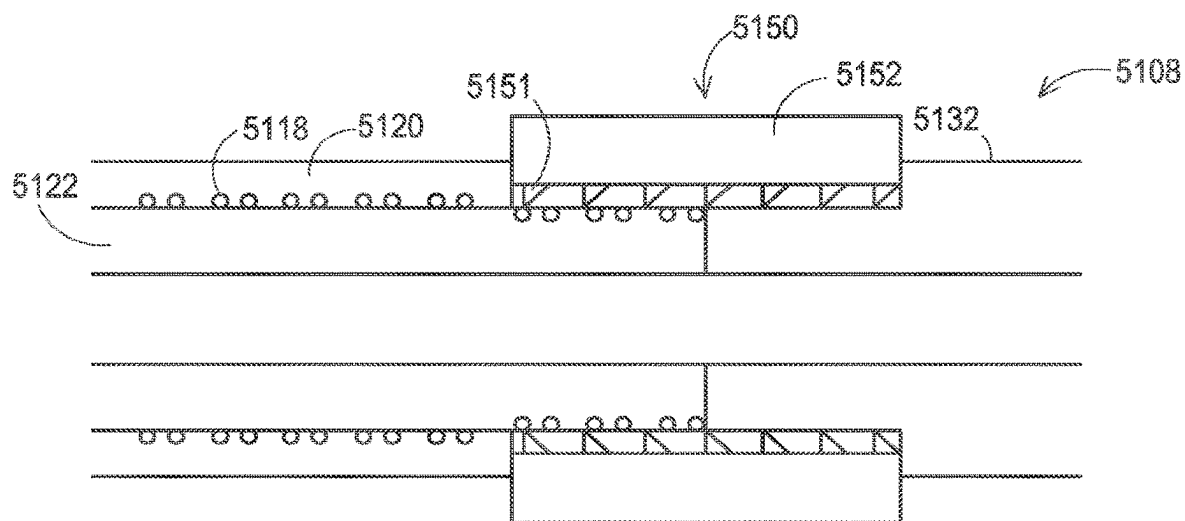


FIG. 28

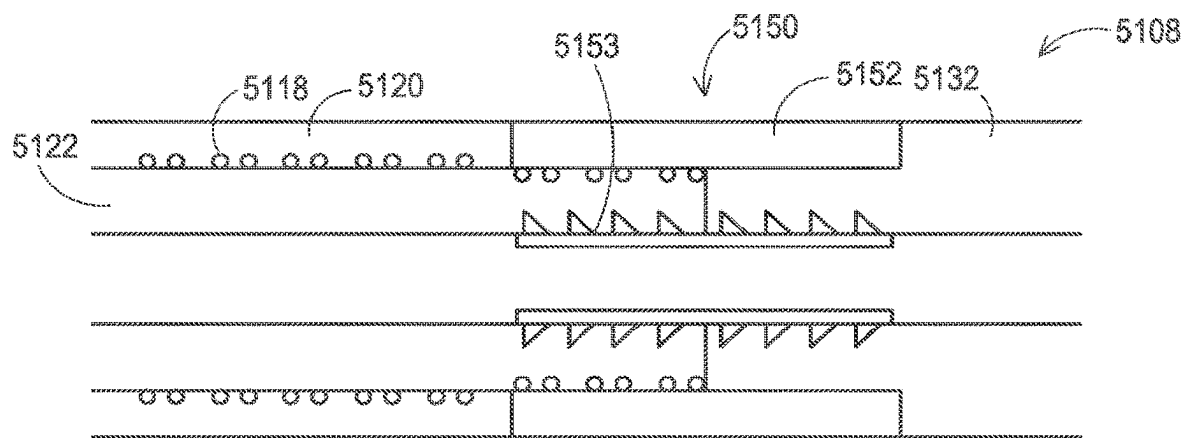


FIG. 29

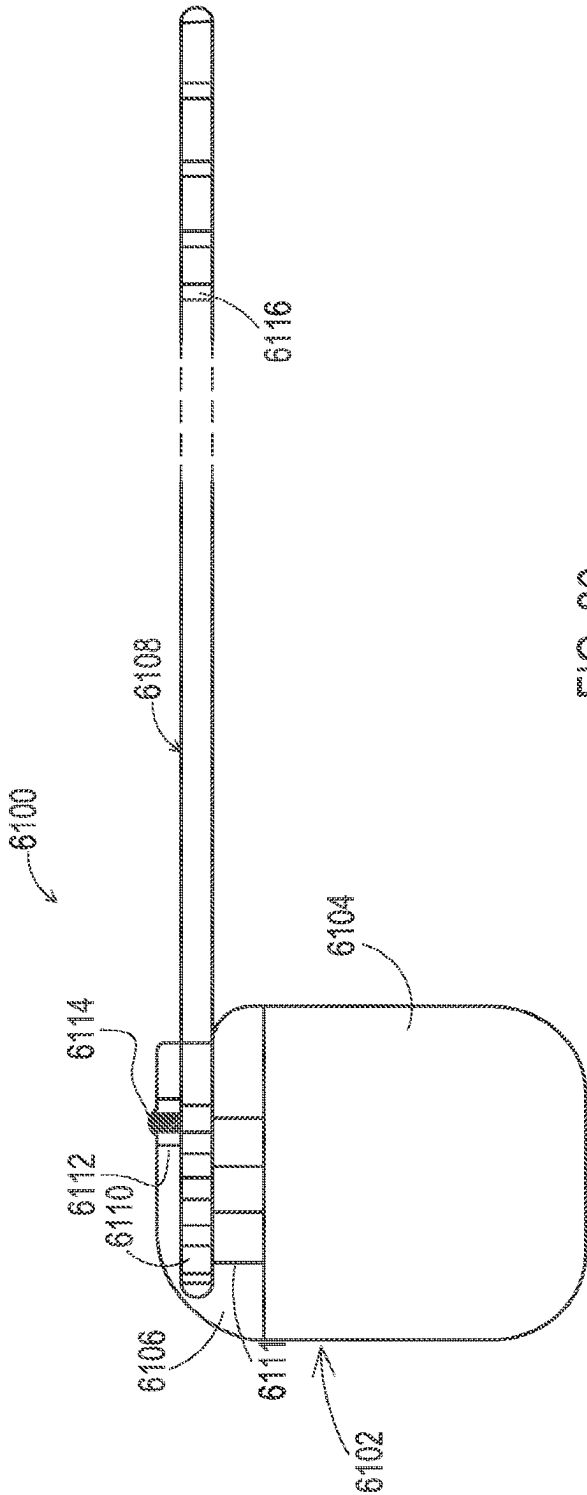


FIG. 30

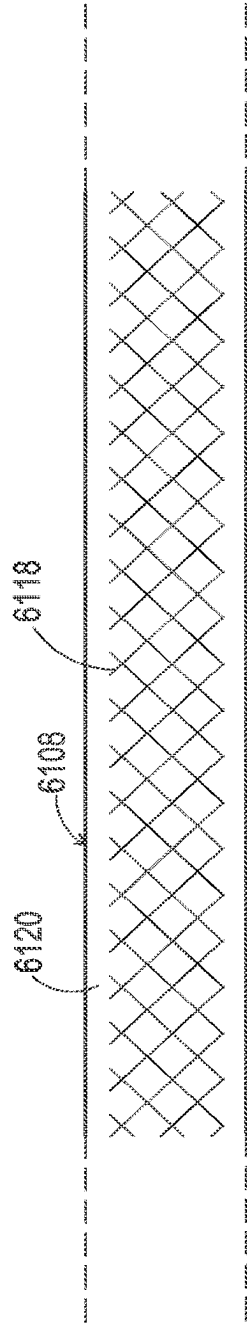


FIG. 31A

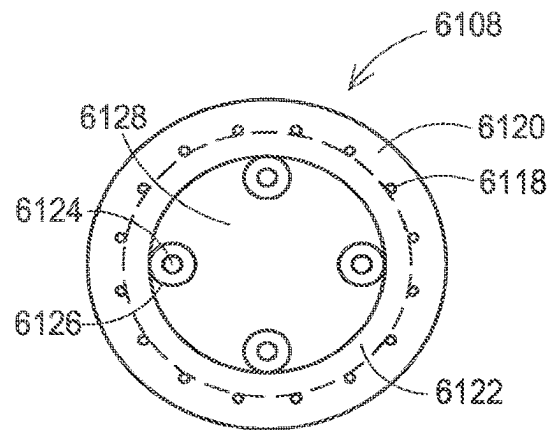


FIG. 31B

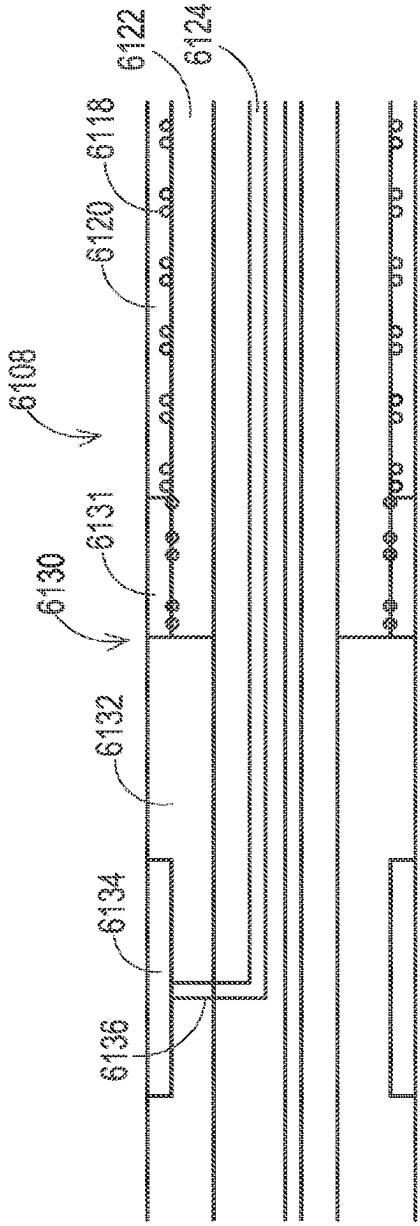
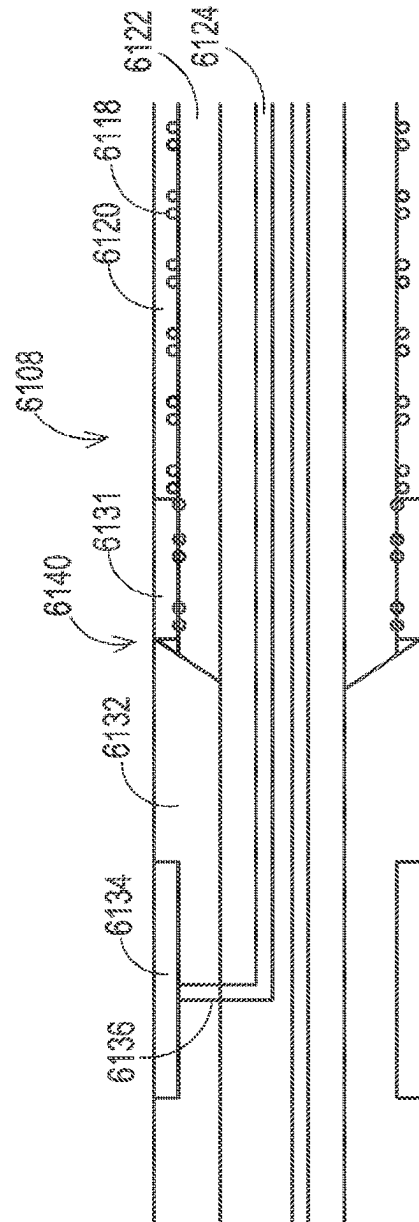


FIG. 32



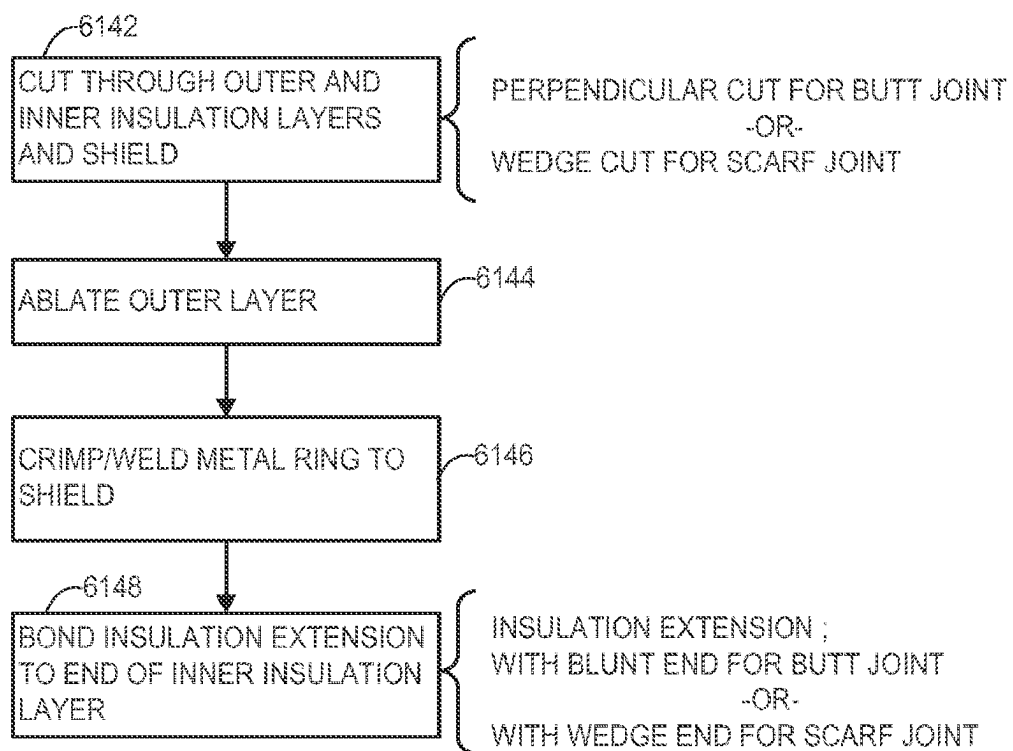


FIG. 34

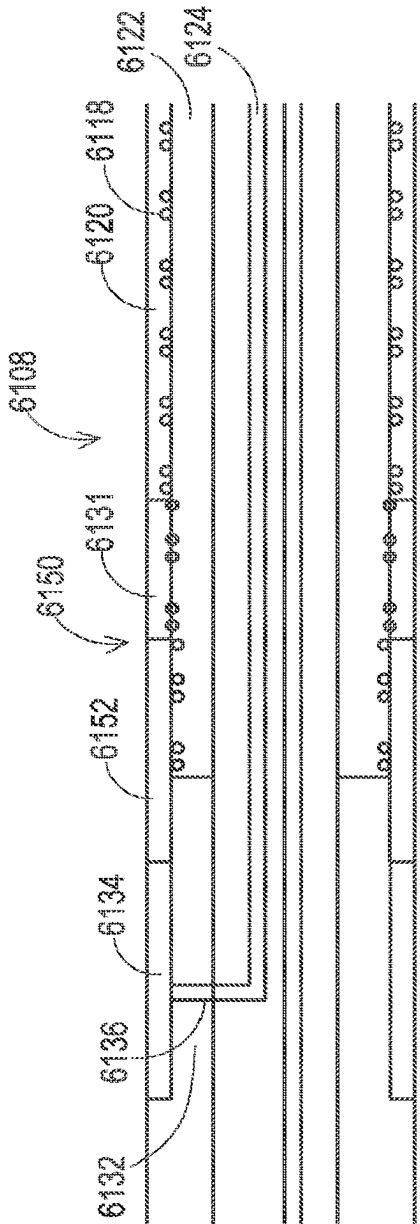


FIG. 35

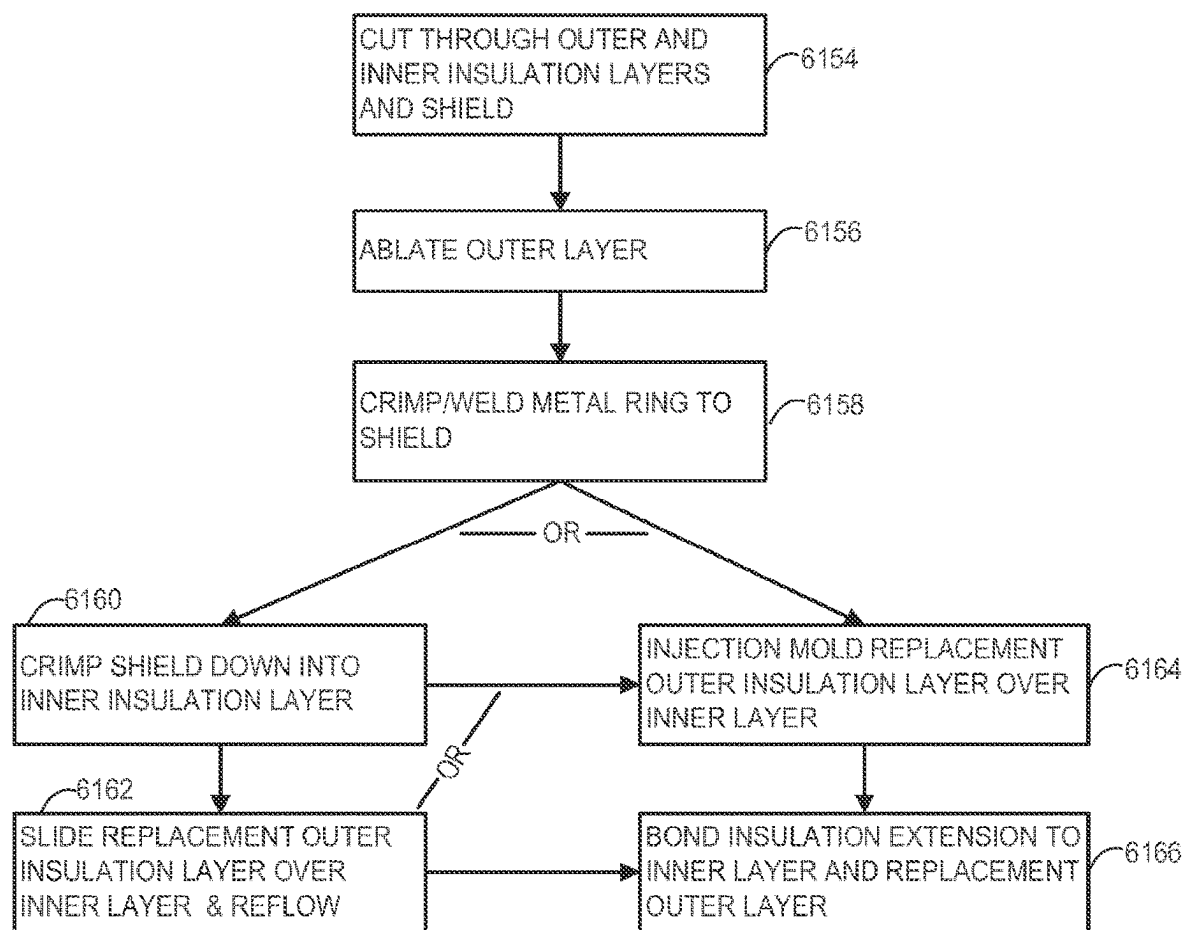


FIG. 36

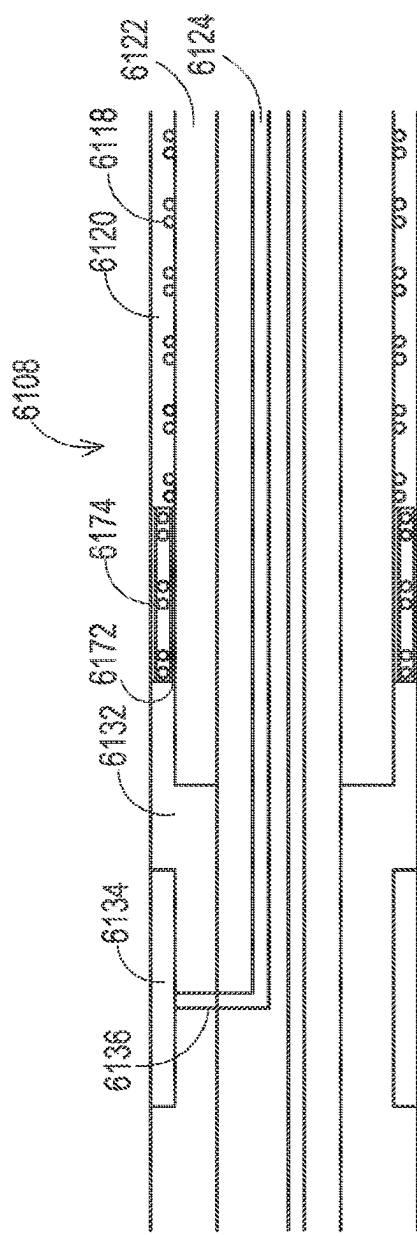


FIG. 37

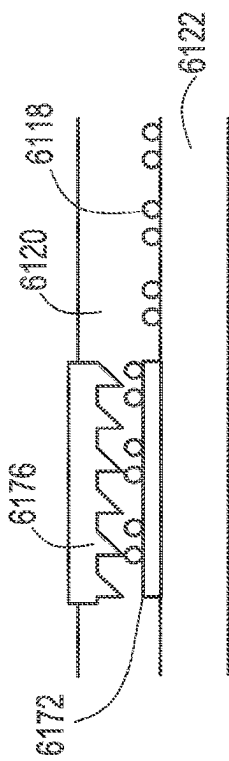


FIG. 38

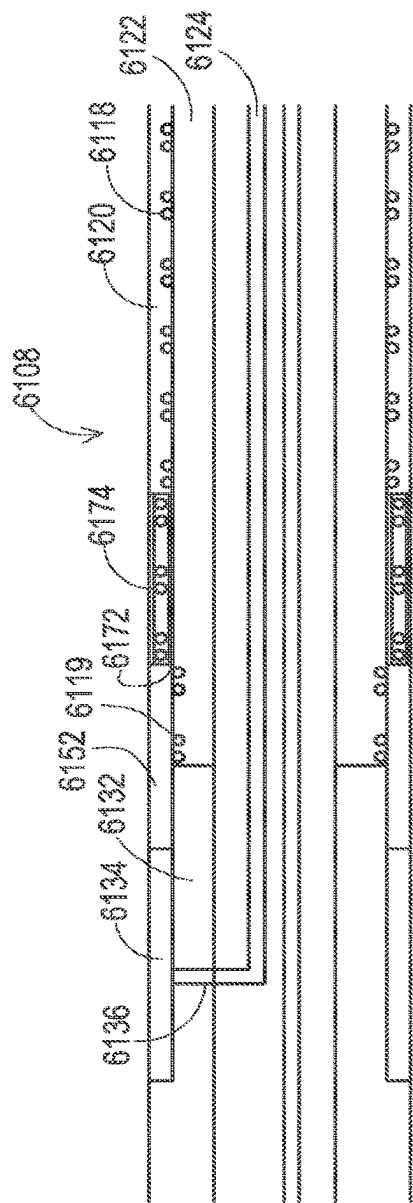


FIG. 39

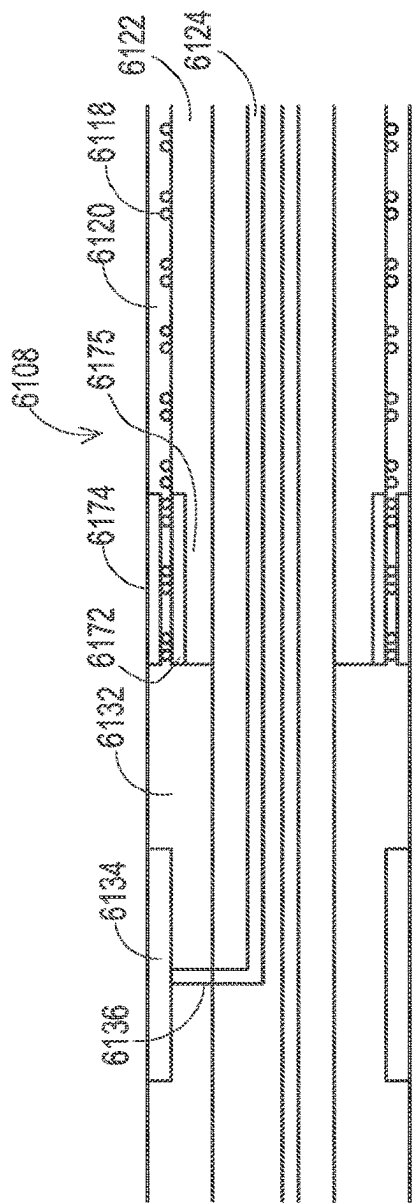


FIG. 40

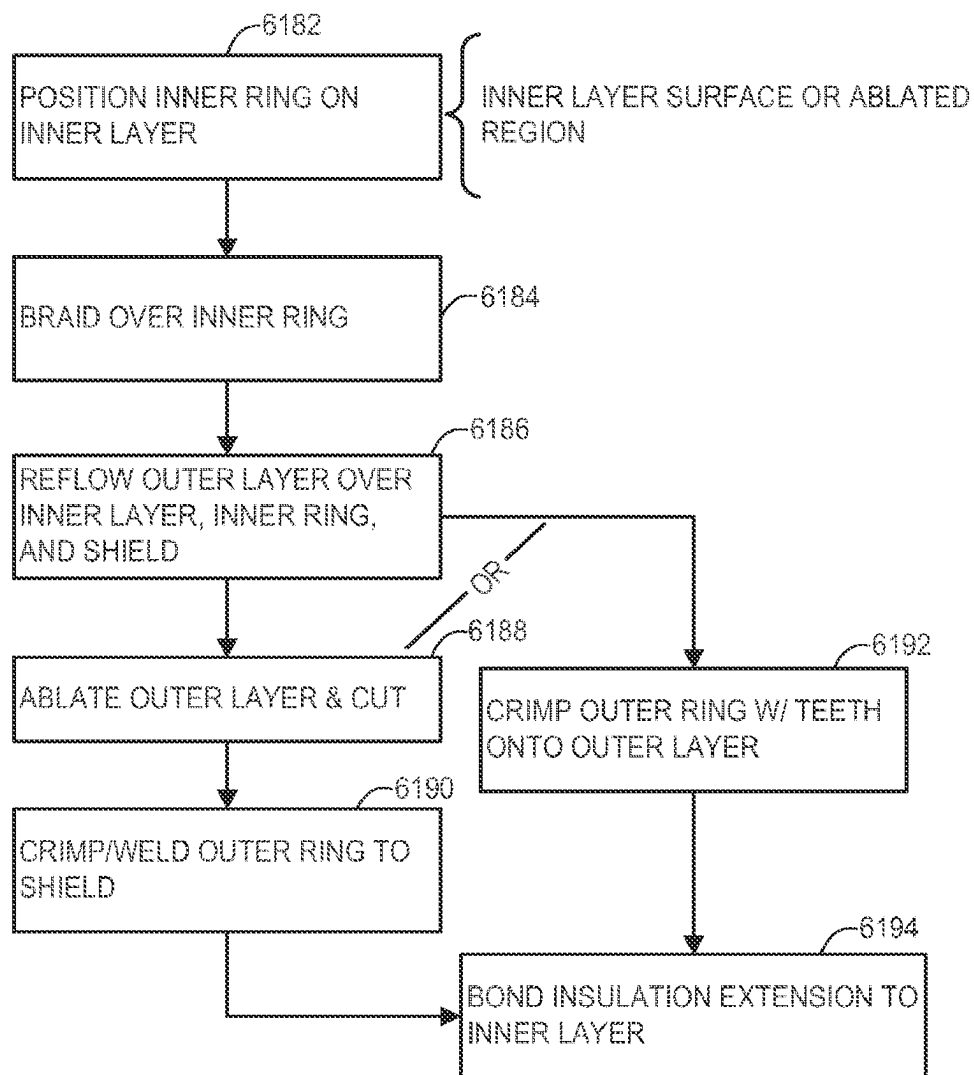
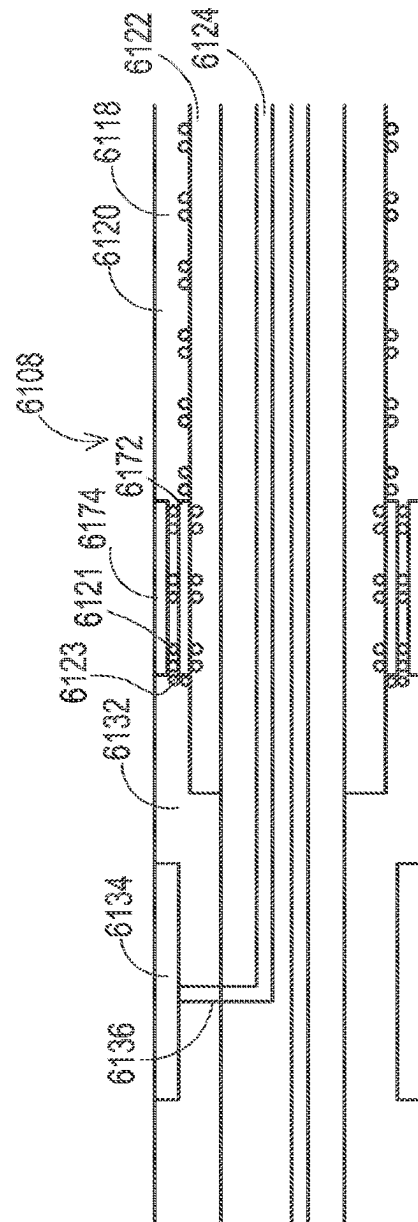


FIG. 41



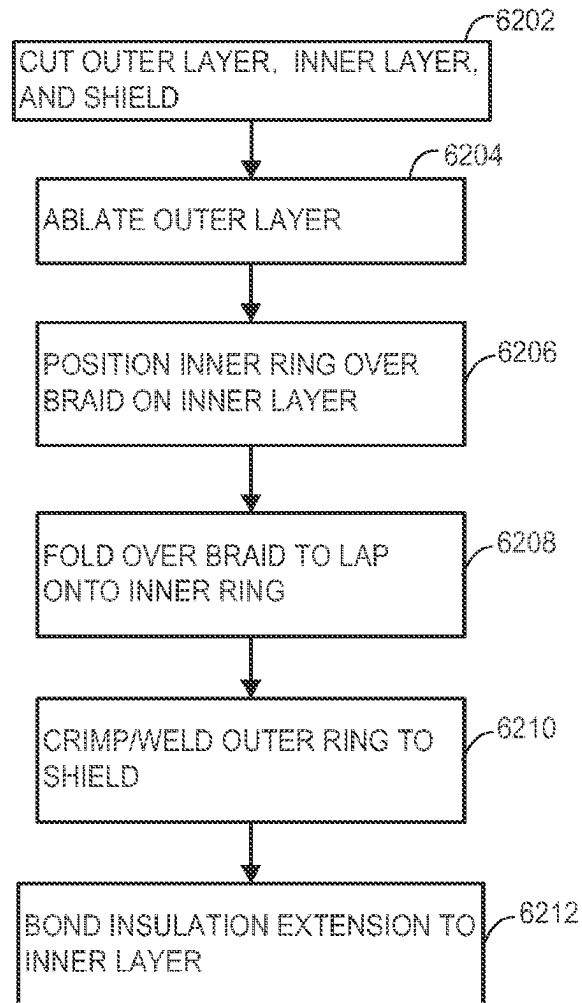


FIG. 43

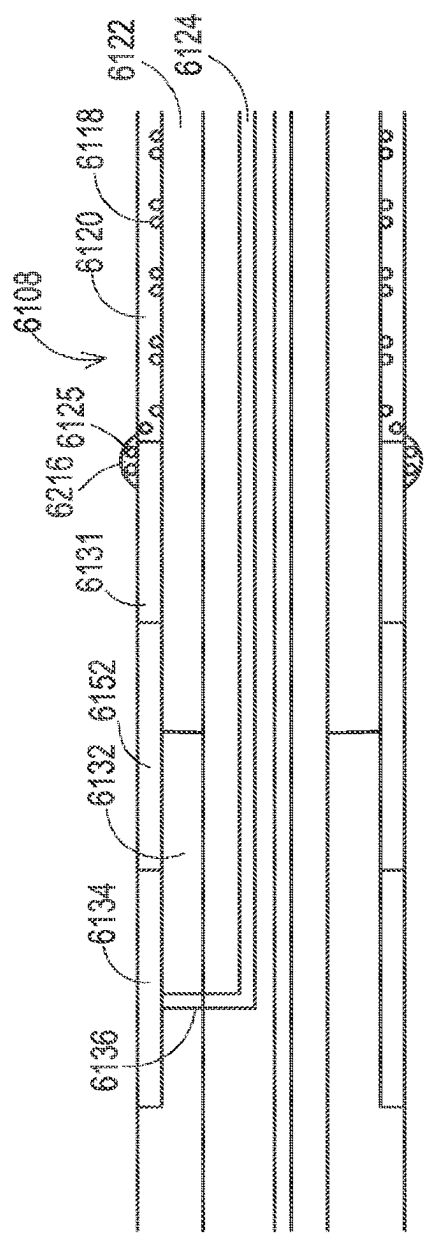


FIG. 44

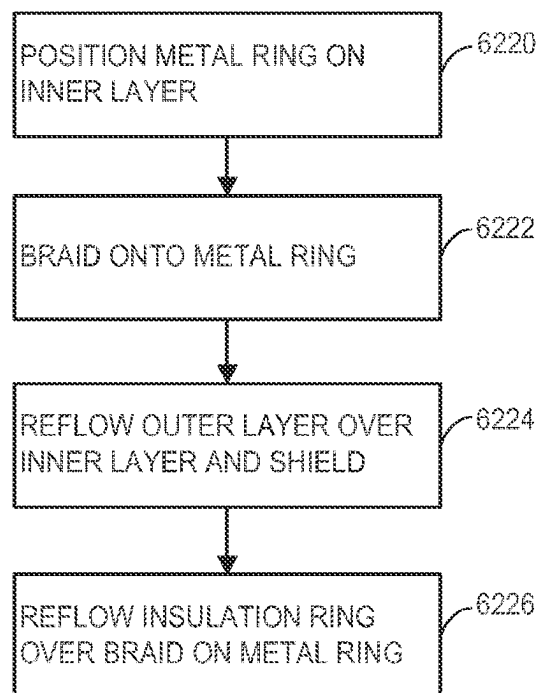


FIG. 45

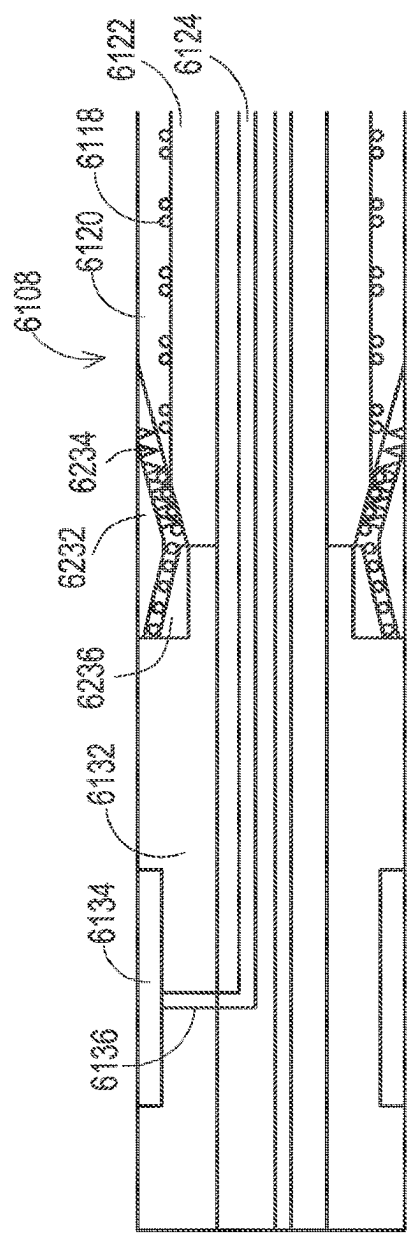


FIG. 46

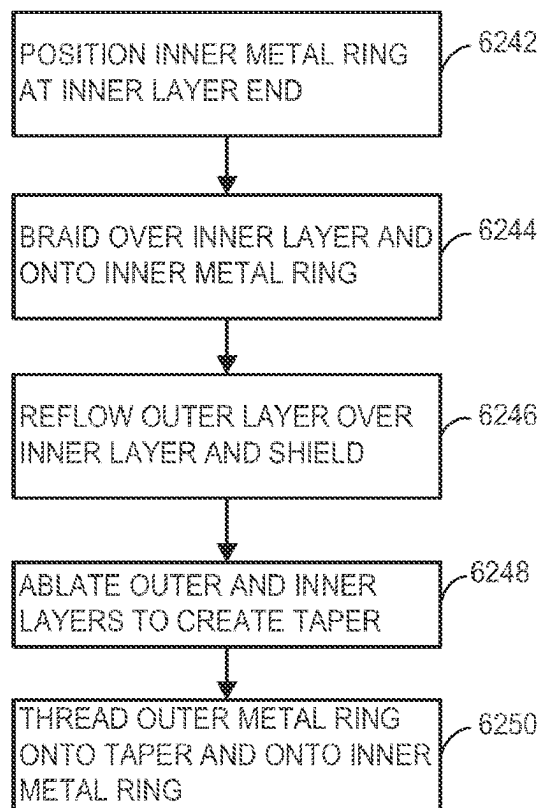


FIG. 47

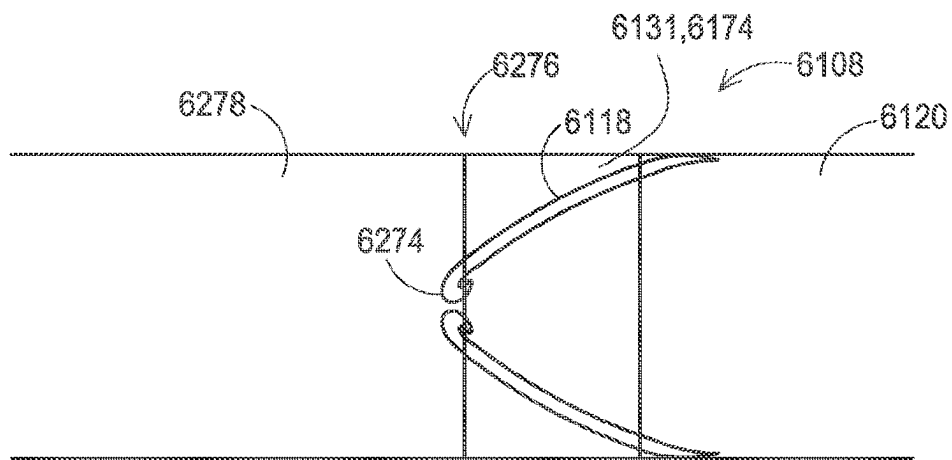
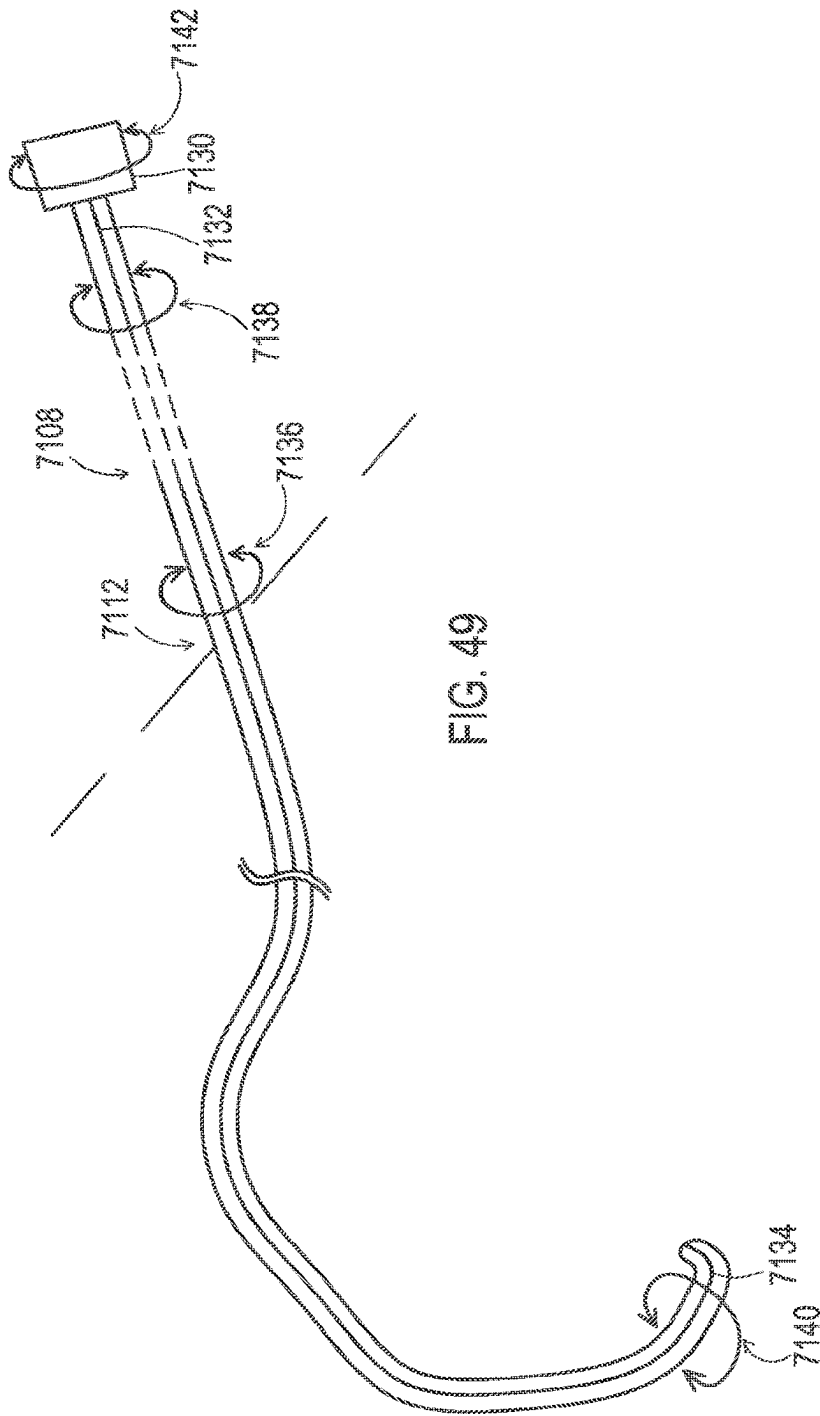


FIG. 48



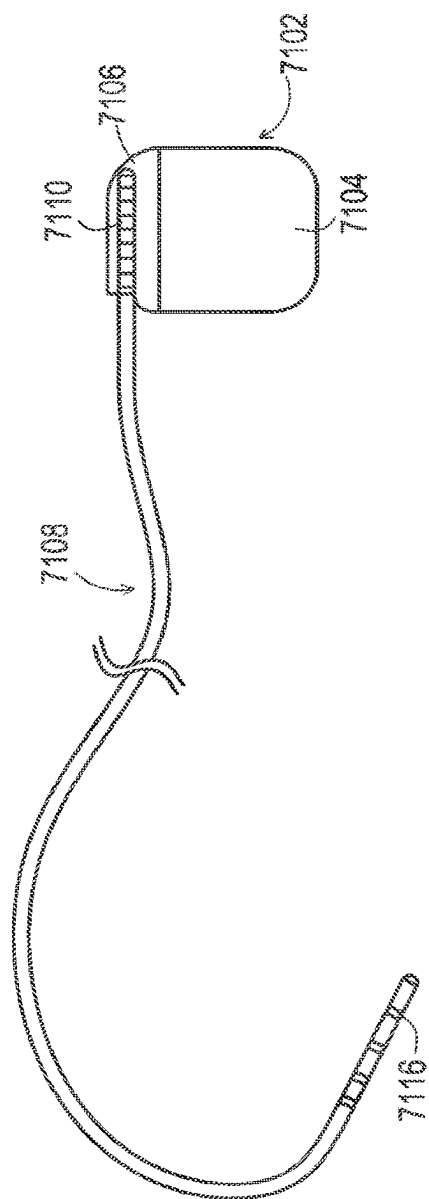


FIG. 50

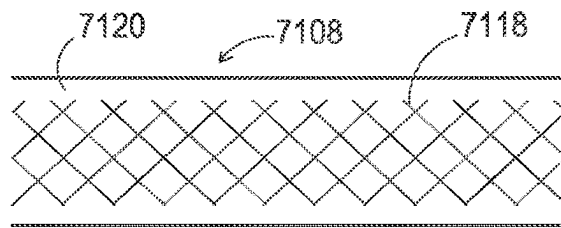


FIG. 51

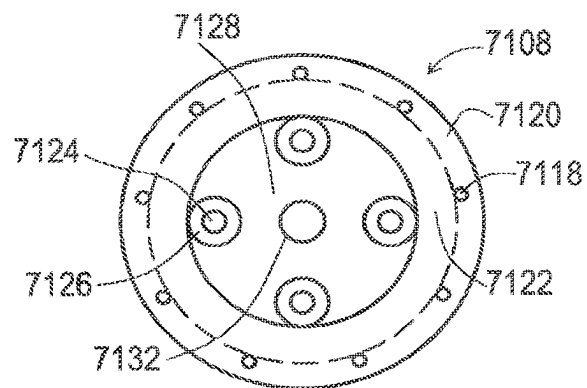


FIG. 52

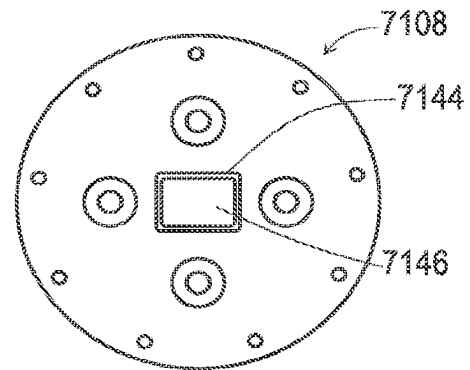


FIG. 53

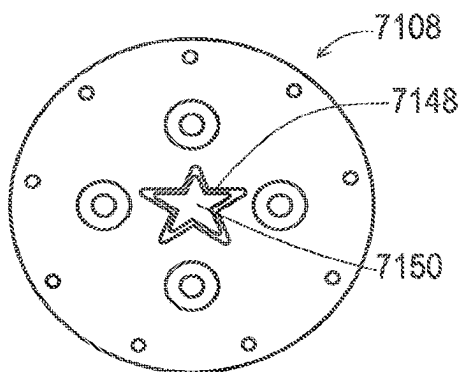


FIG. 54

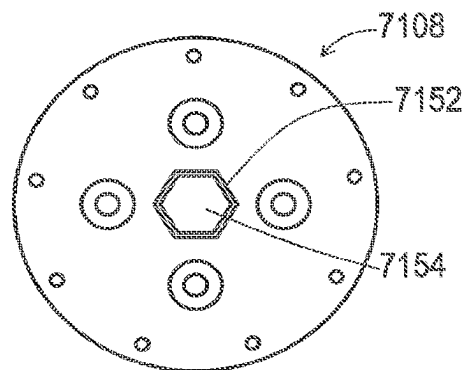


FIG. 55

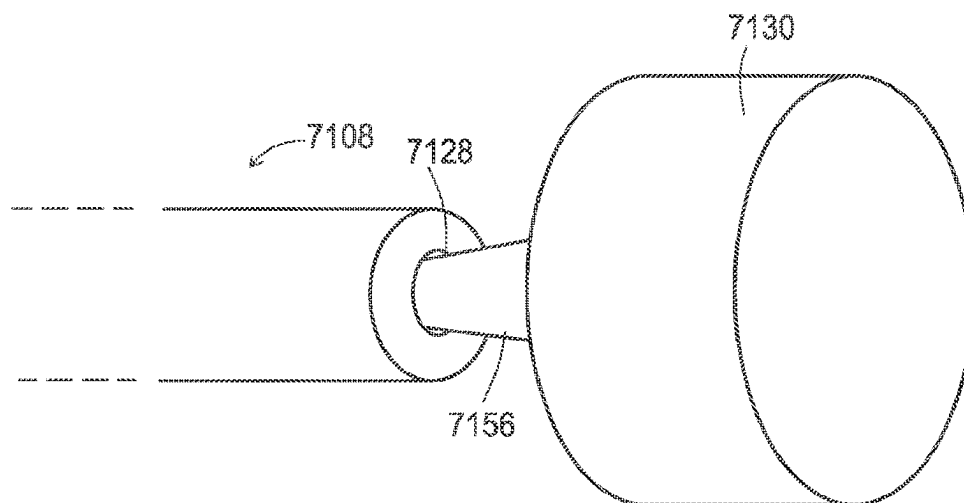


FIG. 56

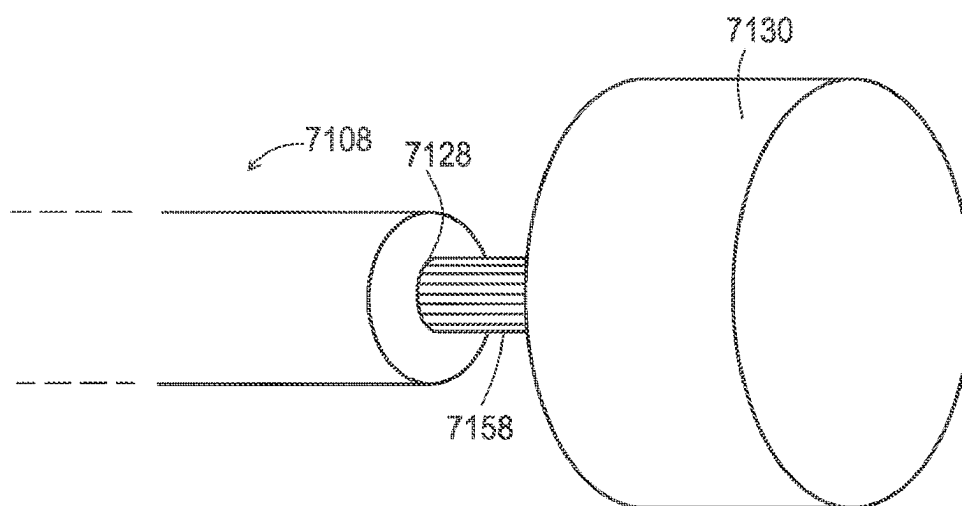


FIG. 57

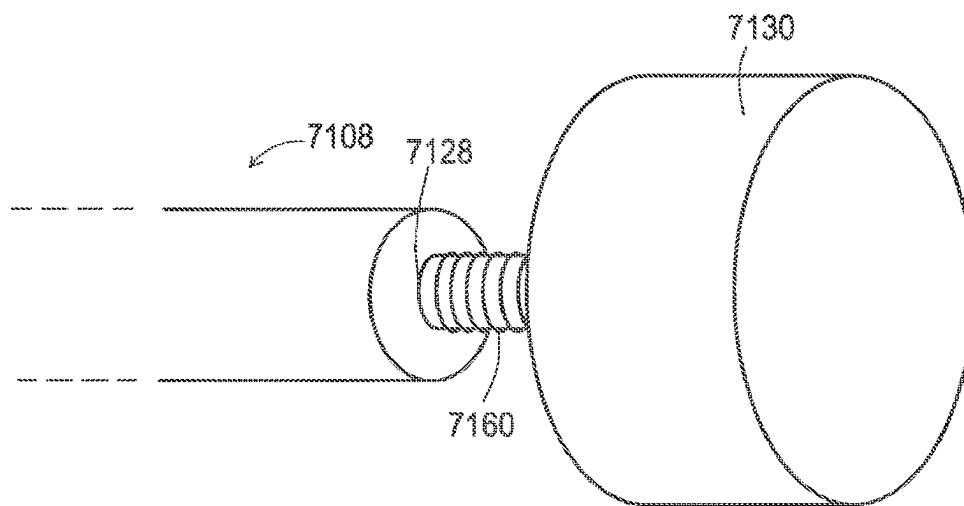
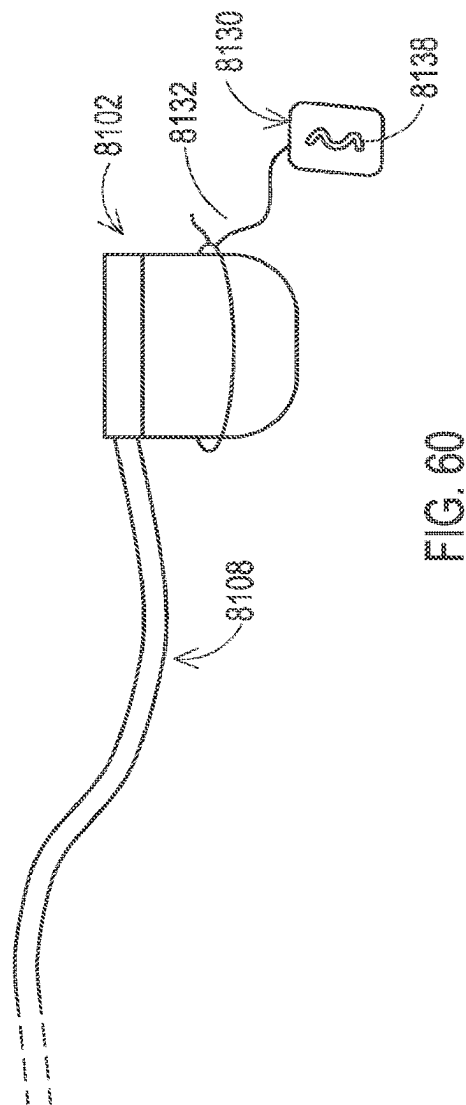
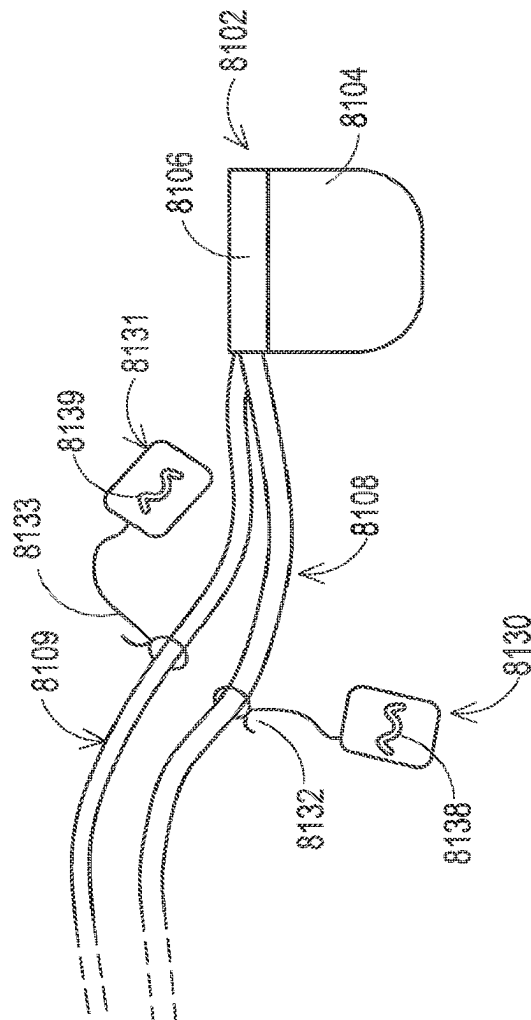


FIG. 58



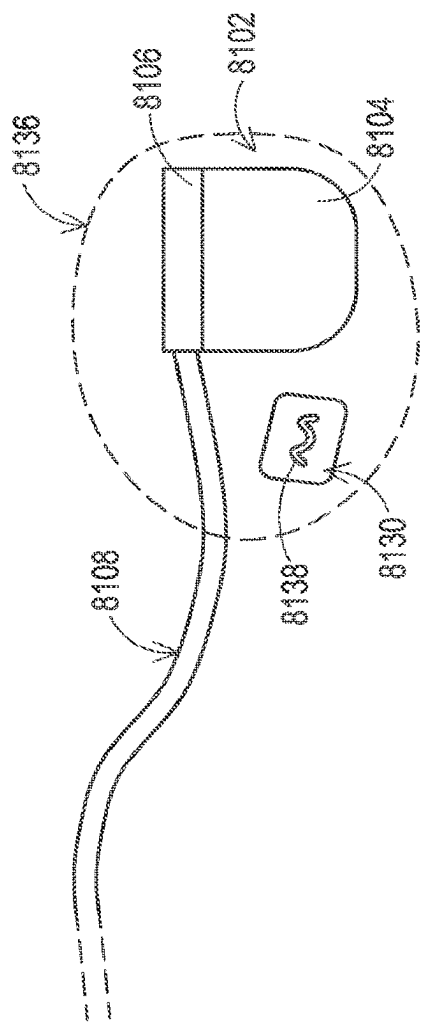


FIG. 61

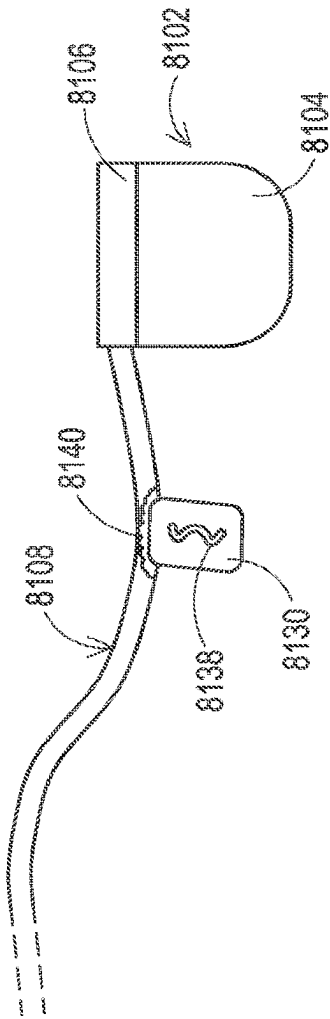


FIG. 62

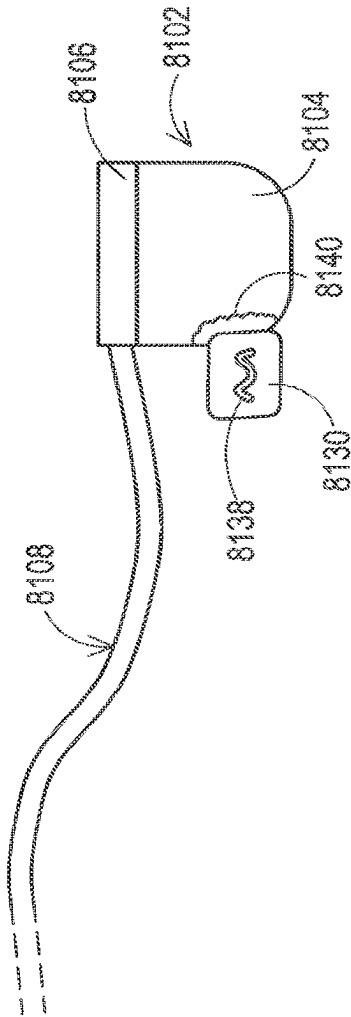


FIG. 63

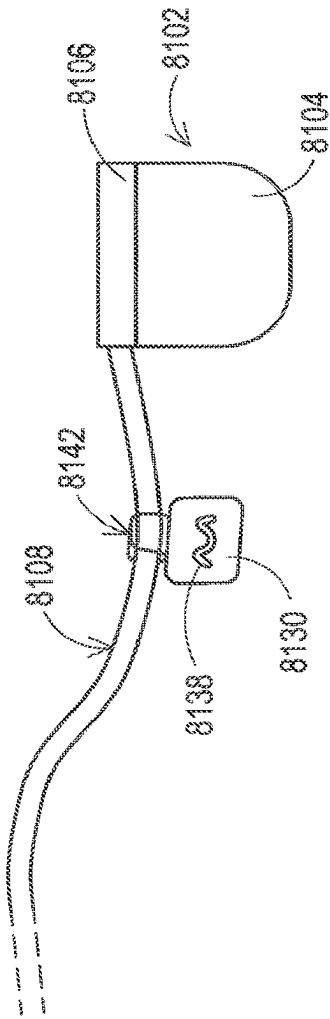


FIG. 64

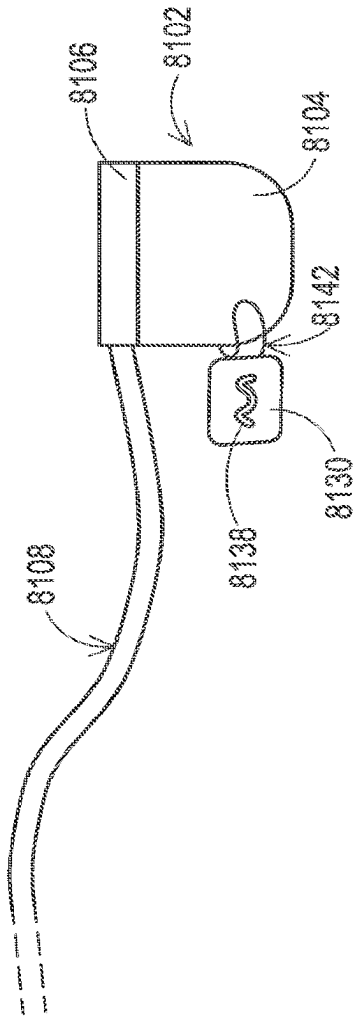
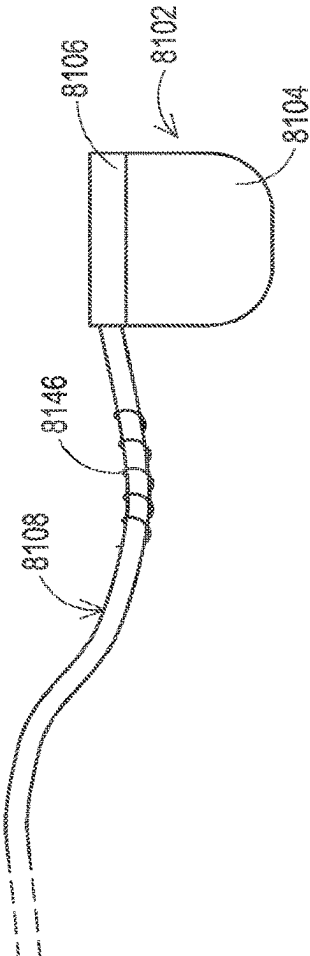
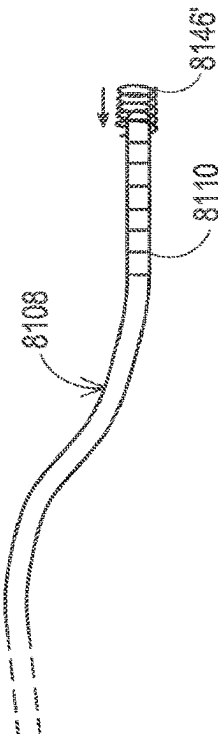
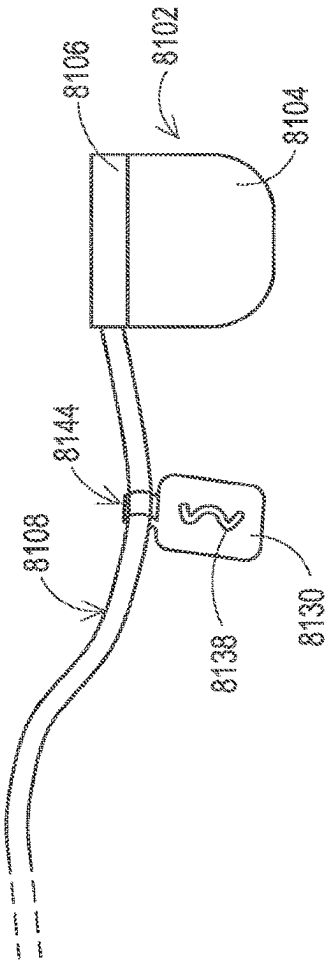


FIG. 65



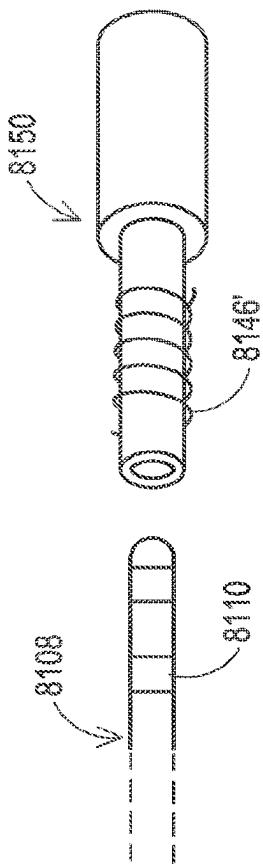


FIG. 68A

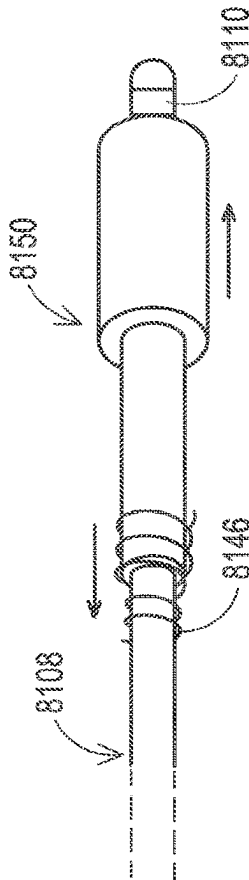


FIG. 68B

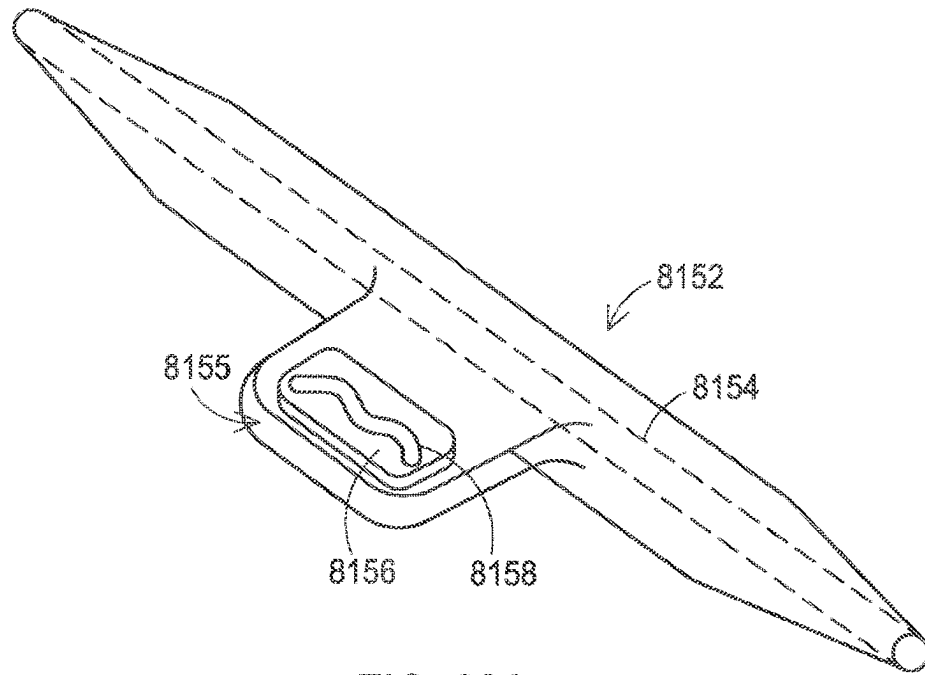


FIG. 69A

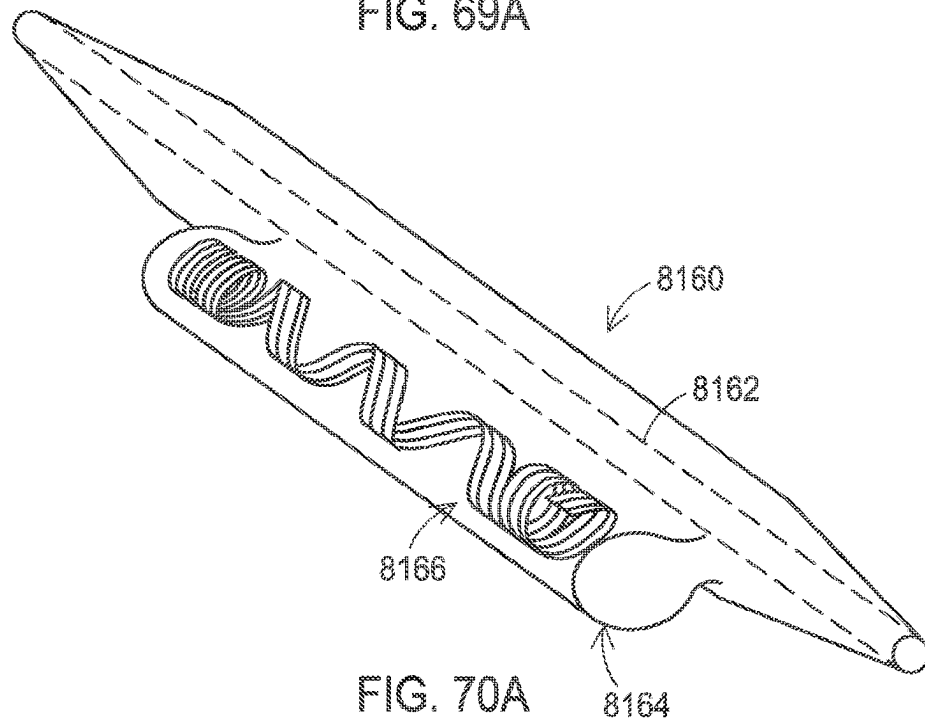
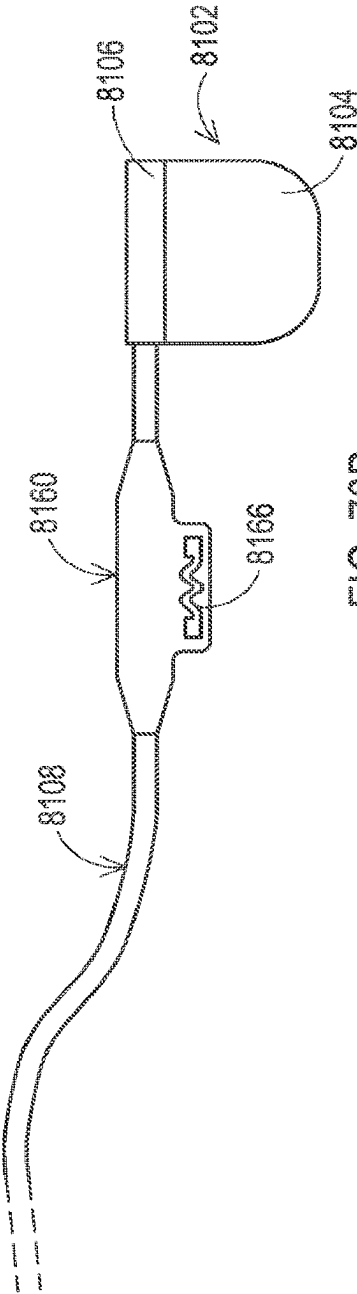
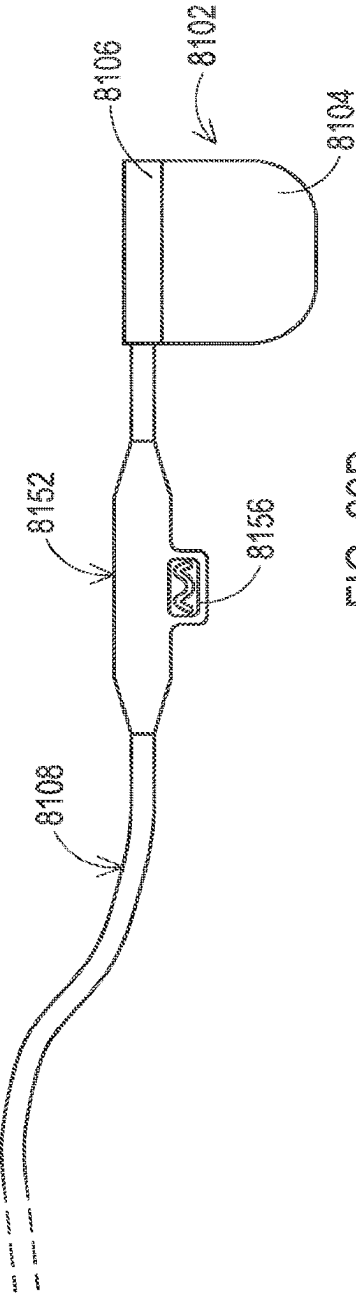
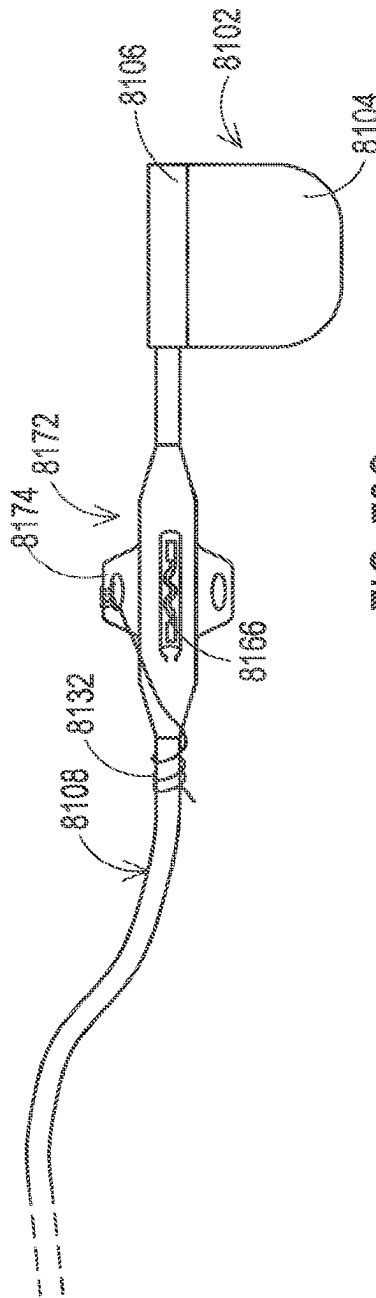
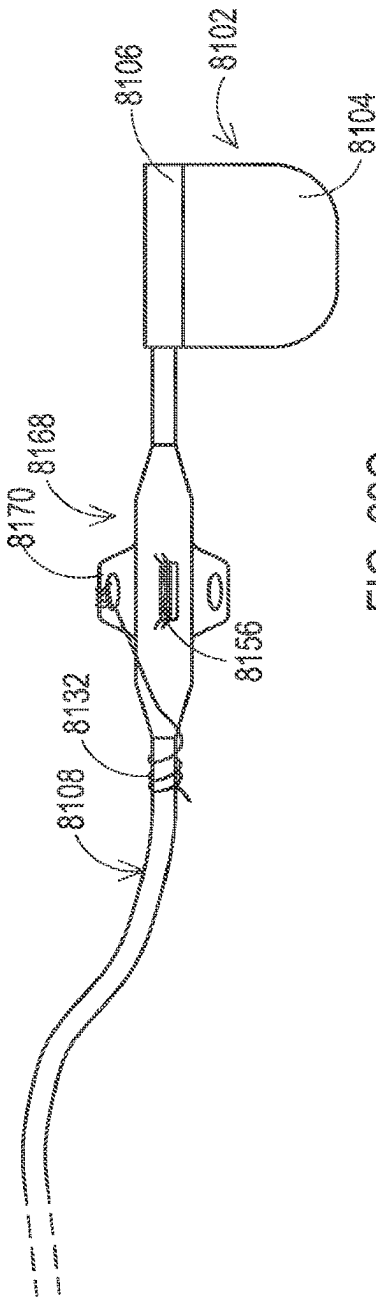


FIG. 70A





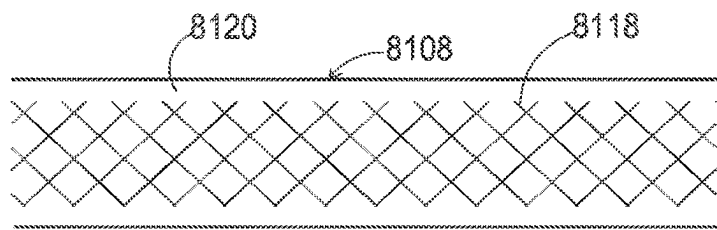


FIG. 71

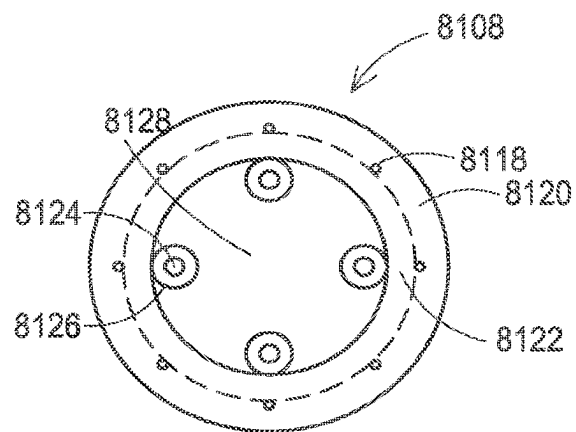
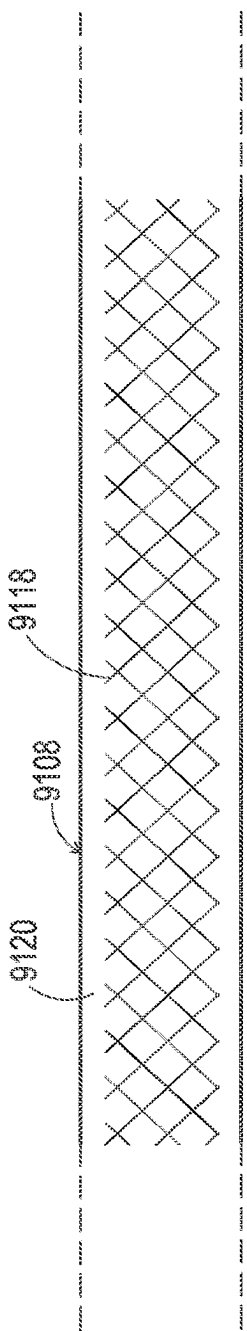
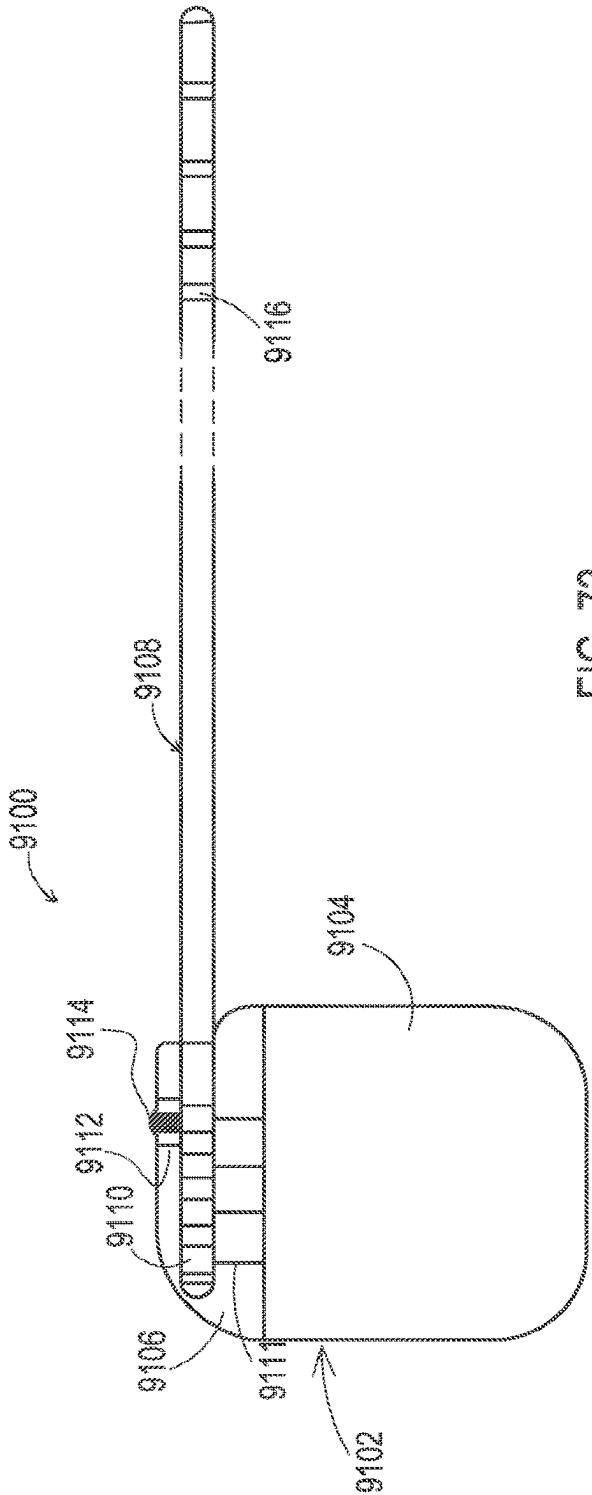


FIG. 72



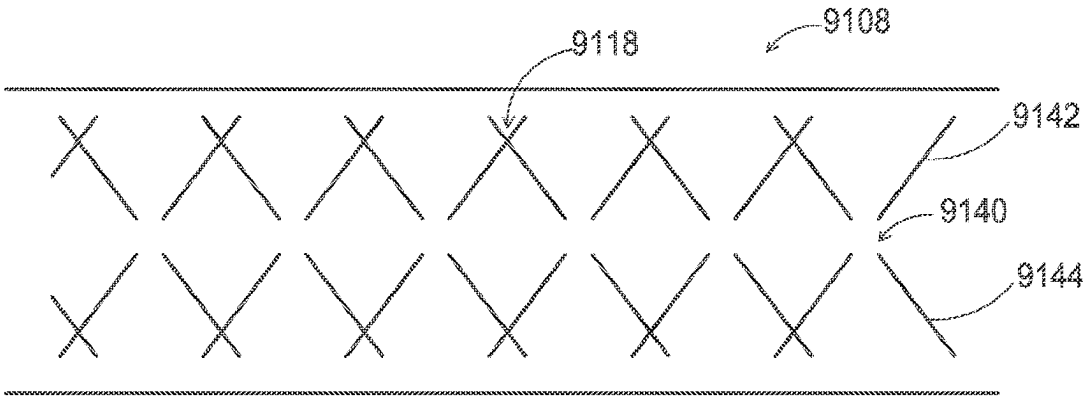


FIG. 74B

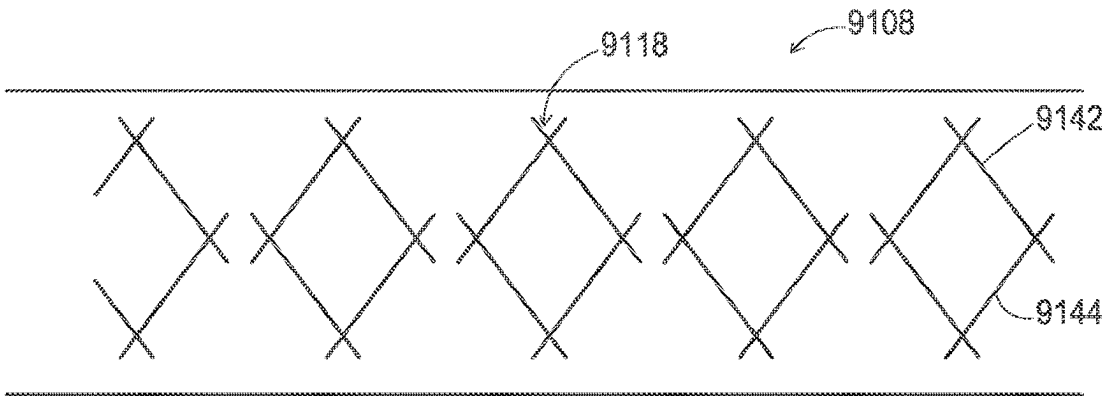


FIG. 74C

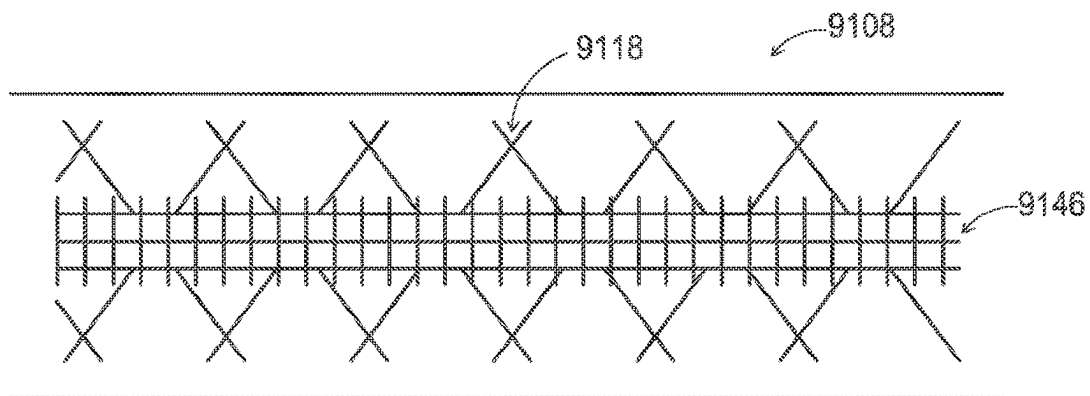


FIG. 74D

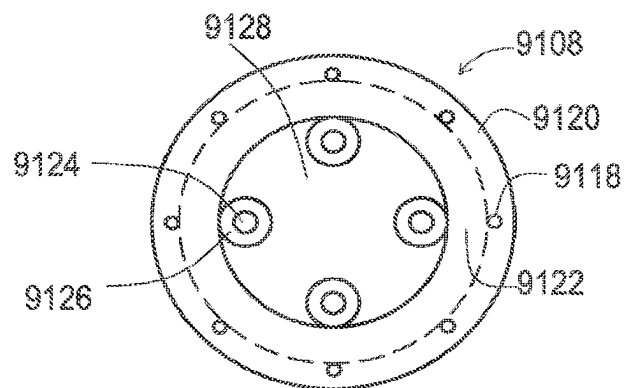


FIG. 75A

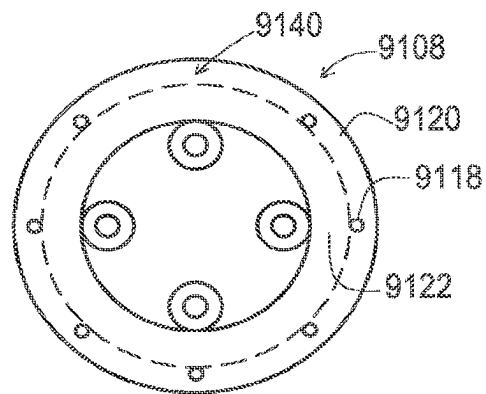


FIG. 75B

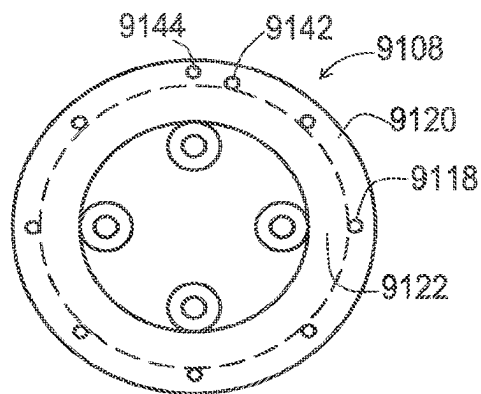


FIG. 75C

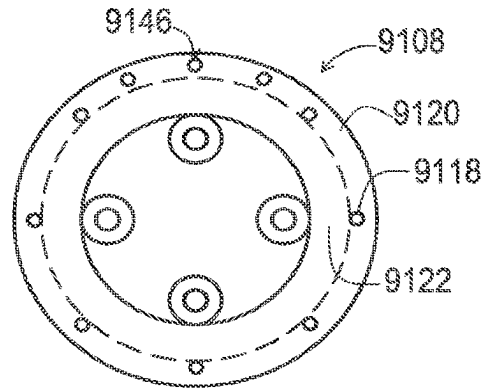


FIG. 75D

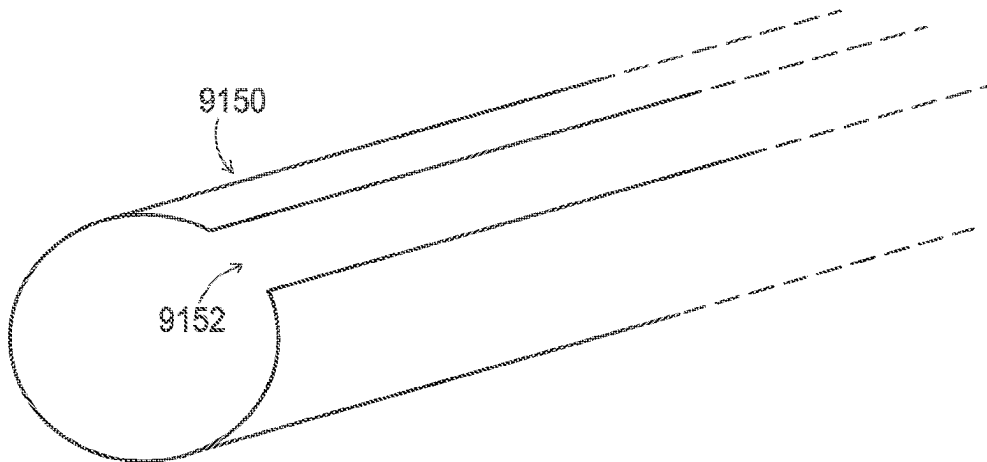


FIG. 76A

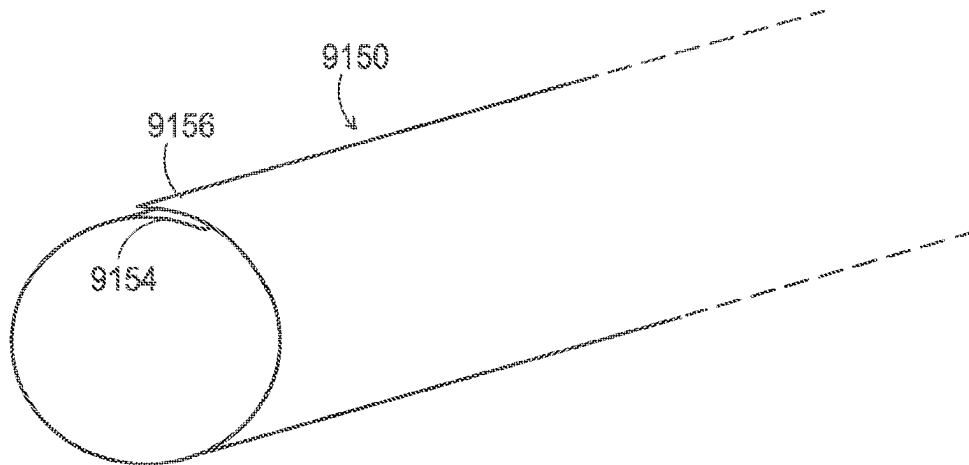


FIG. 76B

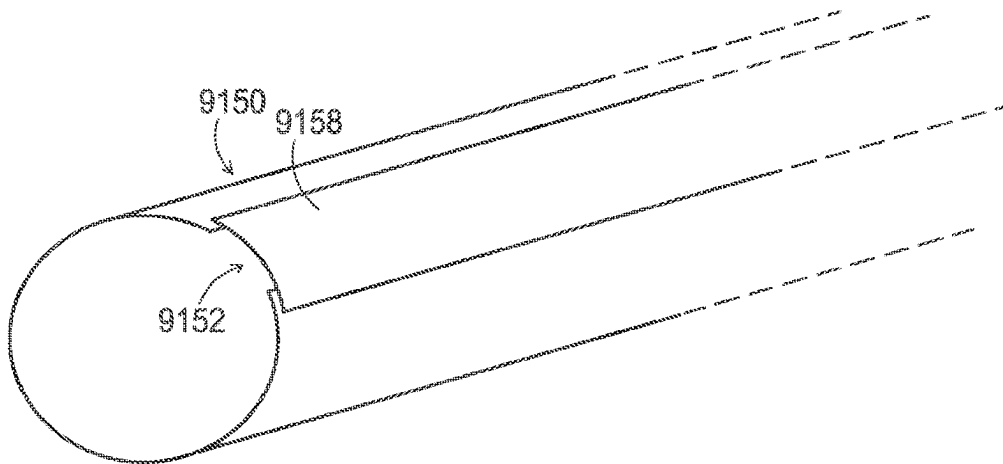


FIG. 76C

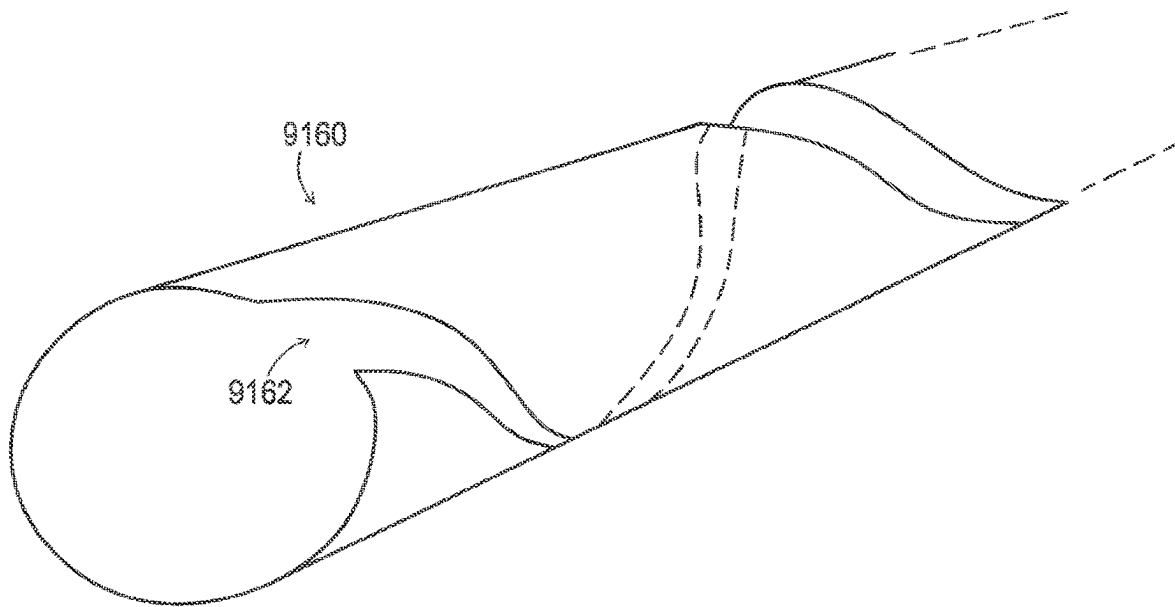
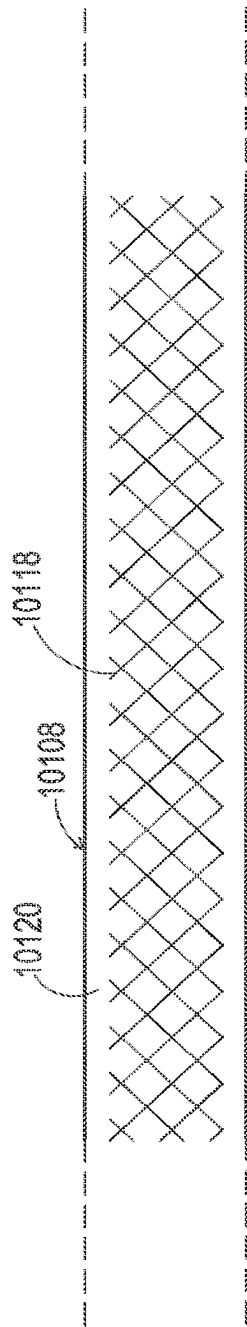
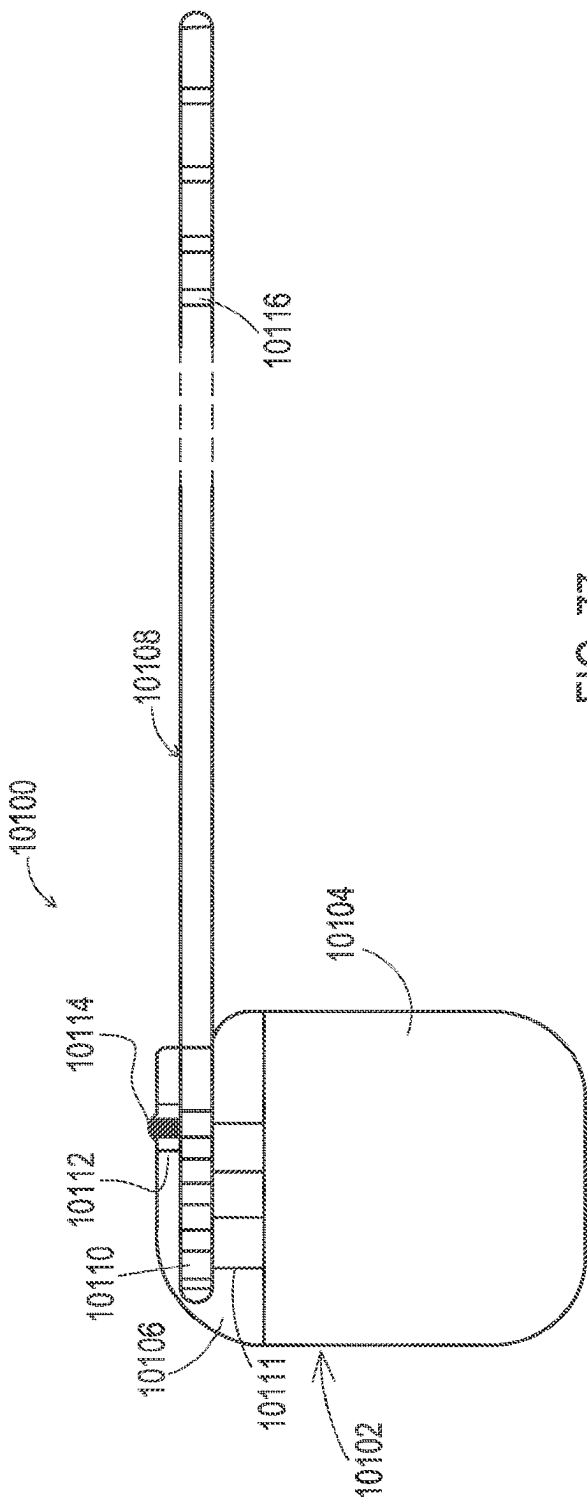


FIG. 76D



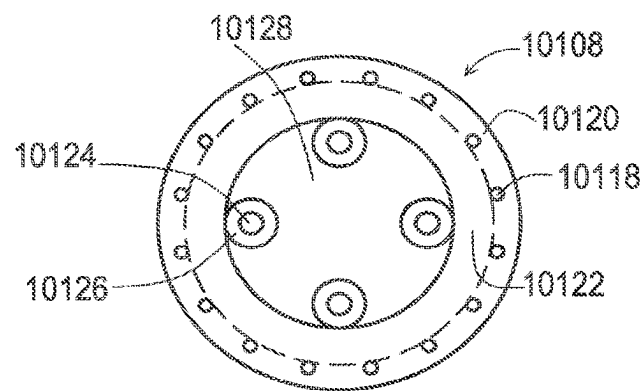


FIG. 78B

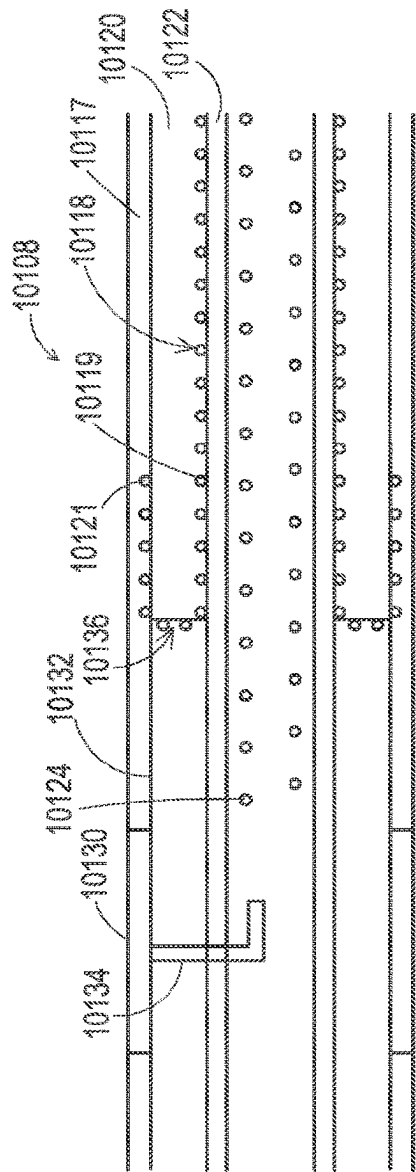


FIG. 79A

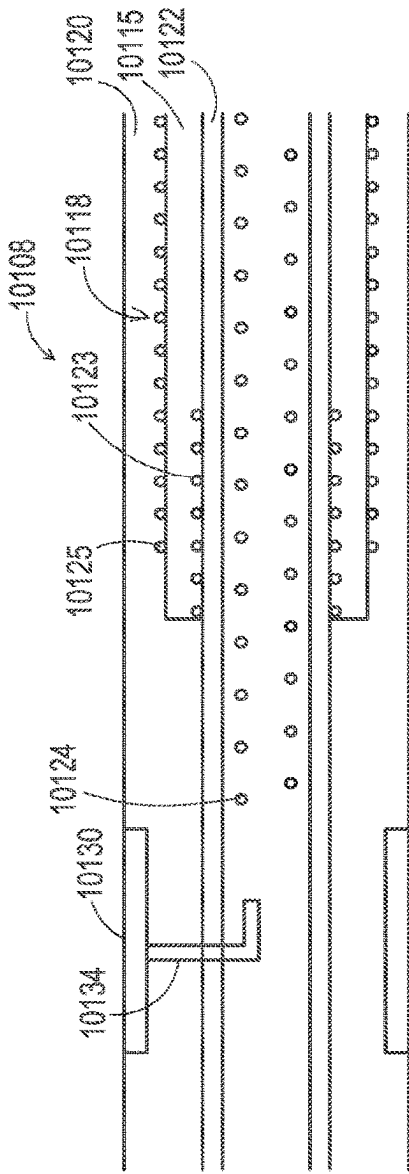
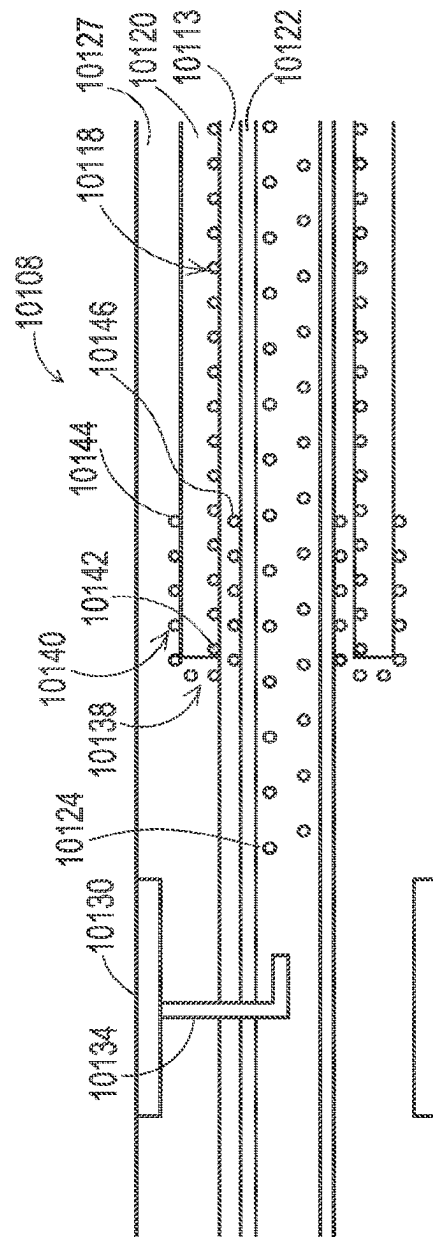


FIG. 79B



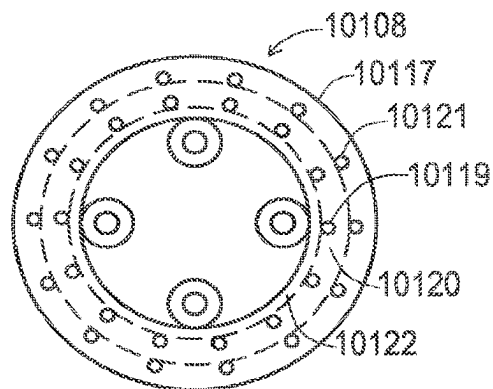


FIG. 80A

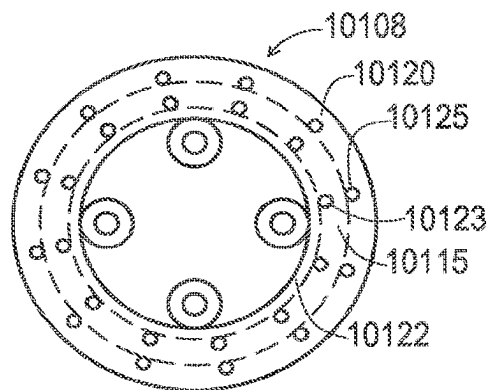


FIG. 80B

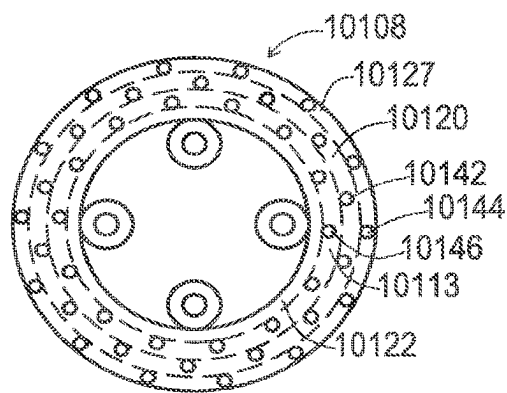


FIG. 80C

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TERMINATION OF A SHIELD WITHIN AN IMPLANTABLE MEDICAL LEAD

RELATED CASES

The present application is a divisional of U.S. Pat. No. 11,213,677, which is a divisional of U.S. Pat. No. 10,201,700, which is a divisional of U.S. Pat. No. 9,452,284, which is a divisional of U.S. Pat. No. 8,788,061, which is a national phase entry of PCT/US10/32526 filed on Apr. 27, 2010, each of which claims priority to and incorporates by reference the following, each as if rewritten herein in its entirety: U.S. Provisional Application 61/174,204 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,216 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,224 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,234 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,247 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,254 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,262 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,276 filed on Apr. 30, 2009; U.S. Provisional Application 61/174,287 filed on Apr. 30, 2009; and U.S. Provisional Application 61/174,296 filed on Apr. 30, 2009.

TECHNICAL FIELD

Embodiments relate to implantable medical leads that include shields. More particularly, embodiments relate to the termination of the shield within implantable medical leads.

BACKGROUND

Implantable medical systems including implantable medical devices (IMD) and associated implantable medical leads provide functions such as stimulation of muscle or neurological tissue and/or sensing of physiological occurrences within the body of a patient. Typically, the IMD is installed in a subcutaneous location that is accommodating and relatively accessible for implantation. For instance, to provide stimulation near the spine or pelvis, the IMD may be installed in a pocket located on the abdomen or upper buttocks region of the patient. The implantable medical lead is installed, either through a percutaneous procedure or a surgical procedure, depending upon the type of lead that is necessary.

Once installed, the lead extends from the stimulation site to the location of the IMD. The separation of the stimulation site to the location of the IMD varies, but may typically range from about 20 cm to about 100 cm. For relatively lengthy separation, if a lead of adequate length is unavailable then a lead extension may be implanted to span from the IMD to a proximal end of the implantable lead.

The implantable medical lead includes connector rings on a proximal end and electrodes on a distal end, and conductive filars interconnecting the electrodes at the proximal end connector rings to the electrodes at a distal end. The lead includes a jacket, often made of a flexible but biocompatible polymer, and the filars are insulated from the body tissue by the jacket. However, the filars are not insulated by the jacket from the presence of electromagnetic radiation. Electromagnetic radiation in the radio frequency (RF) spectrum induces currents into the filars and thus presents current at the electrode that is unintended. In the patient's normal daily experience, the level of RF radiation that is encountered is at a negligible level, and there is no danger of heating of tissue by the unintended current that may result.

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RF radiation poses a risk to tissue in contact with the electrodes when the intensity is significantly higher than the background levels. The surface area of each electrode is relatively small so that a small amount of tissue must dissipate a potentially large amount of induced current. In particular, if the patient is exposed to the RF radiation from a magnetic resonance imaging (MRI) scan, there is a high probability that tissue damage at the stimulation site(s) can occur. This tissue damage may be very dangerous, particularly so for neurological tissue. Therefore, patients with IMDs are typically not permitted to have a body coil MRI scan for at least these reasons.

SUMMARY

Embodiments address issues such as these and others by providing an implantable lead that includes a shield within the jacket that may reduce the amount of current induced on the filars within the lead. The shield may be terminated at an electrical connector in various ways to prevent wires of the shield from losing electrical contact, fraying, and otherwise migrating within the lead.

Embodiments provide a method of terminating a shield within a jacket of an implantable medical lead. The method involves providing an inner insulation layer and providing an inner metal ring located near an end of the inner insulation layer. The method further involves providing the shield between the inner insulation layer of the jacket and an outer insulation layer of the jacket, a portion of the shield lapping onto the inner metal ring and providing an outer metal ring about the inner metal ring with the portion of the shield being positioned between the inner metal ring and the outer metal ring.

Embodiments provide a method of terminating a shield within a jacket of an implantable medical lead. The method involves providing an inner insulation layer having a proximal end and providing an outer metal ring about the inner insulation layer in proximity to the proximal end. The method further involves providing an outer insulation layer that surrounds the inner insulation layer and that terminates prior to the proximal end of the inner insulation layer and providing the shield between the inner insulation layer of the jacket and the outer insulation layer of the jacket, the shield extending beyond the termination of the outer insulation layer with a portion of the shield lapping onto the metal ring.

Embodiments provide a method of terminating a shield within a jacket of an implantable medical lead. The method involves providing the shield between an inner insulation layer of the jacket and an outer insulation layer of the jacket, the shield terminating by the end of the inner insulation layer, the outer insulation layer terminating prior to an end of the shield and the inner insulation layer. The method further involves bonding an insulation extension layer onto the end of the inner insulation layer and positioning a metal ring over the shield and inner insulation layer where the outer insulation layer is not present.

Embodiments provide an implantable medical lead that includes an inner insulation layer, an inner metal ring located near a proximal end of the inner insulation layer, and an outer insulation layer that surrounds the inner insulation layer. The implantable medical lead further includes a shield between the inner insulation layer and the outer insulation layer, a portion of the shield lapping onto the metal ring and an outer metal ring about the inner metal ring with the portion of the shield being positioned between the inner metal ring and the outer metal ring.

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Embodiments provide an implantable medical lead that includes an inner insulation layer having a proximal end and an outer metal ring about the inner insulation layer in proximity to the proximal end. An outer insulation layer surrounds the inner insulation layer and that terminates prior to the proximal end of the inner insulation layer. A shield is located between the inner insulation layer of the jacket and the outer insulation layer of the jacket, the shield extending beyond the termination of the outer insulation layer with a portion of the shield lapping onto the metal ring.

Embodiments provide an implantable medical lead that includes a jacket comprising an inner insulation layer and an outer insulation layer. A shield is located between the inner insulation layer and the outer insulation layer, the shield terminating by the end of the inner insulation layer, the outer insulation layer terminating prior to an end of the shield and the inner insulation layer. An insulation extension layer is bonded onto the end of the inner insulation layer, and a metal ring positioned over the shield and inner insulation layer where the outer insulation layer is not present.

DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 2A shows an embodiment of an implantable lead with the shield revealed.

FIG. 2B shows the embodiment of the implantable lead in cross-section to reveal the shield and filars.

FIG. 2C shows an embodiment of the implantable lead with the shield revealed and with various parameters being specified.

FIG. 2D shows an embodiment of the implantable lead with dual braid wire windings.

FIG. 2E shows an embodiment of the implantable lead with braid wires having a round cross-section.

FIG. 2F shows an embodiment of the implantable lead with braid wires having a rectangular cross-section.

FIG. 2G shows an embodiment of the implantable lead with braid wires having an oval cross-section.

FIG. 2H shows an embodiment of the implantable lead with the lead terminating at a given spacing from a nearest connector and a nearest electrode at the proximal and distal ends.

FIG. 3 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 4A shows an embodiment of an implantable lead with the shield revealed.

FIG. 4B shows the embodiment of the implantable lead in cross-section to reveal the shield and filars.

FIG. 4C shows an example of the implantable lead with a portion of the shield exposed near a proximal end of the implantable lead.

FIG. 4D shows an example of the implantable lead with an external electrode providing a coupling to the shield.

FIG. 5A shows a side view of an embodiment of an implantable medical system where the shield of the lead is grounded to a can of the IMD.

FIG. 5B shows an end view of the embodiment where the lead passes through a connection block having a set screw to ground the shield to the can.

FIG. 5C shows an end view of the embodiment where the lead passes through a connection block having a spring loaded connector to ground the shield to the can.

FIG. 6 shows an example of a spring loaded connector.

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FIGS. 7A-7C show side views of embodiments of an implantable medical system where the shield is grounded with an external wire to the can over a direct current pathway.

FIGS. 8A-8C show side views of embodiments of an implantable medical system where the shield is grounded with an external wire to the can over a capacitively coupled pathway.

FIGS. 9A-9F show side views of embodiments of an implantable medical system where the shield is grounded within a header of the IMD to the can.

FIGS. 10A-10C show side views of embodiments of an implantable medical system where the shield is grounded to a ground plate on the header of the IMD.

FIG. 11 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 12A shows an embodiment of an implantable lead with the shield revealed.

FIG. 12B shows the embodiment of the implantable lead in cross-section to reveal the shield and filars.

FIG. 12C shows an example of the implantable lead with a portion of the shield exposed at a point distant from the distal end of the lead.

FIG. 12D shows an example of the implantable lead with an external electrode providing a coupling to the shield at a point distant from the distal end of the lead.

FIG. 12E shows an example of the implantable lead with a portion of the shield nearly exposed at a point distant from the distal end of the lead.

FIG. 12F shows an example of the implantable lead with a portion of the shield exposed or nearly exposed at a plurality of points distant from the distal end of the lead.

FIG. 12G shows an example of the implantable lead with a plurality of external electrodes providing a coupling to the shield at a plurality of points distant from the distal end of the lead.

FIG. 12H shows the embodiment of the implantable lead in cross-section to reveal the shield and the external metal conductor in contact with the shield.

FIG. 12I shows the embodiment of the implantable lead in cross-section to reveal the exposed shield.

FIG. 12J shows the embodiment of the implantable lead in cross-section to reveal the shield and the metal conductor nearly contacting the shield.

FIG. 12K shows the embodiment of the implantable lead in cross-section to reveal the nearly exposed shield.

FIGS. 13A-13C show metal conductors of various types for attachment to a lead to provide a coupling to the shield.

FIGS. 13D-13F show metal conductors having various configurations of non-conductive coatings for attachment to a lead to provide a coupling to the shield.

FIG. 14A shows the embodiment of the implantable lead in cross-section to reveal the shield and an outer doped jacket layer.

FIG. 14B shows the embodiment of the implantable lead with a plurality of points with the outer doped jacket layer.

FIG. 14C shows an embodiment of the implantable lead in cross-section to reveal the shield and a doped jacket layer at the shield.

FIG. 14D shows the embodiment of the implantable lead with a plurality of points with the doped jacket layer at the shield.

FIG. 15A shows an embodiment of the implantable lead having a lead anchor coupled to a metal conductor to provide the RF pathway to ground.

FIG. 15B shows the embodiment of the implantable lead in cross-section to reveal the shield, the metal conductor, and the lead anchor.

FIG. 15C shows an embodiment of the implantable lead having a lead anchor coupled directly to the shield to provide the RF pathway to ground.

FIG. 15D shows the embodiment of the implantable lead in cross-section to reveal the shield and the lead anchor.

FIG. 15E shows an embodiment of the implantable lead having a lead anchor with a non-conductive coating.

FIG. 16 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to an extension containing a shield which is coupled to a lead containing a shield.

FIG. 17A shows an embodiment of an implantable extension coupled to an implantable lead with the shield of each revealed.

FIG. 17B shows an embodiment of a coupling of the implantable lead and extension in cross-section to reveal the shield, shield electrode, and filars of the lead and a shield connector and jumper wire of the lead extension.

FIG. 17C shows an embodiment of a coupling of the implantable lead and extension in cross-section to reveal the shield, shield electrode, and filars of the lead and a shield connector and shield of the lead extension.

FIG. 17D shows an embodiment of a coupling of the implantable lead and extension in cross-section to reveal the filars, filar jumper, and filar electrode of the lead and a filar connector, filar jumper wire, and shield jumper wire of the lead extension.

FIG. 17E shows an embodiment of a coupling of the implantable lead and extension in cross-section to reveal the filars, filar jumper wire, and filar electrode of the lead and a filar connector, filar jumper wire and shield of the lead extension.

FIG. 17F shows an embodiment of an implantable extension coupled to an implantable lead where a jumper wire interconnects the two shields.

FIG. 17G shows an embodiment of an implantable extension coupled to an implantable lead where the shield of the extension extends to the shield connector of the extension to interconnect the two shields.

FIG. 18 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 19A shows an embodiment of an implantable lead with the shield revealed.

FIG. 19B shows the embodiment of the implantable lead in cross-section to reveal the shield and filars.

FIG. 20 shows an embodiment of the implantable lead where the shield terminates at a butt joint to an insulation extension.

FIG. 21 shows an embodiment of the implantable lead where the shield terminates at a scarf joint to an insulation extension.

FIG. 22 shows one example of a set of steps to create the implantable lead of FIGS. 20 and 21.

FIG. 23 shows an embodiment of the implantable medical lead where the shield terminates at a lap joint to an insulation extension.

FIG. 24 shows one example of a set of steps to create the implantable lead of FIG. 23.

FIG. 25 shows an embodiment of the implantable medical lead where the shield terminates at a ring within a lap joint.

FIG. 26 shows an embodiment of the implantable medical lead where the shield terminates at a ring at a butt joint.

FIG. 27 shows an embodiment of the implantable medical lead where wires of the shield fold over individually at the termination point.

FIG. 28 shows an embodiment of the implantable medical lead where a joint at the shield termination includes a barbed connection to an inner insulation layer.

FIG. 29 shows an embodiment of the implantable medical lead where a joint at the shield termination includes a barbed connection to an inner insulation layer.

FIG. 30 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 31A shows an embodiment of an implantable lead with the shield revealed.

FIG. 31B shows the embodiment of the implantable lead in cross-section to reveal the shield and filars.

FIG. 32 shows an embodiment of the implantable lead where the shield terminates to a metal connector near a butt joint to an insulation extension.

FIG. 33 shows an embodiment of the implantable lead where the shield terminates to a metal connector near a scarf joint to an insulation extension.

FIG. 34 shows one example of a set of steps to create the implantable lead of FIGS. 32 and 33.

FIG. 35 shows an embodiment of the implantable medical lead where the shield terminates to a metal connector near a lap joint to an insulation extension.

FIG. 36 shows one example of a set of steps to create the implantable lead of FIG. 35.

FIG. 37 shows an embodiment of the implantable medical lead where the shield terminates between a pair of metal connectors near a joint to an insulation extension.

FIG. 38 shows an alternative embodiment where a top metal connector of the pair has sharp features to penetrate an outer insulation layer.

FIG. 39 shows an embodiment of the implantable medical lead where the shield terminates between a pair of metal connectors near a lap joint to an insulation extension.

FIG. 40 shows an alternative embodiment of the implantable medical lead where an inner metal connector is mounted flush with an inner insulation layer.

FIG. 41 shows one example of a set of steps to create the implantable lead of FIGS. 37-40.

FIG. 42 shows an embodiment of the implantable medical lead where the shield folds over to terminate between a pair of metal connectors near a joint to an insulation extension.

FIG. 43 shows one example of a set of steps to create the implantable lead of FIG. 42.

FIG. 44 shows an embodiment of the implantable medical lead where the shield laps onto a metal connector near a joint to an insulation extension.

FIG. 45 shows one example of a set of steps to create the implantable lead of FIG. 44.

FIG. 46 shows an embodiment of the implantable medical lead where the shield exits the insulation layers at a taper and terminates between a pair of metal connectors.

FIG. 47 shows one example of a set of steps to create the implantable lead of FIG. 46.

FIG. 48 shows an embodiment of the implantable medical lead where wires of the shield fold over individually at the termination point at a metal connector.

FIG. 49 shows a percutaneous implantation scenario of an embodiment of an implantable medical lead.

FIG. 50 shows an implantable medical system configuration resulting from the percutaneous implantation of FIG. 49.

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FIG. 51 shows an embodiment of the implantable medical lead that has a braided metal shield providing torsional stiffness.

FIG. 52 shows a cross-section of an embodiment of the implantable medical lead where no rotational coupling exists to the stylet.

FIG. 53 shows a cross-section of an embodiment of the implantable medical lead where a square-shaped rotational coupling exists to the stylet.

FIG. 54 shows a cross-section of an embodiment of the implantable medical lead where a star-shaped rotational coupling exists to the stylet.

FIG. 55 shows a cross-section of an embodiment of the implantable medical lead where a hexagonal-shaped rotational coupling exists to the stylet.

FIG. 56 shows a proximal end of an embodiment of the implantable medical lead achieving a rotational coupling with a tapered feature of a stylet hub.

FIG. 57 shows a proximal end of an embodiment of the implantable medical lead achieving a rotational coupling with a splined feature of a stylet hub.

FIG. 58 shows a proximal end of an embodiment of the implantable medical lead achieving a rotational coupling with a threaded feature of a stylet hub.

FIG. 59 shows an embodiment of an implantable medical system including an IMD and two leads, each with a radiopaque marker sutured to the lead.

FIG. 60 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker sutured to the IMD case.

FIG. 61 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker placed loosely in a pocket nearby the IMD and lead.

FIG. 62 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker glued to the lead.

FIG. 63 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker glued to the IMD case.

FIG. 64 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker clamped to the lead.

FIG. 65 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker clamped to the IMD case.

FIG. 66 shows an embodiment of an implantable medical system including an IMD and a lead, with a radiopaque marker crimped to the lead.

FIG. 67A shows an embodiment of an implantable medical system including a lead and a radiopaque coil being placed onto the lead in a radially expanded state.

FIG. 67B shows an embodiment of an implantable medical system including a lead and a radiopaque coil once placed onto the lead in a radially contracted state.

FIG. 68A shows the coil on an installation tool in the radially expanded state prior to placement on the lead.

FIG. 68B shows the coil being placed onto the lead from the tool to achieve the radially contracted state.

FIG. 69A shows an embodiment of a polymer structure that fits axially onto the lead and provides a radiopaque plate.

FIG. 69B shows an embodiment of an implantable medical system where the polymer structure of FIG. 69A is positioned on the lead.

FIG. 69C shows an embodiment of an implantable medical system where an embodiment of a polymer structure that

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has an anchor format including suture wings and a radiopaque plate is sutured in place on the lead.

FIG. 70A shows an embodiment of a polymer structure that fits axially onto the lead and provides a radiopaque coil.

FIG. 70B shows an embodiment of an implantable medical system where the polymer structure of FIG. 70A is positioned on the lead.

FIG. 70C shows an embodiment of an implantable medical system where an embodiment of a polymer structure that has an anchor format including suture wings and a radiopaque coil is sutured in position on the lead.

FIG. 71 shows an embodiment of the lead that includes a shield to provide safety during medical procedures such as an MRI scan.

FIG. 72 shows the embodiment of FIG. 71 in cross-section to reveal the shield, filars, and lumen.

FIG. 73 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 74A shows an embodiment of an implantable lead with the shield revealed.

FIG. 74B shows an embodiment of an implantable lead with the shield having an axial cut that creates a slot.

FIG. 74C shows an embodiment of an implantable lead with the shield having the axial cut but with the edges of the slot brought into an overlapping configuration to close the slot.

FIG. 74D shows an embodiment of an implantable lead with the shield having the axial cut but with a shield patch applied across the slot.

FIG. 75A shows the embodiment of the implantable lead of FIG. 74A in cross-section to reveal the shield and filars.

FIG. 75B shows the embodiment of the implantable lead of FIG. 74B in cross-section to reveal the shield and the slot.

FIG. 75C shows the embodiment of the implantable lead of FIG. 74C in cross-section to reveal the shield having the overlapping edges.

FIG. 75D shows the embodiment of the implantable lead of FIG. 74D in cross-section to reveal the shield and the shield patch.

FIG. 76A shows an embodiment of the shield as an equivalent tube to reveal a linear axial cut that produces a linear slot.

FIG. 76B shows the embodiment of the shield as an equivalent tube to reveal the edges of the slot that overlap to close the slot.

FIG. 76C shows the embodiment of the shield as an equivalent tube to reveal the shield patch that closes the slot.

FIG. 76D shows an embodiment of the shield as an equivalent tube to reveal a helical axial cut forming a helical slot.

FIG. 77 shows an embodiment of an implantable medical system that includes an implantable medical device (IMD) coupled to a lead containing a shield.

FIG. 78A shows an embodiment of an implantable lead with the shield revealed.

FIG. 78B shows the embodiment of the implantable lead of FIG. 78A in cross-section to reveal the shield and filars.

FIG. 79A shows one embodiment of a guard at the termination of the shield.

FIG. 79B shows another embodiment of a guard at the termination of the shield.

FIG. 79C shows another embodiment of a guard at the termination of the shield.

FIG. 80A shows the embodiment of FIG. 79A in cross-section to reveal first and second portions of the continuous shield forming a guard at the termination of the shield.

FIG. 80B shows the embodiment of FIG. 79B in cross-section to reveal first and second portions of the two-piece shield forming a guard at the termination of the shield.

FIG. 80C shows the embodiment of FIG. 79C in cross-section to reveal a first portion including first and second sub-portions and a second portion at the termination of the two-piece shield.

DETAILED DESCRIPTION

Embodiments of implantable medical leads that include shields are disclosed herein. Ten primary subject matter topics are presented, where each new topic begins with reference to FIGS. 1, 3, 11, 16, 18, 30, 49, 59, 73, and 77. However, this detailed description should be read as a whole whereby subject matter of embodiments corresponding to one particular topic is applicable to embodiments corresponding to other topics.

For instance, shield details disclosed in relation to FIGS. 1-2H are also applicable to the shields of all embodiments disclosed in FIGS. 3-80C where such shield details may be desired. The examples of grounding a shield within a lead as disclosed in relation to FIGS. 3-15E are applicable to all embodiments disclosed herein where a grounded shield may be desired. The examples of shielding the extension and interconnecting the shielding of the lead to the extension as disclosed in relation to FIGS. 16-17G are applicable to all embodiments disclosed herein where inclusion of a shielded extension may be desired. The examples of terminating the shield as disclosed in relation to FIGS. 18-48 are applicable to all embodiments disclosed herein where terminating the shield within the lead body may be desired. The examples of rotationally coupling the lead body to a stylet as disclosed in relation to FIGS. 49-58 are applicable to all embodiments disclosed herein where such rotational coupling may be desired. The examples of markers for the lead as disclosed in relation to FIGS. 59-72 are applicable to all embodiments disclosed herein where such a marker may be desired. The examples of breaking the circumferential mechanical continuity of the shield as disclosed in FIGS. 73-76D are applicable to all embodiments disclosed herein where such a lack of continuity may be desired. The examples of guarding the termination of the shield as disclosed in FIGS. 77-80C are applicable to all embodiments disclosed herein where a guarded shield termination may be desired.

Embodiments disclosed in relation to FIGS. 1-2H provide for radio frequency (RF) shielding of an implantable lead that may be connected to an implantable medical device (IMD). A shield is present within the jacket of the implantable lead. The shield is designed to provide RF shielding while also providing various mechanical properties suitable for implantation.

FIG. 1 shows an example of an implantable medical system 1100 that includes an IMD 1102 coupled to a lead 1108. The IMD 1102 includes a metal can 1104, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD 1102 includes a header 1106 typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can 1104. The header 1106 provides a structure for securing the lead 1108 to the IMD 1102 and for establishing electrical connectivity between circuitry of the IMD 1102 and electrodes of the lead 1108.

The lead 1108 includes electrodes 1116 on a distal end that are positioned at a stimulation site within a patient. The lead also includes connector rings 1110 on a proximal end that is

positioned within the header 1106. The connector rings 1110 make physical contact with electrical connections 1111 within the header. The electrical connections 1111 may include a metal contact that the connector ring 1110 rests against upon being inserted into the header 1106 where a wire extends from the metal contact into the can 1104 where the circuitry is housed. Signals applied by the IMD 1102 to the connector rings 1110 are conducted through the lead 1108 to the electrodes 1116 to provide the stimulation therapy to the patient.

The lead 1108 is secured in the header 1106 such as by a set screw block 1112 within the header 1106 that allows at least one set screw 1114 to be tightened against at least one of the connector rings 1110. A shield 1118 such as the one discussed below with reference to FIGS. 2A and 2B is located within the lead 1108. The shield 1118 may or may not be grounded to the metal can 1104 at the IMD 1102 of FIG. 1 or at various points along the length of the lead. The shield 1118 may or may not be grounded through other mechanisms as well. For instance, the shield 1118 may be located within the lead 1108 at a small distance from the surface so that the shield 1118 will effectively capacitively couple to the tissue along the length of the lead to dissipate energy to the tissue over the length.

FIGS. 2A and 2B show an example of the lead 1108, where a shield 1118 is present. An outer jacket layer 1120 is shown transparently in FIG. 2A for purposes of illustrating the shield 1118. The shield 1118 blocks at least some RF energy from directly coupling to conductive filars 1124 that are present within the lead 1108. The conductive filars 1124 extend the length of the lead and interconnect the proximal connector rings 1110 to the distal electrodes 1116 so that stimulation signals are conducted from the proximal end to the distal end of the lead 1108.

As shown in FIG. 2A, the shield 1118 of this example is a braided metal wire. The metal wire may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. The metal braid wire may be a biocompatible metal, particularly for embodiments where a portion of the shield 1118 may be exposed for purposes of grounding. Biocompatible metals ensure that if the shield 1118 is exposed to tissue, either by design or due to wear on the lead 1108, the shield 1118 does not become a toxin to the patient.

As shown in FIG. 2B, the shield 1118 may be embedded within the jacket of the lead 1108. One manner of constructing the lead 1108 with the shield 1118 is to provide a jacket that includes an inner layer of insulation 1122 that isolates an inner region 1121 where the filars 1124 and any additional insulation layer 1126, such as polytetrafluoroethylene (PTFE) that may surround each filar 1124 are located. According to some embodiments, this inner layer 1122 may have a post-assembly thickness 1130 of at least 2 mils and may be significantly larger such as 5 or 6 mils depending upon size constraints for the lead 1108 and/or the size of the outer layer 1120. The shield 1118 may then reside on the outer portion of the inner layer 1122, and the jacket's outer layer of insulation 1120 may then enclose the shield 1118. The outer layer 1120 provides an overall lead diameter 1134. The outer jacket 1120 may be added over the braid 1118, or it may be extruded over the braid.

For embodiments where it is desirable for the shield 1118 to RF couple to tissue, typically as capacitive coupling, either as an alternative to grounding at the can of the IMD or specific points along the length or in addition to grounding at the can or along the length, the entire outer jacket layer

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1120 may be relatively thin, particularly for the portion passing over the braid wires of the shield 1118. According to the various embodiments a post-assembly thickness 1132 for the portion of the outer layer 1120 passing over a single braid wire may be on the order of 0.5 to 5 mils. The thickness of the outer layer 1120 over the shield 1118 is reduced by a braid wire diameter at points where braid wires intersect. Accordingly, the post-assembly thickness 1132 over the single wire may vary depending upon a chosen braid wire diameter so that adequate coverage also exists at the intersection points. Furthermore the thickness may be less than 0.5 mils, particularly where tissue in-growth is not of concern and in that case the outer layer 1120 could be omitted.

This thickness of the outer layer 1120 over the braid wires may also vary depending upon the type of metal used for the braid wires. For instance, it has been found that the thickness of the outer layer 1120 has less of an impact on the heating at the electrode when using a titanium braid wire than when using a tantalum wire with all else being equal. However, with an outer layer 1120 whose post-assembled thickness 1132 is on the lower side of the range, such as 2 mils or less, tantalum braid wires may allow for less heating at the electrodes than if titanium braid wires are used.

Where the shield 1118 grounds at the can 1104 and/or at one or more specific locations along its length, via a direct current coupling or a capacitive coupling, the shield 1118 may be located further from the outer surface of the lead 1108. This increased depth of the shield 1118 within the jacket may provide for a more durable lead 1108 in terms of protecting the braid wires in areas of high flexure and motion, such as in the lumbar spine.

The inner and outer jackets 1122, 1120 may be constructed of the same or similar materials such as various flexible polymers, examples of which are polyurethanes and silicones. Biocompatible materials may be used, especially for the outer layer 1120 when the outer layer 1120 has direct contact with body tissue. A lumen 1128 may be included in an inner region 1121, particularly for percutaneous leads 1108, to allow a stylet to be inserted for purposes of pushing and steering the lead into the desired position within the patient. For leads where an inner region 1121 is filled to define the lumen 1128, such as where filars 1124 are cables rather than the coils as shown, this inner region 1121 may be constructed of materials such as polyurethanes, silicones, polyetheretherketone (PEEK), nylon or other biocompatible polymer material.

FIG. 2C shows a view of the implantable lead 1108 where various parameters related to the braid wires can be seen. The inner layer of insulation 1122, as well as the outer layer 1120, defines an axial dimension 1136 that runs along the length of the lead 1108. Braid wires such as braid wires 1140, 1142 are braided around the inner layer 1122. A first set of braid wires including braid wire 1140 is wound around the inner layer 1122 in a first direction while a second set of braid wires including braid wire 1142 is wound around the inner layer 1122 in a second direction that is opposite the first. The braid wires of the first set and the braid wires of the second set weave together during the braiding with a braid wire of the first set passing over some wires and under others of the second set in a repeating pattern.

The weaving may use a particular pattern, such as passing over one, passing under one, passing over one, and so on or such as passing over two, passing under two, passing over two, and so on. With wires of larger diameter, or where wires are used in pairs, then a pattern of two-over-two-under helps reduce the stress on the wire as it weaves back and forth. If

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the wires are small and single, with a relatively large aperture between braid wires, then one-over-one-under works well. The wire stress is a factor to consider because implant leads flex continually with body motion and typically are expected to last many years.

The braiding has various parameters of interest. A first parameter is the braid angle 1144. Here, the braid angle 1144 is defined as the angle of the braid wire as measured transversely from the axial dimension 1136; however, others sometimes define it relative to the axis of the lead. So, as shown in FIG. 2C, the braid angle 1144 is measured between the braid wire and the transverse dimension 1138. According to various embodiments, the braid angle measured in this way is less than 60 degrees.

This braid angle 1144 has several implications. The braid angle 1144 is one factor in setting the maximum dimension of the braid aperture 1141 shown in FIG. 2C, and hence the degree of coverage formed by the braid wires. This braid angle 1144 is also a factor in relation to the degree of stiffness of the lead in flexure and the tendency of the braid wires to break during flexure. The braid angle is also a factor in the cohesion of the outer layer of insulation 1120 to the inner layer of insulation 1122, because when the aperture is of adequate size, cohesion occurs between the two layers 1120, 1122 through the aperture.

Another parameter of interest as shown in FIG. 2C is the axial spacing 1146 between adjacent wires of a set. According to various embodiments, the axial spacing 1146 has an upper limit equal to the lead diameter 1134. The axial spacing 1146 is also a factor in the aperture size, the axial stiffness, the bending stiffness, and the kink resistance.

Another parameter of interest, which is related to the braid angle 1144 and the axial spacing 1146, is the number of wires in each set. According to various embodiments, the first set of braid wires which are wound in the first direction includes at least three braid wires. Likewise, the second set of braid wires which are wound in the second direction includes at least three braid wires. These two sets of at least three braid wires each ensures that for the various ranges of parameters disclosed herein, the aperture 1141 has a transverse dimension that is sufficiently small to effectively shield the RF energy in the MRI spectrum, which typically spans from 43 MHz to 128 MHz.

The total number of braid wires is limited by the allowable axial and bend stiffness for the braid angle and braid wire size. In some examples, there may be as many as 16 braid wires in each set for a total of 32 braid wires. However, as shown in the example of FIG. 2C, each set includes six braid wires, where braid wire 1140 reappears on a given side of the lead 1108 after five other braid wires are wound. Likewise, braid wire 1142 reappears on the side of the lead 1108 after five other braid wires are wound.

FIG. 2D shows another lead embodiment 1150 that demonstrates another braid wire parameter of interest. In this example, the braid wires are paired so that two braid wires that are in contact wind around the inner layer 1122 instead of a single wire. For instance, dual braid wires 1152 and 1154 of a first set wound in a first direction are in contact as each winds around the inner layer 1122. Dual braid wires 1156 and 1158 of a second set wound in a second direction are in contact as each winds around the inner layer 1122.

The braid wires bundled together in this manner affect the stiffness of the lead 1108 as well as the aperture size. Bundling braid wires in this manner may provide coverage like that of wider dimensioned braid wires, such as rectan-

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gular braid wires, but without the increased bending stresses associated with the corners present on the rectangular braid wire.

FIG. 2E is an enlarged view of a portion of a lead 1108 to illustrate the cross-section of the braid wires. The view is a cross-section where the cut through the lead 1108 is taken at an angle perpendicular to the direction of travel of the topmost braid wire 1142 so as to provide a true cross-section of the topmost braid wire 1142. Here the topmost braid wire 1142 has a round cross-section and provides a braid wire diameter 1148. According to various embodiments, the braid wire diameter ranges from about 0.5 mils to about 2.5 mils. The braid wire diameter is measured as the dimension that faces outward from the inner layer 1122 as shown in FIG. 2E. The round cross-section lacks corners that may otherwise affect the bend stiffness of the lead 1108, but the round cross-section provides less coverage than other cross-sectional shapes that have a same height extending into the outer layer 1120 from the inner layer 1122.

FIG. 2F is an enlarged view of a portion of a lead 1160 to illustrate the cross-section of the braid wires. As in FIG. 2E, the view is a cross-section where the cut through the lead 1160 is taken at an angle perpendicular to the direction of travel of the topmost braid wire 1162 so as to provide a true cross-section of the topmost braid wire 1162. Here the braid wire 1162 has a rectangular cross-section and provides a braid wire width 1168. According to various embodiments, the braid wire width ranges from about 2 mils to about 5 mils. The braid wire width is measured as the dimension that faces outward from the inner layer 1164 as shown in FIG. 2F. The rectangular cross-section has corners that may affect the bend stiffness of the lead 1108 but provides more coverage than a round cross-sectional shape that has a same height extending into an outer layer 1166 from the inner layer 1164.

FIG. 2G is an enlarged view of a portion of a lead 1170 to illustrate the cross-section of the braid wires. As in FIG. 2E, the view is a cross-section where the cut through the lead 1170 is taken at an angle perpendicular to the direction of travel of the topmost braid wire 1172 so as to provide a true cross-section of the topmost braid wire 1172. Here the topmost braid wire 1172 has an oval cross-section and provides a braid wire major axis diameter 1178. According to various embodiments, the braid wire major axis diameter ranges from about 0.5 mils to about 4 mils. The braid wire major axis diameter is measured as the dimension that faces outward from the inner layer 1176 as shown in FIG. 2G. The oval cross-section lacks corners that may affect the bend stiffness of the lead 1108 but provides coverage similar to a rectangular cross-section that has a same height extending into an outer layer 1174 from the inner layer 1176.

In each of the examples of FIGS. 2E-2G, regardless of the cross-sectional shape and the material used, the braid wires have an ultimate tensile strength satisfactory for implantation. According to the various embodiments, this ultimate tensile strength is at least 150,000 pounds per square inch (150 ksi).

FIG. 2H shows the lead 1108 from end to end with the shield 1118 in view to illustrate the termination of the shield 1118 at the proximal end 1105 and the distal end 1107. The shield 1118 terminates prior to reaching the most distal connector 1109 of the proximal end 1105 and prior to reaching the most proximal electrode 1116 of the distal end 1107. Terminating the shield 1118 at a distance 1117 from the connector 1109 and at a distance 1119 from the electrode 1116 reduces the likelihood of RF energy that radiates from the end of the shield, leaking from the shield onto the conductor filars and then to the connector 1109 and/or

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electrode 1116. However, the shield termination distances 1117, 1119 are not too large so that adequate coverage over the filars 1124 is maintained.

The shield termination distance from the distal electrodes and proximal connectors may vary. According to the various embodiments, the distance may range from about 0.5 millimeters to about 10 centimeters depending upon the location of the lead 1108. For instance, if the distal tip is located in the brain or spinal column where intensities of RF energy are lower, then distance from the end of the shield 1118 to the nearest edge of the distal electrode may be from 0.5 mm up to about 10 cm, or from about 2 mm to 2 cm to further reduce electrode coupling and filar exposure. However, in other locations where the entire lead 1108 is just under the skin as for peripheral nerve stimulation, the distance from the end of the shield 1118 to the nearest edge of the distal electrode may be less than about 2 cm to prevent overexposure of the filars 1124. In these cases, the distance may be on the order of 2 mm or more to ensure that excessive RF coupling from the shield 1118 to the electrodes is avoided.

In one particular example, the lead 1108 is provided with a shield 1118 where the total lead diameter is 53.6 mils. The inner insulation layer 1122 has an as assembled inside diameter of 35 mils and an as assembled outside diameter of 50.19 mils for a total thickness of 5.89 mils or 5.39 mils to the inner edge of the braid wire. The outside insulation layer 1120 has an as assembled outside diameter of 53.6 mils and a total thickness of 3.41 mils, with 1.41 mils of thickness existing over braid wire intersection points and while the thickness over a single braid wire approaches 2.66 mils as the single braid wires approaches an intersection point where the single braid wire will pass under an intersecting braid wire. The braid wire is round in cross-section with a diameter of 1.25 mils and being embedded by about 0.5 mils into the inner layer 1122. Two sets of eight braid wires are provided for a total of sixteen braid wires, with the braid wires establishing a braid angle of 22 degrees with an axial spacing between adjacent braid wires of 7.5 mils. The shield 1118 terminates about 2 mm from the nearest edge of the distal electrode and proximal connector.

In another particular example, the lead 108 is provided with the specifications described in the preceding paragraph except that the shield gaps and depth the shield sinks into the inner insulation layer 1122 are different. Here, the shield 1118 terminates about 1 mm from the nearest edge of the distal electrode and proximal connector and the shield sinks 0.25 mil. As a result, the inner insulation thickness to the inner edge of the braid wire is 5.6 mils.

In another particular example, the lead 108 is provided with the specifications described in the preceding paragraph except insulation thicknesses, braid angle, and proximal shield gaps differ. In this example, the braid depth from the outer surface of the outer layer 1120 to the outer edge of a braid wire is about 2 mils at braid wire intersection points while the thickness over the braid wire approaches 3.25 mils as the single braid wire approaches an intersection point where the braid wire passes under an intersecting braid wire. The inner insulation layer 122 has an average thickness of 4.5 mils to the inner edge of the braid wire while the braid wire sinks into the inner insulation layer 1122 by about 0.25 mil. The shield 1118 terminates about 1.27 mm from the nearest edge of the distal electrode and terminates about 10 mm from the nearest edge of the proximal connector. The braid angle is about 23 degrees.

Embodiments disclosed in relation to FIGS. 3-10C provide for radio frequency (RF) grounding of a shield present within an implantable lead. The shield may be grounded in

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various ways such as to a can of an implantable medical device (IMD) or to a ground plate on a header of the IMD. The pathway for grounding may be a direct current pathway or be capacitively coupled. The pathway for grounding the shield may couple to the shield at a point along the lead that is external to the header of the IMD or may couple to the shield at a point within the header.

FIG. 3 shows an example of an implantable medical system **2100** that includes an IMD **2102** coupled to a lead **2108**. The IMD **2102** includes a metal can **2104**, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD **2102** includes a header **2106** typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can **2104**. The header **2106** is shown transparently for purposes of illustration. The header **2106** provides a structure for securing the lead **2108** to the IMD **2102** and for establishing electrical connectivity between circuitry of the IMD **2102** and electrodes of the lead **2108**.

The lead **2108** includes electrodes **2116** on a proximal end that are positioned at a stimulation site within a patient. The lead also includes connector rings **2110** on a proximal end that is positioned within the header **2106**. The connectors **2110** make physical contact with electrical connections **2111** within the header. The electrical connections **2111** may include a metal contact that the electrode **2110** rests against upon being inserted into the header **2106** where a wire extends from the metal contact into the can **2104** where the circuitry is housed. Signals applied by the IMD **2102** to the electrodes **2110** are conducted through the lead **2108** to the electrodes **2116** to provide the stimulation therapy to the patient.

The lead **2108** is secured in the header **2106** such as by a set screw block **2112** within the header **2106** that allows at least one set screw **2114** to be tightened against at least one of the connectors **2110**. The shield **2118** may be grounded by metal contacts provided along the lead to establish a ground pathway from the shield **2118** to the tissue. As another option, the shield **2118** may be located within the lead **2108** at a small distance from the surface so that the shield **2118** will effectively capacitively couple to the tissue along the length of the lead to dissipate energy to the tissue over the length.

FIGS. 4A and 4B show an example of the lead **2108**, where a shield **2118** is present. An outer jacket layer **2120** is shown transparently in FIG. 4A for purposes of illustrating the shield **2118**. The shield **2118** blocks at least some RF energy from directly coupling to conductive filars **2124** that are present within the lead **2108**. The conductive filars **2124** extend the length of the lead and interconnect the proximal connectors **2110** to the distal electrodes **2116** so that stimulation signals are conducted from the proximal end to the distal end of the lead **2108**.

As shown in FIG. 4A, the shield **2118** of this example is a braided metal wire. The metal wire may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield **2118**, particularly for embodiments where a portion of the shield **2118** may be exposed for purposes of grounding. While the shield **2118** is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead **2108** in an overlapping manner or an outer layer **2120** that is heavily doped with conductive particles.

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As shown in FIG. 4B, the shield **2118** may be embedded within the jacket of the lead **2108**. One manner of constructing the lead **2108** with the shield **2118** is to provide an inner jacket **2122** that encloses the filars **2124** and any additional insulation layer **2126**, such as polytetrafluoroethylene (PTFE) that may surround each filar **2124**. The shield **2118** may then reside on the outer portion of the inner jacket **2122**, and the outer jacket **2120** may then enclose the shield **2118**. The outer jacket **2120** maybe added over the braid **2118**, or it may be extruded over the braid.

For embodiments where it is desirable for the shield **2118** to RF couple to tissue, typically as a capacitive coupling, either as an alternative to grounding at the can of the IMD or in addition to grounding at the can, the amount of the outer jacket layer **2120** covering the shield **2118** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **2118** grounds at the can of the IMD and grounding via a capacitive coupling from the shield through the outer jacket **2120** directly to the tissue is of less significance, then the shield **2118** may be located further from the outer surface of the lead **2108**.

The inner and outer jackets **2122**, **2120** may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes, and silicones. A lumen **2128** may be present inside of the inner jacket **2122** around which the insulated filars **2124** are coiled or otherwise positioned. The lumen **2128** may be useful, particularly for percutaneous leads **2108**, to allow a stylet to be inserted for purposes of pushing and steering the lead **2108** into the desired position within the patient.

FIG. 4C shows one example of exposing the shield **2118** for purposes of grounding the shield **2118**. In this example, the outer layer **2120** of the jacket has been removed at first point along the lead **2108** near the proximal end to expose the shield **2118** and the inner jacket **2122**. For example, an excimer laser may be used to ablate the outer layer **2120**. Physical contact may then be established between the shield **2118** and an electrode attached to the lead, a spring loaded connector or a connector block, a wire, or other direct current or capacitive coupling. For instance, a ground wire could be adhesively bonded with glue or tape in contact with the exposed shield **2118**. Depending upon the embodiment, this first point along the lead where the shield **2118** is exposed may be located either inside or outside of the header of the IMD. Furthermore, depending upon the embodiment the coupling to the exposed shield **2118** may be a direct current coupling or a capacitive coupling, either providing a pathway for RF current to pass to ground.

FIG. 4D shows another example of providing a pathway to ground the shield **2118**. Here, an electrode **2130** is attached at the first point along the lead **2108** near the proximal end to provide a robust physical connection to a spring loaded connector, a connector block, a wire, or other direct current or capacitive coupling. Depending upon the embodiment, this first point along the lead where the electrode **2130** is positioned may be located either inside or outside of the header of the IMD. Furthermore, depending upon the embodiment a coupling to the electrode **2130** may be a direct current coupling or a capacitive coupling, either providing a pathway for RF current to pass to ground.

FIGS. 5A-5C show embodiments of grounding the shield to the can of the IMD by using a connector block mounted on the IMD and coupling a grounding path to the shield outside of a header of the IMD. The implantable medical system **2150** includes an IMD **2152** having a metal can **2154**

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and a header **2156**. One or more leads **2164** extend from the header and pass through a connector block **2158** that is mounted to the can **2154**.

The connector block **2158** includes features to ground the shield of the lead **2164** to the can **2154**, such as a connector **2160** and a can attachment **2162**. For instance, the connector block **2158** may be constructed of a biocompatible plastic or other non-conductor while the connector **2160** provides conduction to the can attachment **2162**. The can attachment **2162** may be of various forms. For example, a wire that extends from the connector **2160** to the can **2154** where the can attachment **2162** is welded or otherwise affixed to the can **2154**. As another example, the connector block **2158** may include a metal plate that contacts the metal can **2154** via a weld or other attachment.

FIG. 5B is a side view showing a pair of pass-through features of the connector block **2158** and a pair of leads **2164** having shields to be grounded. The connector block **2158** is shown in a cross-section so that a set screw **2168** is visible. The electrode or other contact for the shield of the lead **2164** is positioned within the pass-through **2166** such that the set screw **2168** and the electrode or other contact for the shield are aligned. The set screw **2168** is tightened against the electrode or other contact to establish the ground to the can **2154**. The pass-through **2166** may be a slot through the connector block **2158** so that the lead **2164** can be lowered into the slot. As another option, the pass-through **2166** may be a bore through the connector block **2158** and the lead **2164** is fed through the bore.

FIG. 5C is a side view showing a pair of pass-through features of another embodiment of the connector block **2158** and a pair of leads **2164** having shields to be grounded. The connector block **2158** includes spring loaded connectors **2172**. The electrode or other contact for the shield of the lead **2164** is positioned within the pass-through **2170** such that the spring loaded connector **2172** and the electrode or other contact for the shield are aligned. The electrode or other contact to the shield is forced within the spring loaded connector **2172** to establish the ground to the can **2154**. As with the embodiment of FIG. 5B, the pass-through **2170** of this embodiment may be a slot through the connector block **2158** so that the lead **2164** can be lowered into the slot or may be a bore through the connector block **2158** where the lead **2164** is fed through the bore.

FIG. 6 shows one example of a spring loaded connector **2174**. The spring loaded connector **2174** can open slightly when forced by insertion of the lead **2164** and then is biased back against the electrode or other contact of the lead **2164** once the lead is seated within the spring loaded connector **2174**. Other spring loaded connector designs are also applicable.

FIG. 7A shows an implantable medical system **2180** where the shield of a lead **2188** is being grounded to a metal can **2184** of an IMD **2182** externally of the header **2186**. Here, a direct current pathway is being provided between the shield and the metal can **2184**. A spring loaded connector **2192** contacts an electrode **2190** on the lead **2188** where the electrode **2190** is in contact with the shield. A wire **2194** may be made from materials such as titanium, tantalum, platinum, stainless steel, nickel chromium, and alloys, and serves as a ground conductor. This wire **2194** is attached to the spring loaded connector **2192** by a weld or other bond. The wire **2194** extends from the spring loaded connector **2192** to the metal can **2184** where a weld **2196** or other bond such as with glue or tape attaches the wire **2194** to the metal can **2184**.

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FIG. 7B shows an implantable medical system **2200** where the shield of a lead **2208** is being grounded to a metal can **2204** of an IMD **2202** externally of the header **2206**. Here, a direct current pathway is also being provided between the shield and the metal can **2204**. A metal connector block **2210** having a set screw **2212** contacts an electrode on the lead **2208** where the electrode is in contact with the shield. A wire **2214** serving as a ground conductor is attached to the connector block **2210** by a weld or other bond. The wire **2214** extends from the connector block **2212** to the metal can **2204** where glue **2216**, such as a conductive epoxy or carbon filled polymer adhesive, or other bond such as a weld or tape attaches the wire **2214** to the metal can **2204**.

FIG. 7C shows an implantable medical system **2220** where the shield of a lead **2228** is being grounded to a metal can **2224** of an IMD **2222** externally of the header **2226**. Here, a direct current pathway is also being provided between the shield and the metal can **2224**. A coupling **2230** such as a ring electrode is in contact with the shield. A wire **2232** serving as a ground conductor is attached to the coupling **2230** by a weld or other bond. The wire **2232** extends from the coupling **2230** to the metal can **2224** where a crimp connector **2234** or other bond such as a weld or tape attaches the wire **2232** to the metal can **2224**.

For the examples of FIGS. 7A-7C, various examples of connecting the grounding wire to the lead and to the can are disclosed. It will be appreciated that any combination of these and other examples of connections of the ground wire may be used to provide the direct current pathway that ultimately provides an RF ground from the shield to the metal can.

FIG. 8A shows an implantable medical system **2240** where the shield of a lead **2248** is being grounded to a metal can **2244** of an IMD **2242** externally of the header **2246**. Here, a capacitively coupled pathway is being provided between the shield and the metal can **2244**. A coupling **2250** such as a spring loaded connector or a ring electrode contacts the lead **2248** and is in contact with the shield. A wire **2252** serving as a ground conductor is attached to the coupling **2250** by a weld or other bond. The wire **2252** extends from the coupling **2250** to nearby the metal can **2244** where a piece of tape **2254** or other tab affixed to the can **2244** attaches to the wire **2252**. The tape **2254**, such as double-sided tapes, epoxies, or polymer based adhesive, or other tab holds the wire in proximity to the metal can **2244** to establish a capacitive coupling between the wire **2252** and the can **2244**.

FIG. 8B shows an implantable medical system **2260** where the shield of a lead **2268** is being grounded to a metal can **2264** of an IMD **2262** externally of the header **2266**. Here, a capacitively coupled pathway is being provided between the shield and the metal can **2264**. A piece of tape **2270** or other tab contacts the lead **2268** at a point where the shield is present. A wire **2272** serving as a ground conductor is attached to the tab **2270** and is held nearby the lead **2268** and shield to establish a capacitive coupling between the wire **2272** and the shield. The wire **2272** extends from the tab **2270** to the metal can **2264** and is affixed to the metal can **2264** with a weld **2274** or other bond.

FIG. 8C shows an implantable medical system **2280** where the shield of a lead **2288** is being grounded to a metal can **2284** of an IMD **2282** externally of the header **2286**. Here, a capacitively coupled pathway is being provided between the shield and the metal can **2284**. A coupling **2290** such as a spring loaded connector or a ring electrode contacts the lead **2288** and is in contact with the shield. A

wire **2292** serving as a ground conductor is attached to the coupling **2290** by a weld or other bond and extends from the coupling **2290** to nearby the metal can **2284**. Non-conductive glue or another non-conductive bond **2294** to the can **2284** is present to adhere to the wire **2292** and hold the wire in proximity to the metal can **2284** to establish a capacitive coupling between the wire **2292** and the can **2284**.

For the examples of FIGS. 8A-8C, various examples of connecting the grounding wire to the lead and to the can are disclosed, using combinations of direct current couplings and capacitive couplings. It will be appreciated that any combination of these and other examples of direct current coupling and capacitive coupling connections of the ground wire may be used to provide the capacitively coupled pathway that ultimately provides an RF ground from the shield to the metal can.

FIG. 9A shows an implantable medical system **2300** where the shield of a lead **2308** is being grounded to a metal can **2304** of an IMD **2302** within the header **2306**. Proximal electrodes **2310** of the lead **2308** are electrically connected via wires **2312** to the IMD **2302**. Here, a direct current coupled pathway is being provided between the shield and the metal can **2304**. A coupling **2314** such as a spring loaded connector or a ring electrode contacts the lead **2308** and is in contact with the shield. A set screw **2316** may be present to further hold the proximal end of the lead **2308** in place within the header **2306**. A wire **2318** serving as a ground conductor is attached to the coupling **2314** by a weld or other bond and extends from the coupling **2314** to the metal can **2304** where a weld **2320** or other bond holds the wire **2318** to the can **2304**.

FIG. 9B shows an implantable medical system **2330** where the shield of a lead **2338** is being grounded to a metal can **2334** of an IMD **2332** within the header **2336**. Proximal electrodes **2340** of the lead **2338** are electrically connected via wires **2342** to the IMD **2332**. Here, a direct current coupled pathway is being provided between the shield and the metal can **2334**. A coupling **2344** such as a spring loaded connector or a ring electrode contacts the lead **2338** and is in contact with the shield. A wire **2346** serving as a ground conductor is attached to the coupling **2344** by a weld or other bond and extends from the coupling **2344** to the metal can **2334** where a weld **2348** or other bond holds the wire **2346** to the can **2334**.

FIG. 9C shows an implantable medical system **2350** where the shield of a lead **2358** is being grounded to a metal can **2354** of an IMD **2352** within the header **2356**. Proximal electrodes **2360** of the lead **2358** are electrically connected via wires **2362** to the IMD **2352**. Here, a capacitively coupled pathway is being provided between the shield and the metal can **2354**. A coupling **2364** such as a spring loaded connector or a ring electrode contacts the lead **2358** and is in contact with the shield. A wire **2366** serving as a ground conductor is capacitively coupled to the coupling **2364** within the header **2356** by the header structure holding the wire in proximity to the coupling **2364**. The wire **2366** extends from the capacitive coupling to the metal can **2354** where a weld **2368** or other bond holds the wire **2366** to the can **2354**.

FIG. 9D shows an implantable medical system **2370** where the shield of a lead **2378** is being grounded to a metal can **2374** of an IMD **2372** within the header **2376**. Proximal electrodes **2380** of the lead **2378** are electrically connected via wires **2382** to the IMD **2372**. Here, a capacitively coupled pathway is being provided between the shield and the metal can **2374**. A coupling **2384** such as a spring loaded connector or a ring electrode contacts the lead **2378** and is

in contact with the shield. A wire **2386** serving as a ground conductor is capacitively coupled to the coupling **2384** within the header **2376** by the header structure holding the wire in proximity to the coupling **2384**. The wire **2386** extends from the capacitive coupling toward the metal can **2374** and is capacitively coupled to the can **2374** within the header **2376** by the header structure holding the wire in proximity to the can **2374**.

FIG. 9E shows an implantable medical system **2390** where the shield of a lead **2398** is being grounded to a metal can **2394** of an IMD **2392** within the header **2396**. Proximal electrodes **2402** of the lead **2398** are electrically connected via wires **2404** to the IMD **2392**. Here, a capacitively coupled pathway is being provided between the shield and the metal can **2394**. A coupling **2406** such as a spring loaded connector or a ring electrode contacts the lead **2398** and is in contact with the shield. A shunt plate such as a tab **2408** or similar structure serving as a ground conductor extends from the coupling **2406** toward the can **2394** and is capacitively coupled to the can **2394** within the header **2396** by the header structure holding the tab **2408** in proximity to the can **2394**.

FIG. 9F shows an implantable medical system **2410** where the shield of a lead **2418** is being grounded to a metal can **2414** of an IMD **2412** within the header **2416**. Proximal electrodes **2420** of the lead **2418** are electrically connected via wires **2422** to the IMD **2412**. Within the can **2414**, filter feed through (FFT) circuits **2424** are present to capacitively couple the wires **2422** to the metal can **2414** while allowing connection of the wires **2422** to stimulation circuits. The FFT circuits **2424** for the electrodes **2420** protects the IMD **2412** from electromagnetic background noise picked up by the filars, albeit potentially less noise due to the presence of the shield.

Here, a capacitively coupled pathway is being provided between the shield and the metal can **2414** also via an FFT circuit **2430**. A coupling **2426** such as a spring loaded connector or a ring electrode contacts the lead **2418** and is in contact with the shield. A wire **2428** serving as a ground conductor extends from the coupling **2426** toward the can **2414** and terminates at the FFT circuit **2430** to provide the capacitive coupling between the shield and the can **2414**.

As shown, the coupling **2426** to the shield may be an existing electrode of the lead **2418** that provides stimulation signals to a filar within the lead **2418**. In that case, the FFT circuit **2430** may provide capacitive coupling to the can for both the filar and the shield. In such a case, it may be desirable to capacitively couple the shield to the coupling **2426** so that relatively low frequency stimulation signals are not present on the shield but induced RF current on the shield has a pathway to the FFT circuit **2430**. For example, the outer jacket may separate the shield from the electrode by a separation on the order of 0.5-5 mils to allow an RF coupling to occur. As an alternative to using the same coupling and FFT circuit for both the shield and the filar, the shield may be provided a dedicated coupling **2426** and a dedicated FFT circuit **2430** that are independent of any electrodes and filars within the lead **2418**.

For the examples of FIGS. 9A-9F, various examples of connecting the grounding conductor to the lead and to the can within the header are disclosed, using combinations of direct current couplings and capacitive couplings. It will be appreciated that any combination of these and other examples of direct current coupling and capacitive coupling connections of the ground conductor may be used to provide the capacitively coupled pathway that ultimately provides an RF ground from the shield within the header to the metal

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can. For instance, a capacitive coupling may be provided in any of the various embodiments at the coupling to the shield as discussed above in relation to FIG. 9F.

FIG. 10A shows an implantable medical system 2440 where the shield of a lead 2448 is being grounded to a metal can 2444 of an IMD 2442 outside of the header 2446. Here, a ground pathway is being provided between the shield and a ground plate 2454 installed on the header 2446. The ground plate provides a relatively large surface area in comparison to an individual electrode and allows for safe dissipation of induced RF current on the shield in the same manner as grounding to the can 2444. A coupling 2450 such as a spring loaded connector or a ring electrode contacts the lead 2448 and is in contact with the shield. A wire 2452 that serves as a ground conductor is attached to the coupling 2450 by a weld or other bond and extends from the coupling 2450 to the ground plate 2454 where a weld or other bond holds the wire 2452 to the ground plate 2454.

FIG. 10B shows an implantable medical system 2460 where the shield of a lead 2468 is being grounded to a metal can 2464 of an IMD 2462 within the header 2466. Here, a ground pathway is also being provided between the shield and a ground plate 2474 installed on the header 2466. A coupling 2470 such as a spring loaded connector or a ring electrode contacts the lead 2468 and is in contact with the shield. A wire 2472 that serves as a ground conductor is attached to the coupling 2470 by a weld or other bond and extends from the coupling 2470 to the ground plate 2474 where a weld or other bond holds the wire 2472 to the ground plate 2474.

FIG. 10C shows an implantable medical system 2480 where the shield of a lead 2488 is being grounded to a metal can 2484 of an IMD 2482 within the header 2486. Here, a ground pathway is being provided between the shield and a connector block 2492 with a relatively large surface area that also acts as a ground plate installed on the header 2486. In this example, the connector block 2492 is a set screw block that uses a set screw 2494 to tighten against a coupling 2490 on the lead 2488. The coupling 2490 such as a ring electrode contacts the lead 2448 and is in contact with the shield. A set screw 2494 extends from the coupling 2490 and through the connector block 2492 and acts as a ground conductor to provide the ground pathway from the shield to the connector block 2492. Other conductive features may also be present within the connector block 2492 to contact the coupling 2490 and provide the RF ground pathway.

For the examples of FIGS. 10A-10C, various examples of connecting the grounding conductor to the lead and to the ground plate are disclosed. It will be appreciated that any combination of direct current coupling and capacitive coupling connections may be used to provide the pathway that ultimately provides an RF ground from the shield to the ground plate. For instance, a capacitive coupling may be provided in any of the various embodiments at the coupling to the shield as shown in FIGS. 10A-10C and as discussed above in relation to FIG. 9F. Likewise, a capacitive coupling may be present between a ground conductor extending from the coupling to the shield and the ground plate.

Embodiments disclosed in relation to FIGS. 11-15E also provide for radio frequency (RF) grounding of a shield present within an implantable lead. The shield may be grounded in various ways such as directly to tissue at one or more points along the lead body. The pathway for grounding may be a direct current pathway or be capacitively coupled. The pathway for grounding may utilize an exposed or nearly exposed shield at one or more points along the lead body,

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metal conductors attached to the lead at one or more points, a jacket with a conductive doping at one more points, and so forth.

FIG. 11 shows an example of an implantable medical system 3100 that includes an IMD 3102 coupled to a lead 3108. The IMD 3102 includes a metal can 3104, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD 3102 includes a header 3106 typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can 3104. The header 3106 is shown transparently for purposes of illustration. The header 3106 provides a structure for securing the lead 3108 to the IMD 3102 and for establishing electrical connectivity between circuitry of the IMD 3102 and electrodes of the lead 3108.

The lead 3108 includes electrodes 3116 on a distal end that are positioned at a stimulation site within a patient. The lead also includes connector rings 3110 on a proximal end that is positioned within the header 3106. The connector rings 3110 make physical contact with electrical connections 3111 within the header. The electrical connections 3111 may include a metal contact that the connector ring 3110 rests against upon being inserted into the header 3106 where a wire extends from the metal contact into the can 3104 where the circuitry is housed. Signals applied by the IMD 3102 to the connector rings 3110 are conducted through the lead 3108 to the electrodes 3116 to provide the stimulation therapy to the patient.

The lead 3108 is secured in the header 3106 such as by a set screw block 3112 within the header 3106 that allows at least one set screw 3114 to be tightened against at least one of the electrodes 3110. With the lead 3108 in place, the shield 3118 of the lead 3108 may then become grounded to the body along one or more points down the length of the lead from the IMD 3102.

FIGS. 12A and 12B show an example of the lead 3108, where a shield 3118 is present. An outer jacket layer 3120 is shown transparently in FIG. 12A for purposes of illustrating the shield 3118. The shield 3118 blocks at least some RF energy from directly coupling to conductive filars 3124 that are present within the lead 3108. The conductive filars 3124 extend the length of the lead and interconnect the proximal electrodes 3110 to the distal electrodes 3116 so that stimulation signals are conducted from the proximal end to the distal end of the lead 3108.

As shown in FIG. 12A, the shield 3118 of this example is a braided metal wire. The metal wire may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield 3118, particularly for embodiments where a portion of the shield 3118 may be exposed for purposes of grounding. While the shield 3118 is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead 3108 in an overlapping manner or an outer layer 3120 that is heavily doped with conductive particles.

As shown in FIG. 12B, the shield 3118 may be embedded within the jacket of the lead 3108. One manner of constructing the lead 3108 with the shield 3118 is to provide an inner jacket 3122 that encloses the filars 3124 and any additional insulation layer 3126, such as polytetrafluoroethylene (PTFE) that may surround each filar 3124. The shield 3118 may then reside on the outer portion of the inner jacket 3122, and the outer jacket 3120 may then enclose the shield 3118.

The outer jacket **3120** maybe added over the braid **3118**, or it may be extruded over the braid.

For embodiments where it is desirable for the shield **3118** to RF couple to tissue, typically as capacitive coupling, either as an alternative to grounding at the can of the IMD or in addition to grounding at the can, the amount of the outer jacket layer **3120** covering the shield **3118** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **3118** grounds at one or more specific locations along its length, via a direct current coupling or a capacitive coupling, the shield may be located further from the outer surface of the lead **3108** with additional features of the lead providing the coupling at the one or more specific locations as discussed below.

The inner and outer jackets **3122**, **3120** may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes, and silicones. A lumen **3128** may be included inside of the inner jacket **3122** around which the insulated filars **3124** are coiled or otherwise positioned. The lumen **3128** may be useful, particularly for percutaneous leads **3108**, to allow a stylet to be inserted for purposes of pushing and steering the lead **3108** into the desired position within the patient.

FIG. 12C shows one example of exposing the shield **3118** at a particular point along the lead **3108** for purposes of grounding the shield **3118**. In this example, the outer layer **3120** of the jacket has been removed at a first point along the lead **3108** distant from the distal end to expose the shield **3118** and the inner jacket **3122**. For example, an excimer laser may be used to ablate the outer layer **3120**. Physical contact may then be established between the shield **3118** and the tissue or between the shield **3118** and an electrode attached to the lead.

FIG. 12D shows another example of providing a pathway to ground the shield **3118**. Here, a metal conductor, specifically a ring electrode **3130**, is attached at the first point along the lead **3108** distant from the distal end to provide a robust physical connection to the tissue while avoiding tissue in-growth that may occur if the shield **3118** is exposed directly. Depending upon the embodiment, a coupling of the shield **3118** to the electrode **3130** may be a direct current coupling or a capacitive coupling, either providing a pathway for RF current to pass to ground. The ring electrode **3130** may be attached by methods such as crimping, clamping, welding, and the like.

FIG. 12E shows an example of nearly exposing the shield **3118** at a particular point along the lead **3108** for purposes of grounding the shield **3118**. In this example, the outer layer **3120** of the jacket has been almost entirely removed at a first point along the lead **3108** distant from the distal end to nearly expose the shield **3118** and the inner jacket **3122**. Only a very thin layer **3120'**, on the order of about 0.5-5 mils, of the outer layer **3120** is remaining. Physical contact between the shield **3118** and the tissue is avoided so that tissue in-growth does not occur, and the shield **3118** capacitively couples to the tissue to provide the RF pathway to ground.

FIG. 12F shows an example of exposing, or nearly exposing, the shield **3118** at a plurality of points **3202** along the lead. At these points **3202**, the outer layer **3120** has been at least partially ablated or otherwise removed to place the shield **3118** in closer proximity to the body tissue so that an RF pathway to ground is established. Where the shield **3118** is exposed, the RF pathway is a direct current coupling to the tissue. Where the shield **3118** is nearly exposed, the RF pathway is a capacitive coupling to the tissue.

Where multiple points of the RF pathway to ground are present, a particular separation of the multiple points is provided. A nearest edge-to-nearest edge distance between one point and an adjacent one is shown by the distance from edge **3204** to edge **3206**. Where the outer layer is removed, the flexibility and strength of the lead is altered for the region including those points and this distance from edge **3204** to edge **3206** can be used to control the flexibility and strength.

Where multiple points of the RF pathway to ground are direct current couplings, another concern is current induced by the gradient magnetic fields present in a magnetic resonance (MR) scan. If the most proximal and most distal points of the direct current coupling are spaced too far apart, then the magnetic gradient may induce a dangerous current through the shield and produce a significant stimulation of tissue at those ground points along the lead. Therefore, choosing the nearest edge-to-nearest edge separation to fall within an illustrative range of 2 millimeters (mm) or more with a most proximal to most distal separation, such as edge **3204** to edge **3208**, of about 40 centimeters (cm) or less may allow for flexibility of the lead in the region while maintaining small loops that prevent large magnetic gradient induced currents should the shield be exposed at the points **3202**.

FIG. 12G shows an example of coupling the shield **3118** to ground with a plurality of metal conductors such as rings **3130** at a plurality of points **3210** along the lead. At these points **3202**, the outer layer **3120** has been at least partially ablated or otherwise removed to place the shield **3118** in close proximity with the metal conductors **3130** so that an RF pathway to ground is established through the metal conductors **3130**. Where the shield **3118** is exposed to the metal conductors **3130**, the RF pathway is a direct current coupling to the tissue. Where the shield **3118** is nearly exposed to the metal conductors **3130**, the RF pathway is a capacitive coupling to the tissue.

As with the example of FIG. 12F, where multiple points of the RF pathway to ground are present, a particular separation of the multiple points is provided. A nearest edge-to-nearest edge distance between one point and an adjacent one is shown by the distance from edge **3212** to edge **3214**. The flexibility and strength of the lead is altered for the region including those points, with the metal conductors **3130** limiting the bending in this region to essentially those sections of lead between the metal conductors **3130**. Thus, in one example, a nearest edge-to-nearest edge distance may be maintained at or above 2 mm or 50% of the grounding ring length so that flexibility of the lead is maintained.

Also, where the shield **3118** is direct current coupled to the metal conductors **3130**, a magnetic gradient induced current is of concern because the metal conductors **3130** have a direct current coupling to the tissue. In that case, the separation of the most proximal to the most distal may be kept within a range that prevents a large loop and avoids a large magnetic induced gradient current. In this particular example, the most proximal to the most distal distance, such as from edge **3212** to edge **3216**, may be maintained at or below approximately 40 cm so that magnetic gradient induced currents are insignificant.

FIG. 12H shows a cross-section of the lead **3108** at a particular point where the outer jacket **3120** has been ablated or otherwise removed. In this example, the lead **3108** at this particular point includes a metal conductor **3130** with a direct current coupling to the shield **3118**. The outer layer **3120** of the jacket has been removed to allow the metal

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conductor **3130**, a ground ring as shown, to wrap around the lead and contact the shield **3118**. The filars may be present within the inner jacket **3122** or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket **3122** or any other inner layer as shown in FIG. 12B.

FIG. 12I shows a cross-section of the lead **3108** at a particular point where the outer jacket **3120** has been ablated or otherwise removed. In this example, the lead **3108** at this particular point has the shield **3118** exposed to tissue for a direct current coupling by entirely removing the outer layer **3120**. The filars may be present within the inner jacket **3122** or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket **3122** or any other inner layer as shown in FIG. 12B.

FIG. 12J shows a cross-section of the lead **3108** at a particular point where a portion of the outer jacket **3120** has been ablated or otherwise removed. In this example, the lead **3108** at this particular point includes a metal conductor **3130** with a capacitive coupling to the shield **3118**. The outer layer **3120** of the jacket has been partially removed, with a remaining thickness of about 0.5 mils to 5 mils, to nearly expose the shield **3118**. This allows the metal conductor **3130**, a ground ring as shown, to wrap around the lead and capacitively couple with the shield **3118** at RF frequencies. The filars may be present within the inner jacket **3122** or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket **3122** or any other inner layer as shown in FIG. 12B.

FIG. 12K shows a cross-section of the lead **3108** at a particular point where a portion of the outer jacket **3120** has been ablated or otherwise removed. The outer layer **3120** of the jacket has been partially removed, with a remaining thickness of about 0.5 mils to 5 mils, to nearly expose the shield **3118**. The shield **3118** capacitively couples to the tissue at RF frequencies. The filars may be present within the inner jacket **3122** or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket **3122** or any other inner layer as shown in FIG. 12B.

FIG. 13A shows an example of a ring electrode **3220** that may be attached to a lead **3108** to form the RF pathway to ground from the shield **3118**. The ring electrode may be constructed of platinum, platinum-iridium, titanium, tantalum, stainless steel, and other similar biocompatible metals. The ring electrode **3220** may have a gap **3222**. The ring electrode **3220** may be sprung open to fit around the lead at the particular point where the jacket has been ablated, and the ring electrode **3220** is crimped back into a tightly fitting configuration. As another example, the ring electrode **3220** may be flat and then rolled into the ring shape about the lead. In some examples, the gap **3222** may close upon crimping while in other embodiments the gap **3222** may remain to some degree.

FIG. 13B shows an example of another ring electrode **3224** that may be attached to a lead **3108** to form the RF pathway to ground from the shield **3118**. The ring electrode **3224** includes a tab **3226** that extends away from the lead to provide an additional surface area and extension into the tissue for adding grounding of the shield **3118**. The ring electrode **3220** may be sprung open to fit around the lead at the particular point where the jacket has been ablated and the ring electrode **3220** is crimped back into a tightly fitting configuration. As in the previous example, the ring electrode **3224** may be flat and then rolled into the ring shape about the lead while maintaining a flat portion as the tab **3226**.

FIG. 13C shows an example of another ring electrode **3228** that may be attached to a lead **3108** to form the RF

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pathway to ground from the shield **3118**. The ring electrode **3228** forms a helix. The ring electrode **3228** may be sprung open to fit around the lead at the particular point where the jacket has been ablated and the ring electrode **3228** is crimped back into a tightly fitting helical configuration. As in the previous examples, the ring electrode **3228** may be flat and then rolled into the helical ring shape about the lead.

FIG. 13D shows an example of a ring electrode **3230** that may be attached to a lead **3108** to form the RF pathway to ground from the shield **3118**. The ring electrode **3230** has an outer side **3234** that faces away from the shield **3118** and an inner side **3236** that faces toward the shield **3118** and may directly contact the shield **3118**. In this example, the outer side **3234** has a non-conductive coating **3232** applied so that the outer side **3234** does not have a direct coupling to the tissue. The non-conductive coating may be of various types such as polyurethane, silicone or other biocompatible polymers.

The inner side **3236** may either have a direct current coupling or a capacitive coupling to the shield. With multiple ring electrodes **3230** in place on a lead, magnetic gradient induced current which is at a relatively low frequency is not a concern because the non-conductive coating **3232** prevents the relatively low frequency induced current from flowing to the tissue. Thus, the distance between adjacent electrodes is not limited by induced current concerns. Meanwhile, the high frequency RF induced current does ground to the tissue through the capacitive coupling provided by the non-conductive coating **3232**.

FIG. 13E shows an example of a ring electrode **3240** that may be attached to a lead **3108** to form the RF pathway to ground from the shield **3118**. The ring electrode **3240** has an outer side **3242** that faces away from the shield **3118** and an inner side **3244** that faces toward the shield **3118** and may directly contact the shield **3118**. In this example, the inner side **3244** has a non-conductive coating **3246** applied so that the inner side **3244** does not have a direct coupling to the shield **3118** even if the shield **3118** is entirely exposed to the ring electrode **3240**. The non-conductive coating **3246** may be of the various types discussed above in the previous example.

The outer side **3242** may have a direct current coupling to the tissue. With multiple ring electrodes **3240** in place on a lead, magnetic gradient induced current is not a concern because the non-conductive coating prevents the relatively low frequency induced current from flowing from the shield **3118** to the ring electrode **3240**. Thus, the distance between adjacent electrodes is not limited by induced current concerns. Meanwhile, the high frequency RF induced current does ground through the ring electrode **3240** to the tissue through the capacitive coupling provided by the non-conductive coating **3246**.

FIG. 13F shows an example of a ring electrode **3250** that may be attached to a lead **3108** to form the RF pathway to ground from the shield **3118**. The ring electrode **3250** has an outer side **3252** that faces away from the shield **3118** and may directly contact the tissue and an inner side **3254** that faces toward the shield **3118** and may directly contact the shield **3118**. In this example, both the inner side **3254** and the outer side **3252** have a non-conductive coating **3256** applied. The inner side **3254** does not have a direct current coupling to the shield **3118** even if the shield is entirely exposed to the ring electrode **3250**. The outer side **3252** does not have a direct current coupling to the tissue even if in physical contact with the tissue. The non-conductive coating **3256** may be of the various types discussed above in the previous examples.

With multiple ring electrodes **3250** in place on a lead, magnetic gradient induced current is not a concern because the non-conductive coating prevents the relatively low frequency induced current from flowing from the shield **3118** to the ring electrode **3250**. Thus, the distance between adjacent electrodes is not limited by induced current concerns. Meanwhile, the high frequency RF induced current does ground through the ring electrode **3250** to the tissue through the capacitive couplings on each side of the ring electrode **3250** provided by the non-conductive coating **3256**.

While the examples of FIGS. **13A-13F** show various shapes of ring electrodes, it will be appreciated that various other shapes are also applicable for metal conductors being attached to the lead to provide the RF ground pathway. Furthermore, while FIGS. **13D-13F** show a particular ring electrode shape with a non-conductive coating, it will be appreciated that the non-conductive coating is applicable to either or both sides of any of the metal conductor configurations including those of FIGS. **13A-13C**.

FIG. **14A** shows a cross-section of a lead **3260** that includes an outer jacket layer **3266** that surrounds a shield **3262** and an inner jacket layer **3264**. The outer jacket layer **3266** is doped with conductive particles **3268** at a particular point along the length of the lead. These conductive particles **3268** provide RF conductive qualities for the outer jacket layer **3266**. Thus, the RF energy couples from the shield **3262** to the tissue through the doped outer jacket layer **3266**. Examples of the conductive particles include carbon, tantalum, titanium, platinum, platinum-iridium, and other biocompatible conductive substances. The filars may be present within the inner jacket **3264** or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket **3264** or any other inner layer like that shown in FIG. **12B**.

FIG. **14B** shows the lead **3260** with a plurality of points **3272** along the lead where the conductive particles **3268** are present within the outer layer **3266**. The doped outer layer **3266** is exposed to create the RF pathway to ground from the outer layer **3266** to the tissue.

FIG. **14C** shows a cross-section of a lead **3260** where the outer jacket layer **3266** that surrounds a shield **3262** has been removed via ablation or other technique to expose the shield **3262** and the inner jacket layer **3264**. Here, the inner jacket layer **3264** is doped with conductive particles **3268** at least at the particular point(s) along the length of the lead where the outer layer **3266** has been removed. These conductive particles **3268** provide RF conductive qualities for the outer portion of the inner jacket layer **3264** where the shield **3262** is present. Thus, the RF energy couples from the shield **3262** to the tissue through the doped jacket layer **3264**. The filars may be present within the inner jacket **3264** or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket **3264** or any other inner layer like that shown in FIG. **12B**.

FIG. **14D** shows the lead **3260** with a plurality of points **3274** along the lead where the conductive particles **3268** are present within the inner layer **3264**. The outer layer **3266** is removed at these points **3274** to expose the doped inner layer **3264** and to create the RF pathway to ground from the inner layer **3264** to the tissue.

FIGS. **15A** and **15B** show an example of an implantable medical lead **3108** where a lead anchor **3280** is attached. In this example, the lead anchor **3280** is an RF conductor to the tissue to provide the ground pathway for the shield **3118**. In this particular example, the lead **3108** includes a ring electrode **3130** that is coupled to the shield **3118**, either via a

direct current coupling or a capacitive coupling. The lead anchor **3280** is constructed of metal or other conductor, or at least has a portion that is or conductive and directly contacts or nearly contacts the ring electrode **3130** and the tissue to ground the shield **3118** at RF frequencies. This ground pathway is secured in place via the conventional mounting of the lead anchor **3280** to the lead body and by the wings **3282** being sutured in place to the tissue. The filars may be present within the inner jacket or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket or any other inner layer like that shown in FIG. **12B**.

FIGS. **15C** and **15D** show another example of an implantable medical lead **3108** where a lead anchor **3280** is attached. In this example, the lead anchor **3280** is an RF conductor to the tissue to provide the ground pathway for the shield **3118**. In this particular example, the lead **3108** does not have a ring electrode **3130** coupled to the shield **3118**. However, as seen in FIG. **15D**, the lead anchor has gripping teeth **3284** that sink into the outer layer of the jacket and either directly contact or nearly contact the shield **3118**. The filars may be present within the inner jacket or any other inner layer as shown or within the lumen created by the inside wall of the inner jacket or any other inner layer like that shown in FIG. **12B**.

Directly contacting the shield **3118** creates a direct current coupled RF pathway while nearly contacting the shield **3118** creates a capacitively coupled RF pathway. As with the previous example, the lead anchor **3280** is constructed of metal or other conductor, or at least has a portion that is conductive and contacts or nearly contacts the tissue to ground the shield **3118** at RF frequencies. This ground pathway is secured in place via the conventional mounting of the lead anchor to the lead body and by the wings **3282** being sutured in place to the tissue.

In these embodiments, the anchor may capacitively couple to the shield **3118** without teeth or rings being present, particularly where the depth of the shield within the outer layer **3120** is relatively small. For example, for depth of the shield **3118** of about 5 mils or less, the anchor may reside on the outer layer **3120** and capacitively couple to the shield **3118** to provide the RF pathway to ground.

FIG. **15E** shows an example of an implantable medical lead **3108** where a lead anchor **3290** is attached. In this example, the lead anchor **3290** is an RF conductor to the tissue to provide the ground pathway for the shield **3118**. However, in this particular example, lead anchor **3290** provides a capacitive coupling to ground by utilizing a non-conductive outer material or coating **3292** to contact the tissue. The lead anchor **3290** may have either a direct current coupling or capacitive coupling to the shield **3118**, or ring electrode **3130** if any. The capacitive coupling to the tissue prevents the lead anchor **3290** from becoming a magnetic gradient induced current electrode, such as where other shield electrodes are present at other points along the lead **3108**.

Utilizing an anchor to provide an RF pathway to ground, as shown in FIGS. **15A-15E**, may also be useful considering that the typical mounting location of an anchor is at a point where the intensities of the RF fields change. For instance, an anchor may be positioned near the entry hole of the cranium where the lead **3108** is used for brain stimulation. The intensities of the field may change from one side of the entry hole to the other and providing the RF pathway to ground via an anchor near the entry hole may assist in dissipating energy received by the shield **3118** externally of the entry hole to prevent such energy from traveling through

the shield **3118** and through the entry hole toward the shield termination which is closer to the stimulation electrodes.

Embodiments disclosed in relation to FIGS. **16-17G** provide for shielding of both an implantable medical lead and an implantable lead extension. The two shields are interconnected with a radio frequency (RF) conductive path to maintain a continuity of the shielding along the length between the implantable medical device (IMD) and the stimulation site.

FIG. **16** shows an example of an implantable medical system **4100** that includes an IMD **4102** coupled to a lead **4108**. The IMD **4102** includes a metal can **4104**, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD **4102** includes a header **4106** typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can **4104**. The header **4106** is shown transparently for purposes of illustration. The header **4106** provides a structure for securing the lead extension **4142** to the IMD **4102** and for establishing electrical connectivity between circuitry of the IMD **4102** and distal connectors of the lead extension **4142** that are located in a distal housing **4140**.

The extension **4142** also includes ring connectors **4110** on a proximal end that is positioned within the header **4106**. The ring connectors **4110** make physical contact with electrical connections within the header. The electrical connections may include a metal contact that the ring connector **4110** rests against upon being inserted into the header **4106** where a wire extends from the metal contact into the can **4104** where the circuitry is housed. Signals applied by the IMD **4102** to the ring connectors **4110** are conducted through the extension **4142** to the connectors within the housing **4140** to provide the stimulation signals to the lead **4108**. The extension **4142** is secured in the header **4106** such as by a set screw block **4112** within the header **4106** that allows at least one set screw **4114** to be tightened against at least one of the ring connectors **4110**.

The lead **4108** includes electrodes **4116** on a distal end that are positioned at a stimulation site within a patient. The lead **4108** also includes ring connectors on a proximal end that is positioned within the housing **4140**. The ring connectors make physical contact with electrical connections within the housing **4140**. The electrical connections may include a metal contact such as a Bal Seal® connector of the Bal Seal Engineering, Inc. of Foothill Ranch, CA, another spring loaded connector, or a set screw block that the electrode rests against upon being inserted into the housing **4140**. A wire extends from the metal contact of the housing **4140** into the extension **4142** to connect with the filars of the extension **4142**. Signals applied by the IMD **4102** to the ring connectors **4110** are conducted through the extension **4142** and lead **4108** to the electrodes **4116** to provide the stimulation therapy to the patient.

The lead **4108** is secured in the housing **4140** such as by a set screw block within the housing **4140** that allows at least one set screw to be tightened against at least one of the electrodes. A shield **4144** of the extension and a shield **4118** of the lead **4108** that are discussed below with reference to FIGS. **17A-17G** are present to prevent the induced RF current on the filars. The shields **4118**, **4144** may be grounded at the IMD **4102** of FIG. **16** or at various grounding points established along the extension **4142** and/or lead **4108**. As another option, the shield **4144** of the extension **4142** and/or the shield **4118** of the lead **4108** may be located within the extension **4142** or lead **4108** at a small distance from the surface so that the shields **4118**, **4144** will effec-

tively capacitively couple to the tissue along the length of the lead to dissipate energy to the tissue over the length. In any of these cases, continuity may be maintained between the shields **4118** and **4144** as discussed herein

FIGS. **17A-17G** show examples of the extension **4142** and lead **4108** where shields **4118**, **4144** are present. The lead **4108** is inserted through an opening **4146** in the housing **4140** on the distal end of the extension **4142**. Outer jacket layers **4120**, **4141** for the lead **4108** and extension **4142** are shown transparently in FIG. **17A** for purposes of illustrating the shields **4118**, **4144**. The shields **4118**, **4144** block at least some RF energy from directly coupling to conductive filars that are present within the lead **4108** and extension **4142**. The conductive filars extend the length of the extension **4142** and lead **4108** and interconnect the proximal connector rings **4110** of the extension **4142** to the distal electrodes **4116** of the lead **4108** so that stimulation signals are conducted to the stimulation site.

As shown in FIG. **17A**, the shields **4118**, **4144** of this example are braided metal wires. The metal wire may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desirable to utilize a biocompatible metal for the shields **4118**, **4144**, particularly for embodiments where a portion of the shields **4118**, **4144** may be exposed for purposes of grounding. While the shield **4118** is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead **4108** in an overlapping manner or an outer layer **4120** that is heavily doped with conductive particles.

FIG. **17A** also shows a set screw block **4143** present on the housing **4140**. The set screw block **4143** may be used to fix the proximal end of the lead **4108** in place within the opening **4146** of the housing **4140** where a set screw is tightened against a connector ring on the lead **4108**. Other manners of fixing the lead **4108** within housing **4140** may also be used.

FIG. **17B** shows a coupling of the lead **4108** to the housing **4140** as a cross-section taken through the coupling of a shield connector **4132** to a shield electrode **4130**. The shield **4118** of the lead **4108** may be embedded within the jacket of the lead **4108**. One manner of constructing the lead **4108** with the shield **4118** is to provide an inner jacket **4122** that encloses the filars **4124** and any additional insulation layer **4126**, such as polytetrafluoroethylene (PTFE) that may surround each filar **4124**. The shield **4118** may then reside on the outer portion of the inner jacket **4122**, and the outer jacket **4120** may then enclose the shield **4118**. The outer jacket **4120** maybe added over the braid **4118**, or it may be extruded over the braid.

For embodiments where it is desirable for the shield **4118** to RF couple to tissue, typically as a capacitive coupling, either as an alternative to or in addition to grounding at the can of the IMD or elsewhere, the amount of the outer jacket layer **4120** covering the shield **4118** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **4118** grounds at the can of the IMD and grounding via a capacitive coupling from the shield through the outer jacket **4120** directly to the tissue is of less significance, then the shield **4118** may be located further from the outer surface of the lead **4108**.

The inner and outer jackets **4122**, **4120** may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes, and silicones. A lumen **4128** may be included inside of the inner jacket **4122** around which the insulated

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filars **4124** are coiled or otherwise positioned. The lumen **4128** may be useful, particularly for percutaneous leads **4108**, to allow a stylet to be inserted for purposes of pushing and steering the lead **4108** into the desired position within the patient.

To provide a robust connection for the shield **4118**, the shield electrode **4130** such as an electrode ring may be wrapped around the outer layer **4120** to contact the shield **4118** and provide a direct current coupling to the shield **4118**. A direct current coupling between the shields avoids large variations in the characteristic impedance for the shielding from the extension **4142** to the lead **4108**. Avoiding variations in the characteristic impedances may reduce the degree of RF reflection that occurs within the shield **4118**, which in turn reduces the amount of RF heating that may occur via the stimulation electrodes.

The housing **4140** includes the shield connector **4132**, such as a set screw block, a Bal Seal® connector, or another spring loaded connector. The shield connector **4132** of this embodiment is enclosed within a housing layer **4134** and contacts the shield electrode **4130** of the lead **4108**. The housing layer **4134** may be constructed of various non-conductive materials such as polyurethane, polysulfone, nylon, silicone and polyetheretherketone (PEEK) and provides a relatively rigid structure similar to that provided by the header **4106** of the IMD **4102**.

A shield jumper wire **4136** is included in this embodiment within the housing layer **4134**. The shield jumper wire **4136** contacts the shield connector **4132** and extends from the shield connector **4132** into the housing layer **4134** and extends proximally to the shield **4144** within the extension **4142**. The shield jumper wire **4136** may be welded, crimped, or otherwise affixed to the shield conductor **4132** and the shield **4144**.

FIG. 17C shows an example similar to the example of FIG. 17B. The lead **4108** is constructed in the same manner. However, the housing **4140** utilizes a different construction. In the housing **4140**, a housing shield **4138** is present and extends to the shield conductor **4132** where the housing shield **4138** contacts the shield connector **4132**. No jumper wire is needed because the housing shield **4138** establishes continuity of the shielding from the shield connector **4132** to the shield **4144** present within the extension **4142**.

The housing shield **4138** may be affixed to the shield connector **4132** in various ways. For instance, the housing shield **4138** may be welded or crimped to the shield connector **4132** to provide a direct current coupling. In some embodiments, the shield connector **4132** is distal relative to stimulation connectors of the housing **4140**. In those cases, extending the housing shield **4138** through the housing **4140** to the shield connector **4132** provides additional shielding protection from RF induced currents.

FIG. 17D shows the lead **4108** coupled to the housing **4140** with a cross-section through a stimulation electrode coupling for embodiments of the housing **4140** that include a shield jumper wire **4136**. The lead **4108** includes a stimulation connector **4150** and a stimulation jumper wire **4156** that interconnects the filar **4124** to the stimulation connector **4150**. The housing **4140** includes a stimulation connector **4152** that contacts the stimulation connector **4150** to form a direct current coupling. A stimulation jumper wire **4154** of the housing **4140** contacts the stimulation connector **4152** and extends through the housing layer **4134** in the proximal direction to a corresponding filar within the extension **4142**.

As shown in FIG. 17D, both the shield jumper wire **4136** and the stimulation jumper wire **4154** are present within the

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housing layer **4134**. Separation between them is provided to avoid transferring significant RF energy being captured by the shields **4118**, **4144** from the shield jumper wire **4136** to the stimulation jumper wire **4154**. For instance, the separation may be in the range of 0.1 millimeters (mm) to 2.0 mm where the housing layer **4134** is constructed of polyurethane, polysulfone, nylon, and PEEK or has a dielectric property of between about 2 and 10.

FIG. 17E shows the lead **4108** coupled to the housing **4140** with a cross-section through a stimulation connector coupling for embodiments of the housing **4140** that include the housing shield **4138** extending through the housing **4140**. The lead **4108** includes the stimulation connector **4150** and the stimulation jumper wire **4156** that interconnects the filar **4124** to the stimulation connector **4150**. The housing **4140** includes the stimulation connector **4152** that contacts the stimulation connector **4150** to form a direct current coupling. The stimulation jumper wire **4154** of the housing **4140** contacts the stimulation connector **4152** and extends in the proximal direction to a corresponding filar within the extension **4142**.

As shown in FIG. 17E, the housing shield **4138** and the stimulation jumper wire **4154** are present within different layers of housing material. The stimulation jumper wire **4154** is present within a housing inner layer **4162** while the housing shield **4138** is present about the housing inner layer **4162**. A housing outer layer **4160** surrounds the housing shield **4138** and the housing inner layer **4162**. The housing inner layer **4162** and the housing outer layer **4160** may be constructed of various non-conductive materials such as those discussed above for the housing layers **4134**, and these layers **4160**, **4162** may be the same or different non-conductive materials. Separation like that discussed above between the stimulation jumper wire **4154** and the housing shield **4138** is provided to avoid transferring significant RF energy being captured by the shields **4118**, **4144** from the housing shield **4138** to the stimulation jumper wire **4154**.

FIG. 17F shows an embodiment of the housing **4140** where the shield jumper wire **4136** is present. The shield jumper wire **4136** has an attachment point **4172** such as a weld or crimp to the shield connector **4132** where the shield electrode **4130** is seated. The shield jumper wire **4136** has another attachment point **4184** such as a weld or crimp to the shield **4144** of the extension **4142**.

The stimulation jumper wires **4154**, **4182** have attachment points **4174**, **4178** such as a weld or crimp to the stimulation connectors **4152** where the stimulation electrodes **4150**, **4176** are seated. Only two stimulation jumper wires **4154**, **4182** are shown for purposes of clarity, and it will be appreciated that any number of stimulation connectors and corresponding stimulation jumper wires may be present within the housing **4140**. As shown, separation is provided between the shield jumper wire **4136** and the stimulation jumper wires **4154**, **4182** to avoid transferring RF energy to the stimulation jumper wires **4154**, **4182**.

FIG. 17G shows an embodiment of the housing **4140** where the housing shield **4138** is present. The housing shield **4138** has attachment points **4186** such as welds or crimps to the shield connector **4132** where the shield electrode **4130** is seated. The housing shield **4138** continues through the housing out layer **4160** while surrounding the housing inner layer **4162**. The housing shield **4138** transitions to the extension shield **4144** upon reaching the junction of the housing **4140** to the body of the extension **4142**. In other embodiments, the housing shield **4138** may be attached to the shield **4144** via welds or crimps rather than transitioning into the body of the extension **4142** as the shield **4144**.

The stimulation jumper wires **4154**, **4182** have attachment points **4174**, **4178** to the stimulation connectors **4152** where the stimulation electrodes **4150**, **4176** are seated. As shown, separation is provided between the housing shield **4138** and the stimulation jumper wires **4154**, **4182** to avoid transferring RF energy to the stimulation jumper wires **4154**, **4182**.

Embodiments as disclosed in relation to FIGS. **18-29** provide for termination of a radio frequency (RF) shield present within an implantable medical lead for use with an implantable medical device (IMD). The shield may be terminated in various ways such as by terminating at an edge of a butt, scarf, or lap joint to an insulation extension. Furthermore, the shield termination may include features such as a ring attached at the shield termination point within the insulation, shield wires with folded over ends, or barbs between the insulation layers.

FIG. **18** shows an example of an implantable medical system **5100** that includes an IMD **5102** coupled to a lead **5108**. The IMD **5102** includes a metal can **5104**, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD **5102** includes a header **5106** typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can **5104**. The header **5106** is shown transparently for purposes of illustration. The header **5106** provides a structure for securing the lead **5108** to the IMD **5102** and for establishing electrical connectivity between circuitry of the IMD **5102** and electrodes of the lead **5108**.

The lead **5108** includes electrodes **5116** on a distal end that are positioned at a stimulation site within a patient. The lead also includes connector rings **5110** on a proximal end that is positioned within the header **5106**. The connector rings **5110** make physical contact with electrical connections **5111** within the header. The electrical connections **5111** may include a metal contact that the connector ring **5110** rests against upon being inserted into the header **5106** where a wire extends from the metal contact into the can **5104** where the circuitry is housed. Signals applied by the IMD **5102** to the connector rings **5110** are conducted through the lead **5108** to the electrodes **5116** to provide the stimulation therapy to the patient.

The lead **5108** is secured in the header **5106** such as by a set screw block **5112** within the header **5106** that allows at least one set screw **5114** to be tightened against at least one of the connector rings **5110**. A shield **5118** as shown in FIGS. **19A** and **19B** may be grounded to the body along one or more points down the length of the lead from the IMD **5102** via ground rings and/or the shield **5118** may be grounded at the can **5104** of the IMD **5102** of FIG. **18**. As another option, the shield **5118** may be located within the lead **5108** at a small distance from the surface so that the shield **5118** will effectively capacitively couple to the tissue along the length of the lead to dissipate energy to the tissue over the length.

Regardless of the manner of grounding, the shield **5118** terminates on one end near the proximal end and on the opposite end near the distal end of the lead **5108**. At the termination point, shields having multiple metal wires such as braided shields are subject to fraying and shield wire migration. Preventing the shield wire from fraying and/or migrating to the tissue or to stimulation conductors within the lead **5108** may be desirable to prevent RF energy captured by the shield **5118** from being directed onto a small area of tissue via an electrode or exposed shield wire.

FIGS. **19A** and **19B** show an example of the lead **5108**, where a shield **5118** is present. An outer insulation layer **5120** of a lead jacket is shown transparently in FIG. **19A** for purposes of illustrating the shield **5118**. The shield **5118**

blocks at least some RF energy from directly coupling to conductive filars **5124** that are present within the lead **5108**. The conductive filars **5124** extend the length of the lead and interconnect the proximal electrodes **5110** to the distal electrodes **5116** so that stimulation signals are conducted from the proximal end to the distal end of the lead **5108**.

As shown in FIG. **19A**, the shield **5118** of this example is a braided collection of metal wires. The metal wires may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield **5118**, particularly for embodiments where a portion of the shield **5118** may be exposed for purposes of grounding. While the shield **5118** is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead **5108** in an overlapping manner or an outer layer **5120** that is heavily doped with conductive particles.

As shown in FIG. **19B**, the shield **5118** may be embedded within the jacket of the lead **5108**. One manner of constructing the lead **5108** with the shield **5118** is to provide an inner insulation layer **5122** of the jacket that encloses the filars **5124** and any additional insulation layer **5126**, such as polytetrafluoroethylene (PTFE) that may surround each filar **5124**. The shield **5118** may then reside on the outer portion of the inner insulation layer **5122**, and the outer insulation layer **5120** may then enclose the shield **5118**. The outer jacket **5120** may be added over the braid **5118**, or it may be extruded over the braid.

For embodiments where it is desirable for the shield **5118** to RF couple to tissue, typically as a capacitive coupling, either as an alternative to grounding at the can **5104** of the IMD **5102** or at specific points along the lead **5108** or in addition to such grounds, the amount of the outer jacket layer **5120** covering the shield **5118** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **5118** grounds at one or more specific locations along its length, via a direct current coupling or a capacitive coupling, the shield **5118** may be located further from the outer surface of the lead **5108** with additional features of the lead providing the coupling at the one or more specific locations as discussed below.

The inner and outer insulation layers **5122**, **5120** of the jacket may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethane, and silicones. A lumen **5128** may be included inside of the inner jacket **5122** around which the insulated filars **5124** are coiled or otherwise positioned. The lumen **5128** may be useful, particularly for percutaneous leads **5108**, to allow a stylet to be inserted for purposes of pushing and steering the lead **5108** into the desired position within the patient.

FIG. **20** shows an embodiment of an implantable medical lead **5108** in cross-section with a cut taken down an axial centerline. The lead **5108** terminates at a butt joint **5130** where the inner insulation layer **5122**, shield **5118**, and outer insulation layer **5120** terminate. At this butt joint **5130**, an insulation extension **5132** abuts and is bonded via RF heating, thermal, reflow, or similar processes to the blunt ends of the inner insulation layer **5122**, outer insulation layer **5120**, and shield **5118**.

As shown in this example, the shield **5118** terminates at the butt joint **5130** rather than farther back within the jacket formed by the inner and outer insulation layers **5122**, **5120**. The insulation extension **5132** in this example extends the remainder of the lead **5108** where ring electrodes **5134** are

located. The filars **5124** jumper to their respective ring electrodes via a filar jumper **5136**. The lumen may be present with the filars **5124** being located about the lumen.

The material for the insulation extension **5132** may be selected to provide more or less stiffness than the inner and outer insulation layers **5122**, **5120**, depending upon which end of the lead the butt joint **5130** is located. For instance, where the electrode **5134** is on the proximal end of the lead **5108** and is being positioned within the header **5106** of the IMD **5102**, the insulation extension **5132** may be constructed of a stiffer material. Where the electrode **5134** is on the distal end of the lead **5108** and is being steered to the stimulation site within the body, the insulation extension **5132** may be constructed of a more flexible material.

Using a stiffer material as the insulation extension **5132** on the proximal end aids in the insertion of the proximal end into the header **5106**. As a particular example, the outer insulation **5120** may be constructed of polyurethane having a durometer 55 D or similar rating while the insulation extension **5132** may be constructed of a polyurethane having a durometer 80 A or similar rating.

Using a less stiff material as the insulation extension **5132** on the distal end aids in the positioning of the distal end at the stimulation site. As a particular example, the outer insulation **5120** may be constructed of polyurethane having a durometer 55 D or similar rating while the insulation extension **5132** may be constructed of polyurethane having a durometer 80 A or similar rating.

The gap between the termination of the shield **5118** at the butt joint **5130** and the nearest edge of the electrode **5134** is selected to avoid RF problems. In particular, the distance is selected so that RF coupling is avoided while the unshielded region of the filars **5124** is not overly exposed to RF. For MRI frequencies that typically range from 43 MHz to 128 MHz, a spacing of from 0.5 mm to 10 cm may be acceptable for these embodiments.

FIG. **21** shows another embodiment of an implantable medical lead **5108** in cross-section with a cut taken down an axial centerline. The lead **5108** terminates at a scarf joint **5140** where the inner insulation layer **5122**, shield **5118**, and outer insulation layer **5120** terminate at a wedged cut. At this scarf joint **5140**, an insulation extension **5132** that has a complementary wedged cut abuts and is bonded to the wedged end of the inner insulation layer **5122**, outer insulation layer **5120**, and shield **5118** via RF heating, thermal, reflow or similar processes.

The scarf joint **5140** may be used rather than the butt joint **5130** of FIG. **20** because the scarf joint **5140** increases the bonding area. As shown in this example, the shield **5118** terminates at the scarf joint **5140** rather than farther back within the jacket formed by the inner and outer insulation layers **5122**, **5120**. The insulation extension **5132** in this example extends the remainder of the lead **5108** where ring electrodes **5134** are located. The filars **5124** jumper to their respective ring electrodes via a filar jumper **5136**.

Similar to the previous embodiment of FIG. **20**, the material for the insulation extension **5132** in this embodiment of FIG. **4** may be selected to provide more or less stiffness than the inner and outer insulation layers **5122**, **5120**, depending upon which end of the lead the scarf joint **5140** is located. For instance, where the electrode **5134** is on the proximal end of the lead **5108** and is being positioned within the header **5106** of the IMD **5102**, the insulation extension **5132** may be constructed of a stiffer material such as polyurethane with a durometer 75 D. Where the electrode **5134** is on the distal end of the lead **5108** and is being steered to the stimulation site within the body, the insulation exten-

sion **5132** may be constructed of a more flexible material such as polyurethane with a durometer 80 A.

The gap between the termination of the shield **5118** at the butt joint **5130** and the nearest electrode **5134** is selected to avoid RF problems. In particular, the distance is selected so that RF coupling is avoided while the unshielded region of the filars **5124** is not overly exposed to RF. For MRI frequencies that typically range from 43 MHz to 128 MHz, spacing between the edge of the electrode **5134** nearest the scarf joint **5140** and the termination of the shield **5118** at the scarf joint **5140** may range from 0.5 mm to 10 cm for these embodiments. With the scarf joint **5140** of FIG. **21**, the spacing between the termination of the shield **5118** and the electrode **5134** varies for different locations around the circumference of the scarf joint **5140**, but the shortest spacing is maintained at 0.5 mm or above and the longest spacing is maintained at 10 cm or below.

FIG. **22** shows a set of steps to create the embodiments of FIGS. **20** and **21**. Initially, a structure including the inner insulation layer **5122**, outer insulation layer **5120**, and shield **5118** may be provided. The shield **5118** has been braided over the inner insulation layer **5122** and then the outer insulation layer **5120** has been positioned and reflowed or otherwise bonded over the inner insulation layer **5122** and the shield **5118**. To begin construction of the lead **5108** and the butt joint **5130** or scarf joint **5140**, the structure is cut to size by making a cut through the insulation layers **5120**, **5122** and the shield **5118** at a cutting step **5142**. For a butt joint **5130**, the cut is perpendicular to the axial dimension to create the blunt end. For a scarf joint **5140**, the cut is at angle other than 90 degrees to the axial dimension to create the wedged end.

The insulation extension **5132** is also provided with a complementary end to bond to the lead **5108** to form the butt joint **5130** or scarf joint **5140**. For the butt joint **5130**, the insulation extension **5132** is cut perpendicular to the axial dimension to create the blunt end. For the scarf joint **5140**, the insulation extension **5132** is cut at an angle other than 90 degrees to the axial dimension to create the wedged end. The two blunt ends for the butt joint **5130** are brought together and bonded at a bonding step **5144**. Likewise, the two wedged ends for the scarf joint **5140** are brought together and bonded at the bonding step **5144**.

FIG. **23** shows another embodiment of an implantable medical lead **5108** in cross-section with a cut taken down an axial centerline. The lead **5108** terminates at a lap joint **5150**. The lap joint **5150** involves removing an end portion of the outer insulation layer **5120** and applying a replacement outer insulation layer **5152** onto the area of the shield **5118** and inner insulation layer **5122** where the outer insulation layer **5120** is missing. The replacement outer insulation layer **5152** also laps over a section of the insulation extension **5132** and may extend to the nearest electrode **5134**.

As shown, the shield **5118** has been crimped down into the inner insulation layer **5122** at the region where the outer insulation layer **5120** has been removed. Doing so prevents the shield **5118** from bunching together during installation of the outer replacement insulation layer **5152**. This may be especially the case where the replacement outer insulation layer **5152** is in the form of tubing that slides into place over the shield **5118** and inner insulation layer **5122** prior to attaching the insulation extension **5132**. Where the replacement outer insulation layer **5152** is tubing, once being slid into place, it is reflowed or otherwise bonded to the inner insulation layer **5122**. As an alternative, the replacement outer insulation layer **5152** may be injection molded into place.

As shown in this example, the shield **5118** terminates at the lap joint **5150** rather than farther back within the jacket formed by the inner and outer insulation layers **5122**, **5120**. The insulation extension **5132** in this example extends the remainder of the lead **5108** where ring electrodes **5134** are located. The filars **5124** jumper to their respective ring electrodes via a filar jumper **5136**. The lumen may be present in some embodiments with the filars **5124** being located about the lumen.

In this embodiment the replacement outer insulation layer **5152** may be constructed of a material that differs in stiffness from the outer insulation layer **5120** depending upon which end of the lead **5108** the lap joint **5150** is located. For instance, where the electrode **5134** is on the proximal end of the lead **5108** and is being positioned within the header **5106** of the IMD **5102**, the replacement outer insulation layer **5152** may be constructed of a stiffer material such as durometer 75 D polyurethane. Where the electrode **5134** is on the distal end of the lead **5108** and is being steered to the stimulation site within the body, the replacement outer insulation layer **5152** may be constructed of a more flexible material such as 80 A polyurethane.

In this embodiment, like that of the previous ones, the material for the insulation extension **5132** may also be selected to provide more or less stiffness than the inner and outer insulation layers **5122**, **5120**, depending upon which end of the lead the lap joint **5150** is located. For instance, where the electrode **5134** is on the proximal end of the lead **5108** and is being positioned within the header **5106** of the IMD **5102**, the insulation extension **5132** may be constructed of a stiffer material such as durometer 75 D polyurethane. Where the electrode **5134** is on the distal end of the lead **5108** and is being steered to the stimulation site within the body, the insulation extension **5132** may be constructed of a more flexible material such as 80 A polyurethane.

The gap between the termination of the shield **5118** at the lap joint **5150** and the nearest electrode **5134** is also selected to avoid RF problems. For MRI frequencies, a spacing of from 0.5 mm to 10 cm may be acceptable for these embodiments.

FIG. 24 shows one example of a set of steps that create the lap joint **5150** of FIG. 23. Initially, a structure including the inner insulation layer **5122**, outer insulation layer **5120**, and shield **5118** may be provided. The shield **5118** has been braided over the inner insulation layer **5122** and then the outer insulation layer **5120** has been positioned and reflowed or otherwise bonded over the inner insulation layer **5122** and the shield **5118**. To begin construction of the lead **5108** and the lap joint **5150**, the structure is cut to size by making a cut through the insulation layers **5120**, **5122** and the shield **5118** at a cutting step **5154**. For a lap joint **5150**, this first cut is perpendicular to the axial dimension to create a blunt end.

Once cut to size, the outer insulation layer **5120** is then ablated by some distance to expose the shield **5118** and the inner insulation layer **5122** at an ablating step **5156**. Ablation may be done using tools such as an excimer laser which can very precisely ablate to expose the shield **5118**. The length of the outer insulation layer **5120** to be ablated may vary, but an illustrative range is from 0.25 centimeters (cm) to 5 cm.

Once ablation is complete, the next step may vary. The replacement outer insulation layer **5152** may be installed in various manners such as by reflowing tubing or by injection molding. If by injection molding, then the next step may be either a crimping step **5158** or an injecting step **5162**. If by reflowing tubing, then it may be helpful to proceed to the crimping step **5158** after ablating.

At the crimping step **5158**, the shield **5118** is crimped so as to sink down into the inner insulation layer **5122** at the area where the outer insulation layer **5120** has been removed. If a ring or other tool is used to crimp the shield **5118** into the inner insulation layer **5122**, the ring or other tool may then be removed. Where the replacement outer insulation layer **5152** is being installed as tubing that is reflowed, then the next step is tubing step **5160**. Where the replacement outer insulation layer **5152** is being installed by injection molding, then the next step is injecting step **5162**.

At the tubing step **5160**, the tubing is slid onto the inner insulation layer **5122** and over the shield **5118** at the area where the outer insulation layer **5120** has been removed and where the shield **5118** has been crimped down. The tubing extends beyond the end of the inner insulation layer **5122** so that it may eventually be bonded to the insulation extension **5132**. The tubing is reflowed, RF heated, etc. to bond to the inner insulation layer **5122** and to the end of the outer insulation layer **5120** where the ablating stopped to form the replacement outer insulation layer **5152**. Contemporaneously or sequentially, the insulation extension **5132** is bonded in place at the blunt end of the inner insulation layer **5122** and to the tubing of the replacement outer insulation layer **5152** that extends beyond the inner insulation layer **5122** at a bonding step **5164**. This tubing may be reflowed, RF heated, etc. onto the insulation extension **5132**.

Returning to the injecting step **5162**, in the scenario where the replacement outer insulation layer **5152** is to be injection molded, then the injecting step **5162** takes place either after the ablating step **5156** or after the crimping step **5158**. Material such as the desired polyurethane is injected onto the inner insulation layer **5122** and the shield **5118** to form the replacement outer insulation layer **5152**. Contemporaneously, the insulation extension **5132** is bonded to the inner insulation layer **5122** and to the replacement outer insulation layer **5152** at the bonding step **5164**.

Alternative manners of creating the lap joint **5150** may also be used. For instance, the structure of the outer insulation layer **5120**, inner insulation layer **5122**, and shield **5118** may be bonded to the insulation extension **5132** via a butt joint. Then, the area where the replacement outer insulation layer **5152** will be positioned that is currently occupied by the outer insulation layer **5120** is ablated. The insulation extension **5132** is also ablated at the same or similar depth as the outer insulation layer **5120**. The replacement outer insulation layer **5152** may then be injection molded or shrunk into position at the ablation site.

FIG. 25 shows an embodiment with an additional feature that may be included for the lap joint **5150**. To further protect the termination of the shield **5118** from fraying or migrating, the wire ends of the shield **5118** may be capped with a ring **5166**. The ring **5166** may be metal, plastic, or similar materials. In this example, a region **5168** of the inner insulation layer **5122** has been ablated to allow the ring to be positioned over the ends of the wires of the shield **5118**.

FIG. 26 shows an embodiment with an additional feature that may be included for the butt or scarf joints **5130**, **5140**. To further protect the termination of the shield **5118** from fraying or migrating, the wire ends of the shield **5118** may be capped with a ring **5170**. Similar to the lap joint scenario, the ring **5170** may be metal, plastic, or similar materials. In this example, a region **5172** of the outer insulation layer **5120** has been ablated to allow the ring **5170** to be positioned over the ends of the wires of the shield **5118**, and then this region **5172** may be filled using a reflow or injection molding of the polyurethane or other polymer.

FIG. 27 shows an embodiment with an additional feature that may be included for a joint 5176, which may be of various types such as the butt, scarf, or lap joints 5130, 5140, and 5150. At the joint 5176, the outer insulation layer 5120 of the lead 5108 encounters another layer 5178. This layer 5178 may be the insulation extension 5132 and/or the replacement outer insulation layer 5152. In either case, wires of the shield 5118 may partially extend into the layer 5178. However, prior to bonding the layer 5178 to the layers 5120 or 5122, the ends of the wires of the shield 5118 may be individually folded over as shown in FIG. 27. In this manner, the folded over ends are less likely to fray and migrate.

FIG. 28 shows an embodiment with another feature that may be included for a joint at the termination of the shield 5118 to assist in holding the bond between the inner insulation layer 5122 and the insulation extension 5132 in place. In this example, a lap joint 5150 is shown, but it will be appreciated that this feature may be applicable to other joints as well including butt and scarf joints 5130, 5140. Here, the replacement outer insulation layer 5152 may be tubing that is provided with barbs 5151 that extend toward the inner insulation layer 5122 and the shield 5118.

During reflow, the barbs may sink into the inner insulation layer 5122 as the inner insulation layer 5122 softens more so than the barbs 5151, and the replacement outer insulation layer 5152 descends into position. The barbs 5151 may also sink into the insulation extension 5132 once reflow or other bonding is attempted after the insulation extension 5132 has been inserted to provide extra grip between the replacement outer insulation layer 5152 and the insulation extension 5132. The barbs 5151 then provide extra grip between the inner insulation layer 5122 and the insulation extension 5132 particularly during axial tension. Rather than incorporating the barbs into the replacement outer tubing 5152 of the lap joint example, a separate barbed ring may be positioned on the inner insulation layer 5122 and then the replacement outer insulation layer 5152 is reflowed or otherwise bonded into place.

FIG. 29 shows an embodiment with another feature that may be included for a joint at the termination of the shield 5118 to assist in holding the bond between the inner insulation layer 5122 and the insulation extension 5132 in place. In this example, a lap joint 5150 is shown, but it will be appreciated that this feature may be applicable other joints as well including butt and scarf joints 5130, 5140. Here, a barbed ring 5153 is positioned inside of the inner insulation layer 5122 and is forced to expand radially until the barbs of the barbed ring 5153 sink into the inner insulation layer 5122. The barbs of the barbed ring 5153 may also sink into the inside of the insulation extension 5132. The barbed ring 5153 provides extra grip between the inner insulation layer 5122 and the insulation extension 5132 especially during axial tension. This barbed ring 5153 feature may also be used in conjunction with the barbs 5151 shown in FIG. 28.

Embodiments as disclosed in relation to FIGS. 30-48 also provide for termination of a radio frequency (RF) shield present within an implantable medical lead for use with an implantable medical device (IMD). The shield may be terminated in various ways such as by terminating at a joint to an insulation extension where one or more metal connectors are present in various configurations to provide a ground path for the shield. Furthermore, the shield termination may include features such as a shield wires with folded over ends, or barbs between the insulation layers.

FIG. 30 shows an example of an implantable medical system 6100 that includes an IMD 6102 coupled to a lead 6108. The IMD 6102 includes a metal can 6104, typically

constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD 6102 includes a header 6106 typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can 6104. The header 6106 is shown transparently for purposes of illustration. The header 6106 provides a structure for securing the lead 6108 to the IMD 6102 and for establishing electrical connectivity between circuitry of the IMD 6102 and electrodes of the lead 6108.

The lead 6108 includes electrodes 6116 on a distal end that are positioned at a stimulation site within a patient. The lead also includes ring connectors 6110 on a proximal end that is positioned within the header 6106. The ring connectors 6110 make physical contact with electrical connections 6111 within the header. The electrical connections 6111 may include a metal contact that the ring connector 6110 rests against upon being inserted into the header 6106 where a wire extends from the metal contact into the can 6104 where the circuitry is housed. Signals applied by the IMD 6102 to the ring connectors 6110 are conducted through the lead 6108 to the electrodes 6116 to provide the stimulation therapy to the patient.

The lead 6108 is secured in the header 6106 such as by a set screw block 6112 within the header 6106 that allows at least one set screw 6114 to be tightened against at least one of the ring connectors 6110. A shield 6118 as shown in FIGS. 31A and 31B may be grounded to the body along one or more points down the length of the lead from the IMD 6102 via ground rings and/or the shield 6118 may be grounded at the can 6104 of the IMD 6102 of FIG. 30.

Regardless of the manner of grounding, the shield 6118 terminates on one end near the proximal end and on the opposite end near the distal end of the lead 6108. At the termination point, shields having multiple metal wires such as braided shields are subject to fraying and shield wire migration. Preventing the shield wire from fraying and/or migrating to the tissue or to stimulation conductors within the lead 6108 may be desirable to prevent RF energy captured by the shield 6118 from being directed onto a small area of tissue via an electrode or exposed shield wire.

FIGS. 31A and 31B show an example of the lead 6108, where a shield 6118 is present. An outer insulation layer 6120 of a lead jacket is shown transparently in FIG. 31A for purposes of illustrating the shield 6118. The shield 6118 blocks at least some RF energy from directly coupling to conductive filars 6124 that are present within the lead 6108. The conductive filars 6124 extend the length of the lead and interconnect the proximal ring connectors 6110 to the distal electrodes 6116 so that stimulation signals are conducted from the proximal end to the distal end of the lead 6108.

As shown in FIG. 31A, the shield 6118 of this example is a braided collection of metal wires. The metal wires may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield 6118, particularly for embodiments where a portion of the shield 6118 may be exposed for purposes of grounding. While the shield 6118 is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead 6108 in an overlapping manner or an outer layer 6120 that is heavily doped with conductive particles.

As shown in FIG. 31B, the shield 6118 may be embedded within the jacket of the lead 6108. One manner of constructing the lead 6108 with the shield 6118 is to provide an inner insulation layer 6122 of the jacket that encloses the filars

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6124 and any additional insulation layer 6126, such as polytetrafluoroethylene (PTFE) that may surround each filar 6124. The shield 6118 may then reside on the outer portion of the inner insulation layer 6122, and the outer insulation layer 6120 may then enclose the shield 6118. The outer insulation layer 6120 may be added over the shield 6118 and shrunk in place or may be extruded over the shield 6118. The outer jacket 6120 may be added over the braid 6118, or it may be extruded over the braid.

For embodiments where it is desirable for the shield 6118 to RF couple to tissue, typically as a capacitive coupling, in addition to grounding at the can or along the lead, the amount of the outer jacket layer 6120 covering the shield 6118 may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield 6118 grounds at one or more specific locations along its length, via a direct current coupling or a capacitive coupling, the shield 6118 may be located further from the outer surface of the lead 6108.

The inner and outer insulation layers 6122, 6120 of the jacket may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes and silicones. A lumen 6128 may be included inside of the inner jacket 6122 around which the insulated filars 6124 are coiled or otherwise positioned. The lumen 6128 may be useful, particularly for percutaneous leads 6108, to allow a stylet to be inserted for purposes of pushing and steering the lead 6108 into the desired position within the patient.

FIG. 32 shows an embodiment of an implantable medical lead 6108 in cross-section with a cut taken down an axial centerline. The lead 6108 includes a butt joint 6130 where the inner insulation layer 6122 and shield 6118 terminate. The outer insulation 6120 terminates prior to the butt joint 6130 to expose the shield 6118 and inner insulation layer 6122. A metal connector 6131 is positioned over the shield 6118 and inner insulation layer 6122 and abuts the end of the outer insulation layer 6120. At the butt joint 6130, an insulation extension 6132 abuts and is bonded to the blunt end of the inner insulation layer 6122, shield 6118, and metal connector 6131 such as via reflow or injection molding.

As shown in this example, the shield 6118 terminates at the butt joint 6130 rather than farther back within the jacket formed by the inner and outer insulation layers 6122, 6120. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located on the proximal end at a separate from the nearest connector ring ranging from about 0.5 millimeters to about 10 centimeters. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The lumen may be present in some embodiments with the filars 6124 being located about the lumen.

The material for the insulation extension 6132 may be selected to provide a different amount of stiffness than the inner and outer insulation layers 6122, 6120. For instance, the insulation extension 6132 may be constructed of a stiffer material to aid in the insertion of the proximal end into the header 6106. As a particular example, the outer insulation 6120 may be constructed of polyurethane having a durometer 55 D or similar rating while the insulation extension 6132 may be constructed of a polyurethane having a durometer 75 D or similar rating.

The shield 6118 may be terminated with an exposed metal connector 6131 at a butt joint on the distal end so long as no terminating ground ring is present at the proximal end to thereby avoid stimulation induced by magnetic gradients. In such a case, the insulation extension may have a durometer

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rating similar to the outer layer 6120 but may instead be constructed of polyurethane having a durometer 80 A or similar rating.

The metal connector 6131 is separated from the distal electrode by at least 0.5 mm up to 10 cm for some body locations, to avoid excessive RF coupling to the distal electrode, with 2 mm being one example of spacing that provides adequate filar coverage with insignificant coupling to the distal electrode. Where the distal end is located in a high RF intensity area such as just under the skin for peripheral nerve stimulation, then the distance may be kept smaller, such as less than 2 cm to avoid overexposure of the filars 6124.

FIG. 33 shows another embodiment of an implantable medical lead 6108 in cross-section with a cut taken down an axial centerline. The lead 6108 terminates at a scarf joint 6140 where the inner insulation layer 6122 and shield 6118 terminate at a wedged cut. The outer insulation 6120 terminates prior to the scarf joint 6140 to expose the shield 6118 and inner insulation layer 6122. A metal connector 6131 is positioned over the shield 6118 and inner insulation layer 6122 and abuts the end of the outer insulation layer 6120. At this scarf joint 6140, an insulation extension 6132 that has a complementary wedged cut abuts and is bonded to the wedged end of the inner insulation layer 6122, shield 6118, and metal connector 6131 such as via reflow or injection molding.

The scarf joint 6140 may be used rather than the butt joint 6130 of FIG. 32 because the scarf joint 6140 has an increased bond area. As shown in this example, the shield 6118 terminates at the scarf joint 6140 rather than farther back within the jacket formed by the inner and outer insulation layers 6122, 6120. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located on the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136.

Similar to the previous embodiment of FIG. 32, the material for the insulation extension 6132 in this embodiment of FIG. 33 may be selected to provide a different stiffness than the inner and outer insulation layers 6122, 6120. For instance, the insulation extension 6132 may be constructed of a stiffer material such as polyurethane with a durometer 75 D.

The metal connector 6131 may be included on either the proximal or distal end to terminate the shield 6118 as discussed above. The separation of the metal connector 6131 to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. 34 shows a set of steps to create the embodiments of FIGS. 32 and 33. Initially, a structure including the inner insulation layer 6122, outer insulation layer 6120, and shield 6118 may be provided. The shield 6118 has been braided over the inner insulation layer 6122 and then the outer insulation layer 6120 has been positioned and reflowed or otherwise bonded over the inner insulation layer 6122 and the shield 6118. To begin construction of the lead 6108 and the butt joint 6130 or scarf joint 6140, the structure is cut to size by making a cut through the insulation layers 6120, 6122 and the shield 6118 at a cutting step 6142. For a butt joint 6130, the cut is perpendicular to the axial dimension to create the blunt end. For a scarf joint 6140, the cut is at angle other than 90 degrees to the axial dimension to create the wedged end.

The end portion of the outer insulation layer 6120 is ablated to reveal the shield 6118 at ablating step 6144. The

metal connector **6131**, such as a ring connector, may then be crimped or welded onto the shield **6118** at crimping step **6146**.

The insulation extension **6132** is bonded to the lead **6108** to form the butt joint **6130** or scarf joint **6140** at a bonding step **6148**. For the butt joint **6130**, the insulation extension **6132** is cut perpendicular to the axial dimension to create the blunt end. For the scarf joint **6140**, the insulation extension **6132** is cut at an angle other than 90 degrees to the axial dimension to create the wedged end. The two blunt ends for the butt joint **6130** are brought together and bonded at a bonding step **6148**. Likewise, the two wedged ends for the scarf joint **6140** are brought together and bonded at the bonding step **6148**.

FIG. **35** shows another embodiment of an implantable medical lead **6108** in cross-section with a cut taken down an axial centerline. The lead **6108** includes a lap joint **6150** where the inner insulation layer **6122** and the shield **6118** terminate. The lap joint **6150** involves removing an end portion of the outer insulation layer **6120** sufficient to allow space for the metal connector **6131** and a replacement outer insulation layer **6152** to lap over the area of the shield **6118** and inner insulation layer **6122** where the outer insulation layer **6120** is missing. The metal connector **6131** abuts the end of the outer insulation layer **6120**. The replacement outer insulation layer **6152** abuts the metal connector **6131**, laps over a section of the insulation extension **6132**, and may extend to the nearest electrode **6134**.

As shown, the shield **6118** has been crimped down into the inner insulation layer **6122** at the region where the outer insulation layer **6120** has been removed. Doing so prevents the shield **6118** from bunching together during installation of the outer replacement insulation layer **6152**. This may be especially the case where the replacement outer insulation layer **6152** is in the form of tubing that slides into place over the shield **6118** and inner insulation layer **6122** prior to attaching the insulation extension **6132**. Where the replacement outer insulation layer **6152** is tubing, once being slid into place, it is reflowed or otherwise bonded to the inner insulation layer **6122**. As an alternative, the replacement outer insulation layer **6152** may be injection molded into place.

As shown in this example, the shield **6118** terminates at the lap joint **6150** rather than farther back within the jacket formed by the inner and outer insulation layers **6122**, **6120**. The insulation extension **6132** in this example extends the remainder of the lead **6108** where ring connectors **6134** are located at the proximal end. The filars **6124** jumper to their respective ring connectors via a filar jumper **6136**. The lumen may be present in some embodiments with the filars **6124** being located about the lumen.

In this embodiment the replacement outer insulation layer **6152** may be constructed of a material that differs in stiffness from the outer insulation layer **6120**. For instance, the replacement outer insulation layer **6152** may be constructed of a stiffer material such as durometer 75 D polyurethane. In this embodiment, like that of the previous ones, the material for the insulation extension **6132** may also be selected to provide a different stiffness than the inner and outer insulation layers **6122**, **6120**. For instance, the insulation extension **6132** may also be constructed of a stiffer material such as durometer 75 D polyurethane.

The metal connector **6131** may be included on either the proximal or distal end to terminate the shield **6118** as discussed above. The separation of the metal connector **6131** to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. **36** shows one example of a set of steps that create the lap joint **6150** of FIG. **35**. Initially, a structure including the inner insulation layer **6122**, outer insulation layer **6120**, and shield **6118** may be provided. The shield **6118** has been braided over the inner insulation layer **6122** and then the outer insulation layer **6120** has been positioned and reflowed or otherwise bonded over the inner insulation layer **6122** and the shield **6118**. To begin construction of the lead **6108** and the lap joint **6150**, the structure is cut to size by making a cut through the insulation layers **6120**, **6122** and the shield **6118** at a cutting step **6154**. For a lap joint **6130**, this first cut is perpendicular to the axial dimension to create a blunt end.

Once cut to size, the outer insulation layer **6120** is then ablated by some distance to expose the shield **6118** and the inner insulation layer **6122** at an ablating step **6156**. Ablation may be done using tools such as an excimer laser which can very precisely ablate to expose the shield **6118**. The length of the outer insulation layer **6120** to be ablated is sufficient to allow for the metal connector **6131** as well as the amount of the replacement outer insulation layer **6152** that laps onto the inner insulation layer **6122**. This length of ablation of the outer insulation layer **6120** may vary but an illustrative range is from 0.25 centimeters (cm) to 5 cm.

Once ablation is complete, the metal connector **6131** may then be put in position over the inner insulation layer **6122** and shield **6118**. The metal connector **6131** is crimped or welded to the shield **6118** while abutting the end of the outer insulation layer **6120** at a crimping step **6158**.

Once the metal connector **6131** is installed, the next step may vary. The replacement outer insulation layer **6152** may be installed in various manners such as by reflowing tubing or by injection molding. If by injection molding, then the next step may be either a crimping step **6160** or an injecting step **6164**. If by reflowing tubing, then it may be helpful to proceed to the crimping step **6160** after ablating.

At the crimping step **6160**, the portion of the shield **6118** that is exposed beyond the metal connector **6131** is crimped so as to sink down into the inner insulation layer **6122**. If a ring or other tool is used to crimp the shield **6118** into the inner insulation layer **6122**, the ring or other tool may then be removed. Where the replacement outer insulation layer **6152** is being installed as tubing that is reflowed, then the next step is tubing step **6162**. Where the replacement outer insulation layer **6152** is being installed by injection molding, then the next step is injecting step **6164**.

At the tubing step **6162**, the tubing is slid onto the inner insulation layer **6122** and over the shield **6118** at the area where the outer insulation layer **6120** has been removed and where the shield **6118** has been crimped down. The tubing extends beyond the end of the inner insulation layer **6122** so that it may eventually be bonded to the insulation extension **6132**. The tubing is reflowed or otherwise bonded to the inner insulation layer **6122** and to abut the end of the metal connector **6131**. Contemporaneously, the insulation extension **6132** is bonded in place at the blunt end of the inner insulation layer **6122** and to the tubing of the replacement outer insulation layer **6152** that extends beyond the inner insulation layer **6122** at a bonding step **6166**. This tubing may be reflowed or otherwise bonded onto the insulation extension **6132**.

Returning to the injecting step **6164**, in the scenario where the replacement outer insulation layer **6152** is to be injection molded, then the injecting step **6164** takes place either after the crimping step **6158** or after the crimping step **6160**. Material such as the desired polyurethane is injected onto the inner insulation layer **6122** and the shield **6118** to form the replacement outer insulation layer **6152**. Contemporarily,

neously, the insulation extension 6132 is bonded to the inner insulation layer 6122 and to the replacement outer insulation layer 6152 at the bonding step 6166.

Alternative manners of creating the lap joint 6150 may also be used. For instance, the structure of the outer insulation layer 6120, inner insulation layer 6122, and shield 6118 may be bonded to the insulation extension 6130 via a butt joint. Then, the area where the metal connector 6131 and replacement outer insulation layer 6152 will be positioned that is currently occupied by the outer insulation layer 6120 is ablated. The insulation extension 6132 is also ablated at the same or similar depth as the outer insulation layer 6120. The metal connector 6131 may then be positioned, and the replacement outer insulation layer 6152 may then be injection molded or shrunk into position at the ablation site.

FIG. 37 shows another embodiment of an implantable medical lead 6108 in cross-section with a cut taken down an axial centerline. The lead 6108 includes a joint between the inner insulation layer 6122 and the insulation extension 6132 where the inner insulation layer 6122 and the shield 6118 terminate. An inner metal connector 6172 is positioned around the inner insulation layer and an outer metal connector 6174 is positioned around the inner metal connector 6172. A portion of the shield 6118 is located between the inner metal connector 6172 and the outer metal connector 6174 such that a robust physical and electrical connection is established to the shield 6118.

In this example, the shield 6118 is braided after the inner metal connector 6172 has been positioned so that the braid of the shield 6118 laps over the inner metal connector 6172. The outer insulation layer 6120 terminates short of the end of the shield 6118 and inner insulation layer 6122. This may be achieved by ablating the outer insulation layer 6120 where it has been previously extruded over the shield and inner metal connector 6172.

As shown in this example, the shield 6118 terminates between the metal connectors 6172, 6174 rather than farther back within the jacket formed by the inner and outer insulation layers 6122, 6120. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located at the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The lumen may be present in some embodiments with the filars 6124 being located about the lumen.

The metal connectors 6172, 6174 may be included on either the proximal or distal end to terminate the shield 6118 as discussed above. The separation of the metal connectors 6172, 6174 to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above for connector ring 6131.

FIG. 38 shows an alternative manner of attaching the outer metal connector. Rather than ablate the outer insulation layer 6120 at the area where the shield 6118 and inner metal connector 6172 are located, an outer metal connector 6176 having features such as teeth that can penetrate through the outer insulation layer 6120 is used. The outer metal connector 6176 is crimped in place so that the features penetrate through the outer insulation layer 6120 to reach the shield 6118 and the inner metal connector 6172 and establish the physical and electrical connection.

FIG. 39 shows a similar embodiment to that of FIG. 37 where the lead includes the inner metal connector 6172 and the outer metal connector 6174. However, in this example, the shield 6118 does not terminate between the metal connectors 6172, 6174 but a portion 6119 of the shield 6118

continues beyond those connectors 6172, 6174 to extend over the remaining portion of the inner insulation layer 6122. This portion 6119 of the shield 6118 may be crimped into a sunken position within the inner insulation layer 6122.

A replacement outer insulation layer 6152 may be bonded over the portion 6119 of the shield 6118 to form a lap joint. The insulation extension 6132 may then be bonded to the inner insulation layer 6122 and the replacement outer insulation layer 6152. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located at the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The lumen may be present in some embodiments with the filars 6124 being located about the lumen.

The metal connectors 6172, 6174 may be included on either the proximal or distal end to terminate the shield 6118 as discussed above. The separation of the metal connectors 6172, 6174 as well as any portion of the shield 6118 extending beyond the metal connectors 6172, 6174 to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. 40 shows a similar embodiment to that of FIG. 37 where the lead includes the inner metal connector 6172 and the outer metal connector 6174. However, in this example, the inner metal connector 6172 does not wrap around the outside of the inner insulation layer 6122 but instead is embedded within the inner insulation layer 6122 so as to provide a flush surface for the shield 6118 to be braided upon. The shield 6118 is located between this inner metal connector 6172 and the outer metal connector 6174. In this example, the metal connectors 6172, 6174 together with the inner insulation layer 6122 form a butt joint with the insulation extension 6132.

The insulation extension 6132 is bonded to the inner insulation layer 6122 and abuts the metal connectors 6172, 6174. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located at the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The lumen may be present in some embodiments with the filars 6124 being located about the lumen.

The metal connectors 6172, 6174 may be included on either the proximal or distal end to terminate the shield 6118 as discussed above. The separation of the metal connectors 6172, 6174 to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. 41 shows one example of a set of steps that create the shield termination of FIGS. 37-40. The inner metal connector 6172 is positioned on the inner insulation layer 6122 or embedded at the end at a connector step 6182. The shield 6118 is braided onto the inner insulation layer 6122 and over the inner metal connector 6172 at a braiding step 6184. The outer insulation layer 6120 is bonded by reflow or another process onto the inner insulation layer 6122 over the shield 6118 and over the inner metal connector 6172 such as by a reflowing step 6186.

At this point, preparation is made for the outer metal connector 6174. In one example, the outer insulation layer is ablated at an ablating step 6188 and then the outer metal connector is crimped or welded onto the exposed shield 6118 at the overlap to the inner metal connector 6172 at a crimping step 6190. Alternatively, the inner metal connector 6176 having the sharp features is crimped onto the outer insulation layer 6120 with the sharp features penetrating to the shield 6118 and the inner metal connector 6172 at a

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crimping step 6192. The insulation extension 6132 is then bonded to the inner insulation layer 6122 at a bonding step 6194.

FIG. 42 shows a similar embodiment to that of FIG. 37 where the lead includes the inner metal connector 6172 and the outer metal connector 6174. However, in this example, the inner metal connector 6172 does not wrap around the outside of the inner insulation layer 6122 prior to the shield 6118 being braided. Instead, the shield 6118 is braided over the inner insulation layer 6122 and the inner metal connector 6172 is then crimped or welded onto the shield 6118. The shield 6118 may be sunken into the inner insulation layer 6122 in the area where the inner metal connector 6172 is positioned.

The shield 6118 inverts as a whole at an inversion 6123 so that a portion 6121 of the shield 6118 laps over the inner metal connector 6172. The outer metal connector 6174 may then be crimped or welded in placed about the portion 6121 and the inner metal connector 6172. A robust electrical and physical termination of the shield 6118 occurs between the metal connectors 6172, 6174. The inversion 6123 may provide additional benefits for the shield 6118, such as reducing any RF energy leakage that might otherwise occur at a blunt end of the shield 6118.

The insulation extension 6132 is bonded to the inner insulation layer 6122 and abuts the metal connectors 6172, 6174. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located at the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The lumen may be present in some embodiments with the filars 6124 being located about the lumen.

The metal connectors 6172, 6174 may be included on either the proximal or distal end to terminate the shield 6118 as discussed above. The separation of the metal connectors 6172, 6174 to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. 43 shows one example of a set of steps that create the shield termination of FIG. 42. The outer insulation layer 6120, inner insulation layer 6122, and shield 6118 are cut to form a blunt end at a cutting step 6202. A portion of the outer insulation layer 6120 is then ablated to reveal the shield 6118 and inner insulation layer 6122 at an ablating step 6204. The inner metal connector 6172 is positioned on the shield 6118 and around the inner insulation layer 6122 with a portion of the shield 6118 and the inner insulation layer 6122 extending beyond the metal connector 6172 at a connector step 6206. The shield 6118 is inverted as a whole and lapped onto the inner metal connector at a folding step 6208.

At this point, the outer metal connector is crimped or welded onto the exposed shield 6118 at the overlap to the inner metal connector 6172 at a crimping step 6210. Alternatively, the outer metal connector 6176 having the sharp features is crimped onto the outer insulation layer 6120 with the sharp features penetrating to the shield 6118 and the inner metal connector 6172. The insulation extension 6132 is then bonded to the inner insulation layer 6122 at a bonding step 6212.

FIG. 44 shows another embodiment of an implantable medical lead 6108 in cross-section with a cut taken down an axial centerline. The lead 6108 includes a joint between the inner insulation layer 6122 and the insulation extension 6132 where the inner insulation layer 6122 terminates. In this example, the shield 6118 does not remain braided upon the inner insulation layer 6122. Instead, a metal connector 6131 is positioned on the inner insulation layer 6122 and a

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portion 6125 of the shield 6118 is braided onto the metal connector 6131. The outer insulation layer 6120 is positioned over the braid 6118 up to the metal connector 6131 where the braid 6118 exits the outer insulation layer 6120 when lapping onto the metal connector 6131.

The portion 6125 may be exposed outside of the lead 6108 as a result of lapping onto the metal connector 6131. However, for embodiments where the metal connector 6131 is for insertion into the header 6106 of the IMD 6102, the exposure may occur immediately at the exit to the header 6106 or nearby the header seal. To the extent tissue ingrowth is to be avoided in that area, an insulation ring 6216 of material the same as or similar to the outer insulation layer 6120 may be reflowed or otherwise bonded over the portion 6125.

As shown in this example, a replacement outer insulation layer 6152 may be present to form a lap joint between the inner insulation layer 6122 and the insulation extension 6132. The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located at the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The lumen may be present in some embodiments with the filars 6124 being located about the lumen.

The metal connector 6131 may be included on either the proximal or distal end to terminate the shield 6118 as discussed above. The separation of the metal connector 6131 to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. 45 shows one example of a set of steps that create the shield termination of FIG. 44. The metal connector 6131 is positioned on the inner insulation layer 6122 at a connector step 6220. The shield 6118 is braided onto the inner insulation layer 6122 and over the metal connector 6131 at a braiding step 6222. The outer insulation layer 6120 is bonded by reflow or another process onto the inner insulation layer 6122 over the shield 6118 up to the metal connector 6131 such as by a reflowing step 6224. The insulation ring 6216 may then be reflowed or injection molded over the braid portion 6125 on the metal connector 6131 at a bonding step 6226.

FIG. 46 shows another embodiment of an implantable medical lead 6108 in cross-section with a cut taken down an axial centerline. The lead 6108 includes a joint between the inner insulation layer 6122 and the insulation extension 6132 where the inner insulation layer 6122 terminates. In this example, the shield 6118 does not remain braided upon the inner insulation layer 6122. Instead, a tapered ablation is created through the outer insulation layer 6120 and inner insulation layer 6122 and the shield 6118 exits the outer insulation layer 6120 and separates from the inner insulation layer 6122 at the taper.

A metal connector 6232 with a threaded taper 6234 is threaded onto the taper of the inner and outer insulation layers 6122, 6120. The threaded taper 6234 bites into the inner and outer insulation layers 6122, 6120 to provide a sturdy physical connection. The shield 6118 passes through the metal connector 6232 to an opposite side where an opposite taper is present. There, the shield 6118 terminates while being positioned firmly between the taper of the metal connector 6232 and a taper of an inner metal connector 6236 that is positioned about the insulation extension 6132.

The insulation extension 6132 in this example extends the remainder of the lead 6108 where ring connectors 6134 are located at the proximal end. The filars 6124 jumper to their respective ring connectors via a filar jumper 6136. The

lumen may be present in some embodiments with the filars **6124** being located about the lumen.

The metal connectors **6232**, **6236** may be included on either the proximal or distal end to terminate the shield **6118** as discussed above. The separation of the metal connectors **6232**, **6236** to the distal electrode or proximal connector ring may also be in accordance with the separation as discussed above.

FIG. **47** shows one example of a set of steps that create the shield termination of FIG. **46**. The inner metal connector **6236** is positioned at the end of the inner insulation layer **6122** at a connector step **6242**. The shield **6118** is braided onto the inner insulation layer **6122** and over the inner metal connector **6236** at a braiding step **6244**. The outer insulation layer **6120** is bonded by reflow or another process onto the inner insulation layer **6122** over the shield **6118** such as by a reflowing step **6246**.

The inner and outer insulation layers **6122**, **6120** are ablated to form the taper and expose the shield **6118** at an ablating step **6248**. The outer metal connector **6232** is then placed into position over the inner metal connector **6236** and the taper of the inner and outer insulation layers **6122**, **6120** at a connector step **6250**. Here, the outer metal connector **6232** may be turned relative to the inner and outer insulation layers **6122**, **6120** to sink the threaded taper **6234** into the inner and outer insulation layers **6122**, **6120** while the outer metal connector **6232** firmly contacts the shield **6118** positioned against the inner metal connector **6236**. The outer metal connector **6232** may be crimped or welded into place over the shield **6118** and the inner metal connector **6236**.

FIG. **48** shows an embodiment with an additional feature that may be included for a joint **6276**, which may be of various types such as the butt, scarf, or lap joints **6130**, **6140**, and **6150**. At the joint **6276**, the outer metal connector **6131**, **6174** of the lead **6108** encounters another layer **6278**. This layer **6278** may be the insulation extension **6132** and/or the replacement outer insulation layer **6152**. In either case, wires of the shield **6118** may partially extend into the layer **6278**. However, prior to bonding the layer **6278** to the layer **6122**, the ends of the wires of the shield **6118** may be individually folded over at areas **6274** as shown in FIG. **48**. In this manner, the folded over ends are less likely to fray and migrate.

Embodiments as disclosed in relation to FIGS. **49-58** provide for rotation of a stylet within a lumen of an implantable medical lead by applying rotation directly to the implantable medical lead. The implantable medical lead has torsional stiffness and is rotationally coupled to the stylet. The torsional stiffness may be provided by features within the jacket of the lead body, such as a shield. The rotational coupling of the implantable medical lead to the stylet may be provided via features of the lead and/or stylet.

FIG. **49** shows a scenario where an implantable medical lead **7108** is being implanted within a patient. The lead **7108** enters the patient at an introduction site **7112** where an introduction needle provides a passageway into the body. The lead **7108** is shown transparently for purposes of illustration to reveal a stylet **7132** present within a lumen of the lead **7108**. The stylet **7132**, and specifically the bent tip **7134** of the stylet **7132**, is used to steer the lead **7108** as the lead **7108** is being inserted in order to direct the distal end of the lead **7108** to the stimulation site which may be a significant distance from the introduction site **7112**.

The bent tip **7134** is rotated in position by the stylet **7132** being rotated. The stylet **7132** may include a stylet hub **7130** on the proximal end. This stylet hub **7130** may engage the lead **7108** as discussed below. To rotate the stylet **7132** and

the bent tip **7134**, the doctor may apply rotation **7136** directly to the lead **7108** at the introduction site **7112** rather than reaching back to grasp the stylet hub **7130**. The lead **7108** is torsionally stiff such that the rotation **7136** causes rotation along the length of the lead **7108** including rotation **7138** near the proximal end, rotation **7142** of the hub, and rotation **7140** near the distal end.

The stylet **7132** is rotationally coupled to the lead **7108** at one or more points. The rotational coupling may be near the proximal end or the distal end of the lead **7108**, and this rotational coupling may be done in various ways as described below. Thus, the rotation **7136** being applied to the lead **7108** at the introduction site **7112** causes the stylet **7132** to rotate along the length to the bent tip **7134**.

The stylet **7132** and stylet hub **7130** may be constructed of various materials. For example, the stylet may be constructed of steel, stainless steel, tungsten, beryllium, and their alloys which provides torsional rigidity. The stylet hub may be constructed of various materials such as nylon, polycarbonate, or other rigid engineering plastics.

FIG. **50** shows an implantable medical system in place once the lead **7108** has been directed to the stimulation site. The implantable medical system includes an IMD **7102** having a biocompatible case and a header **7106**. The lead **7108** includes distal electrodes **7116** at the stimulation site that are used to provide the stimulation. The lead also includes proximal connectors **7110** that are fixed by a set screw or other mechanism within the header **7106** and are connected to electrical circuitry of the IMD **7102**. The IMD **7102** produces stimulation signals that are provided to the connectors **7110**. Filars within the lead **7108** carry the stimulation signals from the connectors **7110** to the electrodes **7116**.

FIGS. **51** and **52** show an embodiment of the implantable medical lead **7108** where a shield **7118** is present that provides the torsional rigidity. An outer jacket layer **7120** is shown transparently in FIG. **51** for purposes of illustrating the shield **7118**. The shield **7118** may be included for various reasons in addition to creating the torsional rigidity. For example, the shield **7118** may provide protection from unwanted RF energy. For instance, the lead **7108** may be a magnetic resonance imaging (MRI) safe lead that allows the patient to have an MRI scan without risking tissue damage due to induced RF currents in the filars of the lead **7108**. The conductive filars **7124** extend the length of the lead **7108** and interconnect the proximal connectors **7110** to the distal electrodes **7116** so that stimulation signals are conducted from the proximal end to the distal end of the lead **7108**.

As shown in FIG. **51**, the shield **7118** of this example is a braided metal wire. The metal wire may be constructed of various materials such as titanium, tantalum, platinum, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield **7118**, particularly for embodiments where a portion of the shield **7118** may be exposed for purposes of grounding. While the shield **7118** is shown as a braid, other shield configurations may be chosen such as a metal foil that is wrapped in an overlapping fashion. If shielding is not desired, then the foil may be more loosely wrapped and still provide torsional rigidity.

As shown in FIG. **52**, the shield **7118** may be embedded within the jacket of the lead **7108**. One manner of constructing the lead **7108** with the shield **7118** is to provide an inner jacket **7122** that encloses the filars **7124** and any additional insulation layer **7126** that may surround each filar **7124**. The

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shield 7118 may then reside on the outer portion of the inner jacket 7122, and the outer jacket 7120 may then enclose the shield 7118.

The shield 7118 may ground to tissue via an RF coupling through the outer layer 7120 and/or via grounding to the can 7104 and/or to the tissue via ground rings. For embodiments where it is desirable for the shield 7118 to RF couple to tissue, the outer jacket layer 7120 may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield 7118 grounds at the can of the IMD and grounding via a RF coupling from the shield 7118 through the outer jacket 7120 directly to the tissue is of less significance, then the shield 7118 may be located further from the outer surface of the lead 7108. The outer jacket 7120 may be added over the shield 7118 by shrinking in place or by being extruded over the shield 7118.

The inner and outer jackets 7122, 7120 may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes and silicones. The lumen 7128 is included in the inner jacket 7122, particularly for percutaneous leads 7108, to allow the stylet 7132 to be inserted for purposes of pushing and steering the lead into the desired position within the patient.

As shown in the cross-section of FIG. 52, at this particular point along the lead the stylet 7132 is free within the lumen 7128. The stylet 7132 has clearance relative to the lumen 7128. This clearance may aid in the insertion of the stylet 7132 into the lumen 7128.

FIG. 53 shows a cross-section at a point along an embodiment of the lead 7108 where a rotational coupling is established between the lead 7108 and the stylet 7132. At this point, the lumen 7128 of the lead 7108 has a portion forming a passageway 7144 that has a square cross-sectional shape rather than being round. As one example, this passageway 7144 may be created in the distal tip of the lead 7108, distal to the location of the distal electrodes. The stylet 7132 likewise has a shaft 7146 that has a square cross-sectional shape and that fits within the square shaped passageway 7144 of the lumen 7128. The position of the square shaped passageway 7144 may be such that when the stylet 7132 is fully inserted in the lead 7108, the square shaped shaft 7146 of the stylet 7132 mates to the square shaped passageway 7144 of the lumen 7128. The square shape effectively keys the stylet 7132 to the lead 7108 so that a rotational coupling is achieved.

FIG. 54 shows a cross-section at a point along another embodiment of the lead 7108 where a rotational coupling is established between the lead 7108 and the stylet 7132. At this point, the lumen 7128 of the lead has a particular portion forming a passageway 7148 that has a star cross-sectional shape rather than being round. As one example, this passageway 7148 may be created in the distal tip of the lead 7108, distal to the location of the distal electrodes. The stylet 7132 likewise has a shaft 7150 that has a star cross-sectional shape and that fits within the star shaped passageway 7148 of the lumen 7128. The position of the star shaped passageway 7148 may be such that when the stylet 7132 is fully inserted in the lead 7108, the star shaped shaft 7150 of the stylet 7132 mates to the star shaped passageway 7148 of the lumen 7128. The star shape effectively keys the stylet 7132 to the lead 7108 so that a rotational coupling is achieved.

FIG. 55 shows a cross-section at a point along another embodiment of the lead 7108 where a rotational coupling is established between the lead 7108 and the stylet 7132. At this point, the lumen 7128 of the lead has a particular portion forming a passageway 7152 that has a hexagonal cross-

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sectional shape rather than being round. As one example, this passageway 7152 may be created in the distal tip of the lead 7108, distal to the location of the distal electrodes. The stylet 7132 likewise has a shaft 7154 that has a hexagonal cross-sectional shape and that fits within the hexagonal shaped passageway 7152 of the lumen 7128. The position of the hexagonal shaped passageway 7152 may be such that when the stylet 7132 is fully inserted in the lead 7108, the hexagonal shaped shaft 7154 of the stylet 7132 mates to the hexagonal shaped passageway 7152 of the lumen 7128. The hexagonal shape effectively keys the stylet 7132 to the lead 7108 so that a rotational coupling is achieved.

The square, star, and hexagonal shapes are shown for purposes of illustration. It will be appreciated that any number of shaped engagements may be utilized to establish a rotational coupling between the lead 7108 and the stylet 7132. Furthermore, it will be appreciated that the coupling may occur at any point or multiple points along the lead 7108 where the torsional stiffness is present.

FIG. 56 shows a side view of a proximal end of the lead 7108 with the lumen 7128 engaging an embodiment of the stylet hub 7130 in order to establish a rotational coupling between the stylet 7132 and the lead 7108. The stylet hub 7130 includes a tapered region 7156 that extends from the hub 7130 to the stylet 7132. This tapered region 7156 at a large diameter end has a diameter larger than that of the lumen 7128. As a result, the tapered region 7156 may be press fit into the lumen 7128 of the lead 7108 in order to produce a frictional fit that establishes a rotational coupling.

FIG. 57 shows a side view of a proximal end of the lead 7108 with the lumen 7128 engaging another embodiment of the stylet hub 7130 in order to establish a rotational coupling between the stylet 7132 and the lead 7108. The stylet hub 7130 includes a splined region 7158 that extends from the hub 7130 to the stylet 7132. The diameter created by the splined region 7158 may be greater than the diameter of the lumen 7128. This splined region 7158 may be press fit into the lumen 7128 of the lead 7108 in order to engage the splines with the lumen 7128 to establish a rotational coupling.

FIG. 58 shows a side view of a proximal end of the lead 7108 with the lumen 7128 engaging another embodiment of the stylet hub 7130 in order to establish a rotational coupling between the stylet 7132 and the lead 7108. The stylet hub 7130 includes a threaded region 7160 that extends from the hub 7130 to the stylet 7132. The diameter created by the threaded region 7160 may be greater than the diameter of the lumen 7128. This threaded region 7160 may be screwed into the lumen 7128 of the lead 7108 in order to engage the threads with the lumen 7128 to establish a rotational coupling.

The tapered, splined, and threaded engagements of the hub 7130 to the lumen 7128 are shown for purposes of illustration. It will be appreciated that any number of hub features may be used to engage the lumen 7128 to provide the rotational coupling. It will be further appreciated that similar features may be used to allow the hub 7130 to instead engage the outer layer 7120 of the lead 7108 at the proximal end such as by having a taper, splines, or threads that surround the outer layer 7120 but with a smaller diameter than the outer layer 7120. These features face inward to engage the outer layer 7120 and establish the rotational coupling.

Embodiments as disclosed in relation to FIGS. 59-72 provide radiopaque markers that are added to implantable medical leads or to implantable medical devices (IMD) connected to the leads to identify the leads as being designed

for safe application of a medical procedure such as an MRI scan. The radiopaque markers are visible on an X-ray or during fluoroscopy so that administering personnel can have a visual assurance that the lead is designed for safe application of the medical procedure of interest.

FIG. 59 shows an embodiment of an implantable medical system that includes an IMD 8102 having a can 8104 that houses electronics and a header 8106. In this example, the IMD 8102 provides signals to a pair of implantable medical leads 8108, 8109 which are physically and electrically connected to the IMD 8102 via the header 8106.

Radiopaque markers 8130, 8131 are provided to identify the leads 8108, 8109 as being safe for a given procedure. In this particular example, the radiopaque markers 8130, 8131 are tags that are fixed directly to corresponding leads 8108, 8109. Sutures 8132 of the permanent type hold the tag 8130 to the lead 8108 while sutures 8133 hold the tag 8131 to the lead 8109. By individually tagging both leads 8108, 8109, the administering personnel can be assured that both leads are safe for the given procedure.

The tags 8130, 8131 may be added after the leads 8108, 8109 have been successfully implanted into the patient. For percutaneous leads, this is particularly desirable because the lead 8108, 8109 is inserted into the body of the patient via an introducer needle that lacks clearance for the tags 8130, 8131. Thus, once the leads 8108, 8109 are in position with the proximal ends of the leads being near the incision site and with the introducer needle removed, the tags 8130, 8131 can be inserted into a pocket made for the IMD 8102 and sutured in place by the doctor.

The tags 8130, 8131 may be constructed of a biocompatible material that has a density that is adequately radiopaque by being visible on an X-ray or during fluoroscopy. Examples of such materials include barium, tantalum, platinum, and platinum-iridium. The size of the tags 8130, 8131 may vary but when sized to have a length and width in the range of 0.25 to 5 centimeters and 0.01 to 0.2 inch thickness, the tag 8130, 8131 is adequately visible while being small enough to comfortably fit within or nearby the pocket near the IMD 8102.

When administering personnel wish to perform a given medical procedure such as an MRI, the personnel may take an X-ray or conduct fluoroscopy to look for the radiopaque marker. The IMD 8102 itself may need to also be designed for safety during a given medical procedure and may have its own internal or external radiopaque marker. Thus, placing the tags 8130, 8131 nearby the IMD 8102 may be desirable so that the tags of both the leads 8108, 8109 and the marker of the IMD 8102 are in the same field of view of an X-ray or during fluoroscopy.

In this example of FIG. 59, the tags 8130, 8131 include an aperture 8138, 8139 in the shape of a particular symbol. Due to the aperture 8138, 8139, this shape within the tag 8130, 8131 is visibly distinguishable on the X-ray or during fluoroscopy. Thus, this aperture 8138, 8139 may identify the safety aspects of the lead 8108, 8109 and/or the medical procedures that are safe to conduct. The shape of the apertures 8138, 8139 in FIG. 59 is a wave that represents that the leads 8108, 8109 are safe for an MRI scan conducted within normal operating parameters.

FIG. 60 shows a similar configuration for the radiopaque tag 8130. However, rather than the doctor suturing the tag 8130 to the lead 8108, the doctor connects the proximal end of the lead 8108 to the IMD 8102 that is placed into the pocket and sutures the tag 8130 to the IMD 8102. In the example shown, sutures 8132 extending from the tag 8130 are tied around the can 8104. It will be appreciated that the

sutures 8132 could be tied to the IMD 8102 in other ways or to designated features of the IMD 8102.

FIG. 61 shows another example of placing the tag 8130 in close proximity to the IMD 8102 and lead 8108. However, in this example, the tag 8130 is not tied to either but is instead left loosely positioned within the pocket 8136 where the IMD 8102 is positioned. The pocket 8136 prevents the tag 8130 from migrating away from the position of the IMD 8102 so that the tag 8130 remains in the same field of view as the IMD 8102 and lead 8108 during an X-ray or fluoroscopy.

FIG. 62 shows another example of placing the tag 8130 in close proximity to the IMD 8102 and the lead 8108. In this example, rather than suturing the tag 8130 to the lead 8108, the doctor may have chosen to bond the tag 8130 to the lead using a glue 8140. Examples of a glue suitable for bonding the tag 8130 to the lead 8108 include medical adhesives.

FIG. 63 shows another example of placing the tag 8130 in direct proximity of the IMD 8102 by bonding the tag 8130 to the IMD 8102. Here, the tag 8130 is bonded to the IMD 8102 with the glue 8140. Examples of a glue suitable for bonding the tag 8130 to the IMD 8102 also include medical adhesives.

FIG. 64 shows another example of placing the tag 8130 in close proximity to the IMD 8102 and the lead 8108. In this example, the tag 8130 is attached to a clamp 8142, such as a U-shaped spring-loaded clamp or other clamp structures such as features that mechanically lock including detents. The clamp 8142 tightens against the lead 8108 to hold the tag 8130 in position relative to the lead 8108.

FIG. 65 shows another example of placing the tag 8130 in direct proximity to the IMD 8102. Here, the tag 8130 includes the clamp 8142 which is tightened against the can 8104 of the IMD 8102. The clamp 8142 could tighten against other portions of the IMD 8102 as well such as the header 8106. The clamp 8142 of FIG. 65 may be of the same types discussed above in relation to FIG. 64.

FIG. 66 shows another example of placing the tag 8130 in close proximity to the IMD 8102 and the lead 8108. In this example, the tag 8130 has an extension 8144 that forms a ring shape. Initially, the extension 8144 may be an open ring so that it easily fits onto the lead 8108. The extension 8144 may then be crimped to form a closed or nearly closed ring about the lead 8108 and to fix the tag 8130 relative to the lead 8108.

FIG. 67A shows an example of a radiopaque marker being installed on a lead where the radiopaque marker is not a tag. Instead, the radiopaque marker is a radiopaque coil 8146' that is in a radially expanded state produced by axially compressing the coil 8146'. The radially expanded state allows the coil 8146' to be placed about the lead 8108, with the lead 8108 traveling through the center of the coil in an axial direction. The coil 8146' is placed onto the proximal end of the lead 8108, shown here with connectors 8110, prior to the proximal end being inserted into the header 8106.

FIG. 67B shows the radiopaque coil 8146 in a radially contracted state. Here, once properly positioned along the lead 8108, the coil 8146 has been allowed to naturally expand axially to radially contract until the coil diameter meets that of the lead 8108 to fix the coil 8146 in place on the lead 8108. The lead 8108 is then connected to the IMD 8102, with the coil 8146 being located in proximity to the IMD 8102 in this example so as to be in the same field of view. The coil 8146 itself is the visible shape that indicates that the lead 8108 is safe for a particular medical procedure such as an MRI.

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The radiopaque coil **8146** may be constructed of materials similar to the tag **8130**. For instance, the coil **8146** may be constructed of barium, tantalum, platinum, and platinum-iridium. The size of the coil **8146** may vary but when sized in the range of 0.04 inch to 1.0 inch in length and from 2 mils to 0.10 inch in diameter when radially contracted, the coil **8146** is adequately visible while being small enough to comfortably fit within or nearby the pocket near the IMD **8102**.

FIGS. **68A** and **68B** show an example of a tool **8150** being used to place the coil **8146** in the radially expanded state onto the lead **8108** and to deposit the coil **8146** in the radially contracted state at the desired position on the lead **8108**. The tool **8150** holds the coil **8146** in the radially expanded state by providing a larger diameter than the lead **8108** and while providing a passageway for the lead **8108**.

As shown in FIG. **68B**, the tool **8150** is positioned on the lead **8108** with the lead **8108** passing through the passageway of the tool **8150**. The coil **8146** is pushed off of the tool **8150** until the coil **8146** has a radially contracted end about the lead **8108**. The tool **8150** may then be pulled away from the lead **8108** to allow the remainder of the coil **8146** in the radially expanded state to slide off of the tool **8150** and onto the lead **8108** where the coil **8146** achieves the radially contracted state.

FIG. **69A** shows a polymer structure **8152** that may be used to place a radiopaque marker onto the lead **8108**. FIG. **69B** shows the polymer structure **8152** once positioned on the lead **8108**. This polymer structure **8152** includes a cylindrical aperture **8154** that allows the lead **8108** to pass through. The cylindrical aperture **8154** may stretch to a larger diameter than the lead **8108** such as by using a conventional anchor deployment tool to position the polymer structure onto the lead **8108**. The polymer structure **8152** may then be removed from the anchor tool to allow the polymer structure **8152** to contract onto the lead **8108**.

The polymer structure **8152** includes an offset portion **8155**. Within this offset portion **8155**, a radiopaque plate **8156** is embedded. The radiopaque plate **8156** may include a symbol **8158** or other information to be conveyed to administering personnel. The radiopaque plate **8156** may be constructed of materials similar to the tag **8130**. For instance, the plate **8156** may be constructed of barium, tantalum, platinum, and platinum-iridium. The size of the plate **8156** may vary but when sized in at about 0.040 inch in length/width and about 0.01 to 0.2 inch thick, the plate **8156** is adequately visible while being small enough to be contained within the polymer structure **8152**.

The polymer structure **8152** of FIGS. **69A** and **69B** is similar to a lead anchor. However, this polymer structure **8152** lacks suture wings. Because the contraction of the cylindrical aperture **8154** holds the polymer structure in place, no sutures are needed.

The offset of the portion **8155** where the radiopaque plate **8156** is located provides for ease of removal of the polymer structure **8152** from the lead **8108**. An axial cut can be made along the cylindrical aperture **8154** because the radiopaque marker does not surround the cylindrical aperture **8154**. However, if ease of removal is not of concern, then embodiments may provide the radiopaque plate **8156** centered about the cylindrical aperture **8154**.

FIG. **69C** shows a similar polymer structure **8168**. However, the polymer structure **8168** is in the form of a lead anchor that includes suture wings **8170** while also including the radiopaque plate **8156**. Rather than relying on the cylindrical aperture to contract onto the lead **8108**, the lead anchor **8168** may additionally or alternatively have sutures

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8132 that tie the suture wings **8170** to the lead **8108** to hold the polymer structure **8168** in place in proximity to the IMD **8102**. The radiopaque plate **8156** may be centered about the lead **8108** within the polymer structure **8160** even where ease of removal is desired if the polymer structure **8160** is held in place by the sutures **8132** rather than a contracted state upon the lead **8108**.

FIG. **70A** shows another polymer structure **8160** that may be used to place a radiopaque marker onto the lead **8108**. FIG. **70B** shows the polymer structure **8160** once positioned on the lead **8108**. This polymer structure **8160** includes a cylindrical aperture **8162** that allows the lead **8108** to pass through. The cylindrical aperture **8162** may stretch to a larger diameter than the lead **8108** such as by using a conventional anchor deployment tool to position the polymer structure **8160** onto the lead **8108**. The polymer structure **8160** may then be removed from the anchor tool to allow the polymer structure **8160** to contract onto the lead **8108**.

The polymer structure **8160** includes an offset portion **8164**. Within this offset portion **8164**, a radiopaque coil **8166** is embedded. The radiopaque coil **8166** may form a symbol or other information to be conveyed to administering personnel. The radiopaque coil **8166** may be constructed of materials similar to the coil **8146**. For instance, the coil **8166** may be constructed of barium, tantalum, platinum, and platinum-iridium. The size of the coil **8166** may vary but when sized in the range of 0.04 to 1.0 inch in length and 2 mils to 0.10 inch in wire diameter with an overall diameter of 0.020 to 0.5 inch, the coil **8166** is adequately visible while being small enough to be contained within the polymer structure **8160**.

The polymer structure **8160** of FIGS. **70A** and **70B** is also similar to a lead anchor. However, this polymer structure **8160** lacks suture wings. Because the contraction of the cylindrical aperture **8162** holds the polymer structure **8160** in place, no sutures are needed.

The offset of the portion **8164** where the radiopaque coil **8166** is located provides for ease of removal of the polymer structure **8160** from the lead **8108**. An axial cut can be made along the cylindrical aperture **8162** because the radiopaque marker does not surround the cylindrical aperture **8162**. However, if ease of removal is not of concern, then embodiments may provide the radiopaque coil **8166** centered about the cylindrical aperture **8162**.

FIG. **70C** shows a similar polymer structure **8172**. However, the polymer structure **8172** is in the form of a lead anchor that includes suture wings **8174** while also including the radiopaque coil **8166**. Rather than relying on the cylindrical aperture to contract onto the lead **8108**, the lead anchor **8172** may additionally or alternatively have sutures **8132** that tie the suture wings **8174** to the lead **8108** to hold the polymer structure **8172** in place in proximity to the IMD **8102**. The radiopaque coil **8166** may be centered about the lead within the polymer structure **8172** even where ease of removal is desired if the polymer structure **8172** is held in place by the sutures **8132** rather than a contracted state upon the lead **8108**.

FIGS. **71** and **72** show an embodiment of the implantable medical lead **8108** where a shield **8118** is present. This shield **8118** may provide protection from RF energy that allows the lead **8108** to be conditionally MRI safe and thus eligible to carry the radiopaque marker for an MRI. An outer jacket layer **8120** is shown transparently in FIG. **71** for purposes of illustrating the shield **8118**. The shield **8118** may provide protection from RF energy of an MRI that might otherwise cause tissue damage due to induced RF currents in the filars

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of the lead **8108**. The conductive filars **8124** extend the length of the lead **8108** and interconnect the proximal connectors **8110** to the distal electrodes so that stimulation signals are conducted from the proximal end to the distal end of the lead **8108**.

As shown in FIG. **71**, the shield **8118** of this example is a braided metal wire. The metal wire may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield **8118**, particularly for embodiments where a portion of the shield **8118** may be exposed for purposes of grounding. While the shield **8118** is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead **8108** in an overlapping manner or an outer layer **8120** that is heavily doped with conductive particles.

As shown in FIG. **72**, the shield **8118** may be embedded within the jacket of the lead **8108**. One manner of constructing the lead **8108** with the shield **8118** is to provide an inner jacket **8122** that encloses the filars **8124** and any additional insulation layer **8126** that may surround each filar **8124**. The shield **8118** may then reside on the outer portion of the inner jacket **8122**, and the outer jacket **8120** may then enclose the shield **8118**.

The shield **8118** may ground to tissue via an RF coupling through the outer layer **8120** and/or via grounding to the can **8104** and/or to the tissue via ground rings. For embodiments where it is desirable for the shield **8118** to RF couple to tissue, the outer jacket layer **8120** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **8118** grounds at the can **8104** of the IMD **8102** and grounding via a RF coupling from the shield **8118** through the outer jacket **8120** directly to the tissue is of less significance, then the shield **8118** may be located further from the outer surface of the lead **8108**. The outer jacket **8120** may be added over the shield **8118** by shrinking in place or by being extruded over the shield **8118**.

The inner and outer jackets **8122**, **8120** may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes and silicones. A lumen **8128** may be included inside of the inner jacket **8122** around which the insulated filars **8124** are coiled or otherwise positioned. The lumen **8128** may be useful, particularly for percutaneous leads **8108**, to allow a stylet to be inserted for purposes of pushing and steering the lead **8108** into the desired position within the patient.

Embodiments as disclosed in relation to FIGS. **73-76D** provide for reduced torsional stiffness of a shield present within an implantable medical lead for use with an implantable medical device (IMD). The torsional stiffness of the shield may be reduced in various ways such as by axially cutting the shield to form a slot that breaks the circumferential mechanical continuity of the shield. The slot may then be closed to re-establish the circumferential shielding continuity of the shield and to preserve the shielding function.

FIG. **73** shows an example of an implantable medical system **9100** that includes an IMD **9102** coupled to a lead **9108**. The IMD **9102** includes a metal can **9104**, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD **9102** includes a header **9106** typically constructed of materials such as polysulfone or polyurethane, that is affixed to the metal can **9104**. The header **9106** is shown transparently for purposes of illustration. The header **9106**

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provides a structure for securing the lead **9108** to the IMD **9102** and for establishing electrical connectivity between circuitry of the IMD **9102** and electrodes of the lead **9108**.

The lead **9108** includes electrodes **9116** on a distal end that are positioned at a stimulation site within a patient. The lead also includes connector rings **9110** on a proximal end that is positioned within the header **9106**. The connector rings **9110** make physical contact with electrical connections **9111** within the header. The electrical connections **9111** may include a metal contact that the connector ring **9110** rests against upon being inserted into the header **9106** where a wire extends from the metal contact into the can **9104** where the circuitry is housed. Signals applied by the IMD **9102** to the connector rings **9110** are conducted through the lead **9108** to the electrodes **9116** to provide the stimulation therapy to the patient.

The lead **9108** is secured in the header **9106** such as by a set screw block **9112** within the header **9106** that allows at least one set screw **9114** to be tightened against at least one of the connector rings **9110**. A shield **9118** as shown in FIGS. **74A** and **74B** may be grounded to the body along one or more points down the length of the lead from the IMD **9102** via capacitive coupling through the jacket or via ground rings. The shield **9118** may also be grounded at the can **9104** of the IMD **9102** of FIG. **73**.

FIGS. **74A** and **75A** show an example of the lead **9108**, where a shield **9118** is present. An outer insulation layer **9120** of a lead jacket is shown transparently in FIG. **74A** for purposes of illustrating the shield **9118**. The shield **9118** blocks at least some RF energy from directly coupling to conductive filars **9124** that are present within the lead **9108**. The conductive filars **9124** extend the length of the lead and interconnect the proximal connector rings **9110** to the distal electrodes **9116** so that stimulation signals are conducted from the proximal end to the distal end of the lead **9108**.

As shown in FIG. **74A**, the shield **9118** of this example is a braided collection of metal wires. The metal wires may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield **9118**, particularly for embodiments where a portion of the shield **9118** may be exposed for purposes of grounding. While the shield **9118** is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a coiled configuration, foil strip wrapped about the lead **9108** in an overlapping manner or an outer layer **9120** that is heavily doped with conductive particles.

As shown in FIG. **75A**, the shield **9118** may be embedded within the jacket of the lead **9108**. One manner of constructing the lead **9108** with the shield **9118** is to provide an inner insulation layer **9122** of the jacket that encloses the filars **9124** and any additional insulation layer **9126**, such as polytetrafluoroethylene (PTFE) that may surround each filar **9124**. The shield **9118** may then reside on the outer portion of the inner insulation layer **9122**, and the outer insulation layer **9120** may then enclose the shield **9118**. The outer insulation layer **9120** may be added over the shield **9118** and shrunk in place or may be extruded over the shield **9118**. The outer jacket **9120** maybe added over the braid **9118**, or it may be extruded over the braid.

For embodiments where it is desirable for the shield **9118** to RF couple to tissue, typically as a capacitive coupling, in addition to grounding at the can **9104** or along the lead **9108**, the amount of the outer jacket layer **9120** covering the shield **9118** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **9118** grounds at one or more

specific locations along its length, via a direct current coupling or a capacitive coupling, the shield 9118 may be located further from the outer surface of the lead 9108.

The inner and outer insulation layers 9122, 9120 of the jacket may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes and silicones. A lumen 9128 may be included inside of the inner jacket 9122 around which the insulated filars 9124 are coiled or otherwise positioned. The lumen 9128 may be useful, particularly for percutaneous leads 9108, to allow a stylet to be inserted for purposes of pushing and steering the lead 9108 into the desired position within the patient.

As shown, the shield 9118 has mechanical and shielding continuity about the circumference of the inner insulation layer 9122. This continuity is achieved by the wires of the braided shield 9118 being continuous. The circumferential shielding continuity exists because there are no non-conductive openings large enough to allow RF energy to easily pass through. The mechanical continuity produces a large increase in torsional stiffness over an unshielded lead, which may be beneficial in some respects but is detrimental in other respects.

The detrimental aspects may include difficulties twisting the lead during implant procedures due to high torsional stiffness. Twisting the lead 9108 may be beneficial when guiding the lead 9108 to the stimulation site and to the IMD 9102 and when wrapping excess lengths of the lead 9108 about the IMD 9102. Thus, in some instances, it may be desirable to provide a shielded implantable medical lead with reduced torsional stiffness.

FIGS. 74B and 75B show the lead 9108 once the shield 9118 has been cut in an axial direction to create a slot 9140. A shield portion 9142 creates one edge of the slot 9140 while an opposing shield portion 9144 creates the opposite edge. The slot 9140 may be created by a single cut or by two cuts that are roughly parallel and result in a section of the shield 9118 being removed. The shield 9118 may be cut prior to applying the outer layer 9120 so that the outer layer 9120 does not need to be cut to cut the shield 9118.

The shield 9118 of FIG. 74B now lacks the circumferential mechanical continuity due to the slot 9140, and the torsional stiffness is reduced considerably as a result. However, the slot 9140 also breaks the circumferential shielding continuity because the slot 9140 has an axial dimension that allows RF energy to easily pass through the slot 9140. Therefore, to preserve the RF shielding function of the shield 9118, the slot 9140 is closed in one of various ways that re-establishes the circumferential shielding continuity while allowing the circumferential mechanical continuity to remain broken.

FIGS. 74C and 75C show one embodiment of the lead 9108 where the slot 9140 is closed to re-establish circumferential shielding continuity by closing the slot sufficiently relative to the wavelengths of RF energy so that the RF energy cannot easily penetrate the shield 9118 at the location of the slot 9140. In this example, the shield portion 9144 laps over the shield portion 9142 to close the slot 9140 from a shielding continuity standpoint. The shield portion 9144 may or may not contact the shield portion 9142 such that circumferential electrical continuity may or may not be re-established. However, either way, the shield portion 9144 is not bonded to the shield portion 9142 such that they remain mobile with respect to one another, thereby maintaining the break in the circumferential mechanical continuity.

The shield portion 9144 may be lapped onto the shield portion 9142 as a natural result of cutting the shield 9118, such as where the shield 9118 is loosely braided over the inner insulation layer 9122. The loose braiding provides an uncut shield diameter that is slightly larger than the diameter of the inner insulation layer 9122 such that cutting the shield 9118 to create the slot 9140 allows the shield portion 9144 to collapse onto the shield portion 9142. This collapse of the shield 9118 closes the slot 9140 while the shield diameter is reduced down to the diameter of the inner insulation layer 9122.

The outer insulation layer 9120 is then added over the shield 9118. The outer insulation layer 9120 is a polymer that holds the shield 9118 in place against the inner insulation layer 9122. However, the polymer of the outer insulation layer 9120 is compliant so that the shield portion 9144 can move relative to the shield portion 9142 upon application of torque to the lead 9108.

FIGS. 74D and 75D show another embodiment of the lead 9108 where the slot 9140 is closed to re-establish circumferential shielding continuity by closing the slot 9140 sufficiently relative to the wavelengths of RF energy so that the RF energy cannot easily penetrate the shield 9118 at the location of the slot 9140. In this example, the slot 9140 is closed by application of a shield patch 9146. The shield patch 9146 may be constructed similarly to the braided shield 9118, using the same or similar metal wire. In the example shown, the shield patch 9146 is a grid pattern, but it will be appreciated that other patterns may also be used such as where the grid includes U-shaped portions along the axial wires so that the U-shaped portions allow for axial extension of the lead 9108. The shield patch 9146 overlaps onto the shield 9118 on both sides of the slot 9140. The shield portion 9144 may or may not contact the shield portion 9142 such that circumferential electrical continuity may or may not be re-established. However, either way, the shield patch 9146 is not bonded to the shield 9118 such that the shield patch 9146 can move relative to the shield 9118 on either side of the slot 9140. As a result, the circumferential mechanical continuity of the shield 9118 remains broken.

Because the shield patch 9146 is being added to the lead 9108, the braided shield 9118 may be applied to the inner insulation layer 9122 in a close fitting manner as opposed to loosely braiding the shield 9118. Once the cut is complete, the shield patch 9146 may then be placed into position directly onto the shield 9118 and across the slot 9140.

The outer insulation layer 9120 is then added over the shield 9118 and the shield patch 9146. As in the embodiment of FIGS. 74C and 75C, the outer insulation layer 9120 is a polymer, and the outer insulation 9120 holds the shield 9118 in place against or close to the inner insulation layer 9122 and also holds the shield patch 9146 in place against or close to the shield 9118. However, the polymer of the outer insulation layer 9120 is compliant so that the shield patch 9146 can move relative to the shield 9118 on either or both sides of the slot 9140 upon application of an axial twisting moment to the lead 9108.

FIG. 76A shows a representation of a shield 9150 that may be used in an implantable medical lead 9108. The representation is a tube, and this tube is illustrative for multiple reasons. Where the shield 9150 is a braided shield, such as the shield 9118, the apertures are small relative to the wavelengths of the RF energy such that the braided shield 9118 is effectively a tube from the perspective of the RF energy. Where the shield 9150 is another configuration, such as a foil strip wrapped around the inner insulation layer 9122

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in an overlapping fashion, the foil strip forms a true tube. In either case, a linear axial cut in the shield **9150** reduces torsional stiffness but appears as the slot **9152**, which presents an opening that the RF energy may pass through.

FIG. **76B** shows a tubular representation of the shield **9150** that may correspond to a braided shield or other shield configuration such as an overlapping wrapped foil strip. Here, the shield **9150** uses the overlap technique such as that shown above in FIGS. **74C** and **75C** to close the slot **9152** formed by the linear axial cut. As can be seen, a shield portion **9156** on one side of the slot **9152** overlaps another shield portion **9154** on the opposite side of the slot **9152** and may or may not contact the shield portion **9154**. Thus, the slot **9152** is effectively closed to establish shielding continuity across the slot **9152**, but the shield portions **9154**, **9156** may move relative to one another so that the mechanical continuity remains broken.

FIG. **76C** shows a tubular representation of the shield **9150** that may also correspond to a braided shield or other shield configuration such as an overlapping wrapped foil strip. Here, the shield **9150** uses a shield patch technique such as that shown above in FIGS. **74D** and **75D** to close the slot **9152** formed by the linear axial cut. As can be seen, a shield patch **9158** reaches across the slot **9152** to overlap and may or may not physically contact the shield **9150** on both sides of the slot **9152**. Thus, the slot **9152** is effectively closed to establish shielding continuity across the slot **9152**, but the shield patch **9158** may move relative to the shield **9150** on either or both sides of the slot **9152** so that the mechanical continuity remains broken.

FIG. **76D** shows a tubular representation of the shield **9160** that may correspond to a braided shield or other shield configuration such as an overlapping wrapped foil strip. Here, the shield **9160** has been cut axially using a helical cut rather than a linear cut to create a helical slot **9162**. The slot **9162** breaks the circumferential mechanical continuity so as to reduce the torsional stiffness, but the slot **9162** also breaks the circumferential shielding continuity.

The slot **9162** may be closed using techniques discussed above. A shield patch may be wrapped around the helical slot **9162** to reach across the slot **9162** and achieve circumferential shielding continuity. Or, the shield **9160** may be given a larger diameter than the inner insulation layer upon which it is positioned so that upon creating the slot **9162**, the shield **9160** may collapse to create an overlap along the helical slot **9162** to establish circumferential shielding continuity. This shield patch may be another piece of foil or may be a braided patch.

Embodiments as disclosed in relation to FIGS. **77-80C** provide for guarding a termination of a shield to reduce coupling of RF energy from the termination of the shield to filars present within an implantable medical lead for use with an implantable medical device (IMD). The guarding of the termination of the shield may be done in various ways such as by inverting the shield near the termination such that a first portion of the shield separates the termination of the shield from inner layers of the lead. Other examples may involve including separate pieces of the shield to form first and second portions, where one portion separates the termination of the other portion from the inner layers of the lead.

FIG. **77** shows an example of an implantable medical system **10100** that includes an IMD **10102** coupled to a lead **10108**. The IMD **10102** includes a metal can **10104**, typically constructed of a medical grade titanium, such as grades 1-4, 5 or 9 titanium, or similar other biocompatible materials. The IMD **10102** includes a header **10106** typically constructed of materials such as polysulfone or polyure-

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thane, that is affixed to the metal can **10104**. The header **10106** is shown transparently for purposes of illustration. The header **10106** provides a structure for securing the lead **10108** to the IMD **10102** and for establishing electrical connectivity between circuitry of the IMD **10102** and electrodes of the lead **10108**.

The lead **10108** includes electrodes **10116** on a distal end that are positioned at a stimulation site within a patient. The lead **10108** also includes connector rings **10110** on a proximal end that is positioned within the header **10106**. The connector rings **10110** make physical contact with electrical connections **10111** within the header. The electrical connections **10111** may include a metal contact that the connector ring **10110** rests against upon being inserted into the header **10106** where a wire extends from the metal contact into the can **10104** where the circuitry is housed. Signals applied by the IMD **10102** to the connector rings **10110** are conducted through the lead **10108** to the electrodes **10116** to provide the stimulation therapy to the patient.

The lead **10108** is secured in the header **10106** such as by a set screw block **10112** within the header **10106** that allows at least one set screw **10114** to be tightened against at least one of the connector rings **10110**. A shield **10118** as shown in FIGS. **78A** and **78B** may be grounded to the body along one or more points down the length of the lead from the IMD **10102** via capacitive coupling through the jacket or via ground rings. The shield **10118** may also be grounded at the can **10104** of the IMD **10102** of FIG. **77**.

FIGS. **78A** and **78B** show an example of the lead **10108**, where a shield **10118** is present. An outer insulation layer **10120** of a lead jacket is shown transparently in FIG. **78A** for purposes of illustrating the shield **10118**. The shield **10118** blocks at least some RF energy from directly coupling to conductive filars **10124** that are present within the lead **10108**. The conductive filars **10124** extend the length of the lead and interconnect the proximal connectors **10110** to the distal electrodes **10116** so that stimulation signals are conducted from the proximal end to the distal end of the lead **10108**.

As shown in FIG. **78A**, the shield **10118** of this example is a braided collection of metal wires. The metal wires may be constructed of various materials such as titanium, tantalum, niobium, platinum-iridium alloy, platinum, palladium, gold, stainless steel, and their alloys, or other metals. It may be desired to utilize a biocompatible metal for the shield **10118**, particularly for embodiments where a portion of the shield **10118** may be exposed for purposes of grounding. While the shield **10118** is shown as a braid, other shield configurations may be chosen particularly where flexibility is not an issue such as a foil strip wrapped about the lead **10108** in an overlapping manner.

FIG. **78B** is a cross-section that shows one example of construction of the lead **10108**. In this embodiment, either the guard is not provided or the guard is not present at the area where the cross-section is taken. Thus, FIG. **78B** shows the general construction of the lead **10108** without the specifics of the guard which are discussed below in relation to FIGS. **79A-79C** and **80A-80C**. The shield **10118** may be embedded within the jacket of the lead **10108**. One manner of constructing the lead **10108** with the shield **10118** is to provide an inner insulation layer **10122** of the jacket that encloses the filars **10124** and any additional insulation layer **10126**, such as polytetrafluoroethylene (PTFE) that may surround each filar **10124**. The shield **10118** may then reside on the outer portion of the inner insulation layer **10122**, and the outer insulation layer **10120** may then enclose the shield

10118. The outer insulation layer **10120** may be added over the shield **10118** and shrunk in place or may be extruded over the shield **10118**.

For embodiments where it is desirable for the shield **10118** to RF couple to tissue along, typically as a capacitive coupling, in addition to grounding at the can **10104** or along the lead **10108**, the entire outer jacket layer **10120** may be relatively thin, such as on the order of 0.5 to 5 mils. Where the shield **10118** grounds at one or more specific locations along its length, via a direct current coupling or a capacitive coupling, the shield **10118** may be located further from the outer surface of the lead **10108**.

The inner and outer insulation layers **10122**, **10120** of the jacket may be constructed of the same or similar materials such as various flexible and biocompatible polymers, examples of which are polyurethanes and silicones. A lumen **10128** may be included inside of the inner jacket **10122** around which the insulated filars **10124** are coiled or otherwise positioned. The lumen **10128** may be useful, particularly for percutaneous leads **10108**, to allow a stylet to be inserted for purposes of pushing and steering the lead **10108** into the desired position within the patient.

FIG. 79A shows an embodiment of the implantable medical lead **10108** in an axial cross-section where termination of the shield **10118** is guarded to reduce coupling of RF energy to one or more filars **10124**. FIG. 80A shows a radial cross-section of the same embodiment, with the cross section taken where the shield terminates. A single coiled filar **10124** is shown in this example but additional filars may be included and the filars may be of other forms such as linear cables rather than coils. In this example, the shield **10118** is one continuous shield of braided metal wires, but it will be appreciated that other shields may also be used such as the wrapped foil discussed above.

The shield **10118** of this example has an inversion **10136** near the distal end of the lead **10108**. This inversion **10136** creates two sections to the shield **10118**, a first portion **10119** that extends from the inversion **10136** back to the proximal end of the lead **10108** and a second portion **10121** that forms the distal termination of the shield **10118**. The inversion **10136** creates a guard for the shield termination.

The second portion **10121** is separated from the inner insulation layer **10122** as well as the filars **10124** by the first portion **10119**. The first portion **10119** is braided upon the inner insulation layer **10122** and then may be coated with the outer insulation layer **10120** with the second portion **10121** remaining uncoated. The inversion **10136** may then be created so that the second portion **10121** then laps onto the outer insulation layer **10120** so as to be separated from contact with the first portion **10119**. The second portion **10121** may extend from the inversion **10136** toward the proximal end by various distances, for instance ranging from about 1/8 inch to about 1 inch, such that the second portion **10121** may be axially shorter than the first portion **10119** which extends to the proximal end or the second portion **10121** may also extend to the proximal end. The second portion **10121** may then be covered by an additional outer insulation layer **10117**, made of the same or similar material as the outer insulation layer **10120**, if it is desired that the second portion **10121** be physically isolated from the body tissue.

The thickness of the outer insulation layer **10120** at the inversion **10136** dictates the bend radius of the inversion **10136** where the second portion **10121** laps onto the outer insulation layer **10120**. It may be desirable to have a bend radius that is sufficiently large, such as 0.002 inches, so that the inversion **10136** does not act as a shield termination from

which RF might couple to the filars **10124**. The lead diameter that is allowable for a given application may dictate the relative thicknesses of each of the layers and thus set an upper limit for the bend radius of the inversion **10136**.

Prior to or contemporaneously with the addition of the outer insulation layer **10117**, an extension **10132** to the outer insulation layer **10120** may be created to extend further toward the distal end where electrodes such as electrode **10130** are located. The extension **10132** may be constructed of the same or similar materials as that of the outer insulation layers **10117**, **10120**. The electrode **10130** has a filar jumper wire **10134** or the filar **10124** itself that extends through this extension **10132** and between the electrode **10130** and the filar **10124**. Alternatively, the area where extension **10132** is shown may be created as a continuation of the outer insulation layer **10117**.

FIG. 79B shows another embodiment of the implantable medical lead **10108** in an axial cross-section where termination of the shield **10118** is guarded to reduce coupling of RF energy to one or more filars **10124**. FIG. 80B shows a radial cross-section of the same embodiment, with the cross section taken where the shield **10118** terminates. A quad coiled filar **10124** is shown in this example but additional or fewer filars may be included and the filars may be linear cables rather than coils. In this example, the shield **10118** is two separate pieces forming a first portion **10123** and a second portion **10125** of the shield **10118** made of braided metal wires. It will be appreciated that either or both pieces may be another form of a shield such as wrapped foil as discussed above.

The shield **10118** of this example has the first portion **10123** that is a separate piece that resides at the distal end of the lead **10108** and may extend toward the proximal end for a relatively short distance, for instance, in the range of about 1/8 inch to about 1 inch. The first portion **10123** is wrapped around the inner insulation layer **10122**. An intervening layer of insulation **10115** then surrounds the first portion **10123**.

The second portion **10125** of the shield **10118** is wrapped around the intervening layer of insulation **10115** and is therefore physically isolated from contact with the first portion **10123**. The second portion **10125** then extends on to the proximal end of the lead **10108** and may therefore be axially longer than the first portion **10123**. The outer insulation layer **10120** then surrounds the second portion **10125**. As a result of this configuration, the first portion **10123** is located between the termination point at the second portion **10125** and the inner layers including the inner insulation layer **10122** and filars **10124**.

Because there is no inversion in this embodiment of FIGS. 79B and 80B, the thickness of the layers **10115**, **10120** may not be as large as the thickness of the outer insulation layer **10120** of the embodiment of FIGS. 79A and 80A where that thickness established the bend radius at the inversion **10136**. As a result, the separation between the first portion **10123** and the second portion **10125** may be smaller than the separation between the first portion **10119** and the second portion **10121** of FIGS. 79A and 80A. For instance, the intervening insulation layer **10115** may have a thickness ranging from about 0.002 inches to about 0.006 inches to control the separation between the first portion **10123** and second portion **10125**.

The outer insulation layer **10120** may be continued to extend on toward the distal end of the lead **10108**, including filling the area where the electrode **10130** is located. Alternatively, prior to or contemporaneously with the addition of the outer insulation layer **10120**, an extension layer from the

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outer insulation layer **10120** may be created to extend further toward the distal end where electrodes such as electrode **10130** are located. The extension may be constructed of the same or similar materials as that of the outer insulation layers **10117**, **10120**.

FIG. **79C** shows another embodiment of the implantable medical lead **10108** in an axial cross-section where termination of the shield **10118** is guarded to reduce coupling of RF energy to one or more filars **10124**. FIG. **80C** shows a radial cross-section of the same embodiment, with the cross section taken where the shield **10118** terminates. A quad coiled filar **10124** is shown in this example but additional filars may be included and the filars may be other forms such as linear cables rather than coils. In this example, the shield **10118** is two separate pieces forming a first portion **10140** and a second portion **10142** of the shield **10118** made of braided metal wires, but it will be appreciated that either piece may be another form of a shield such as wrapped foil as discussed above.

The shield **10118** of this example has the first portion **10140** that is a separate piece that resides at the distal end of the lead **10108** and that has an inversion **10138** to establish a first sub-portion **10146** and a second sub-portion **10144**. Both sub-portions **10144**, **10146** may extend toward the proximal end of the lead **10108** for a relatively short distance in the range of about $\frac{1}{8}$ inch to about 1 inch. The first sub-portion **10146** is wrapped around the inner insulation layer **10122**. An intervening layer of insulation **10113** then surrounds the first sub-portion **10146**.

The second portion **10142** of the shield **10118** is wrapped around the intervening layer of insulation **10113** and is therefore physically isolated from contact with the first sub-portion **10146**. The second portion **10142** then extends on to the proximal end of the lead **10108** and is therefore axially longer than the first sub-portion **10146** and the second sub-portion **10144**. The outer insulation layer **10120** then surrounds the second portion **10142**. As a result of this configuration, the first sub-portion **10146** is located between the termination point at the second portion **10142** and the inner layers including the inner insulation layer **10122** and filars **10124**.

The second sub-portion **10144** of the first portion **10140** laps onto the outer insulation layer **10120** as a result of the inversion **10138**. The second sub-portion **10144** may then be covered by an additional outer insulation layer **10127**, made of the same or similar material as the outer insulation layer **10120**, if it is desired that the second sub-portion **10144** be physically isolated from the body tissue.

The thickness of both the intervening layer of insulation **10113** and the outer insulation layer **10120** at the inversion **10138** dictates the bend radius of the inversion **10138** where the second sub-portion **10144** laps onto the outer insulation layer **10120**. It may be desirable to have a bend radius that is relatively large, such as about 0.002 inches, so that the inversion **10136** does not act as a shield termination from which RF might couple to the filars **10124**. The lead diameter that is allowable for a given application may dictate the relative thicknesses of each of the layers and thus set an upper limit for the bend radius similar to the upper limit for the embodiment of FIGS. **79A** and **80A**.

The outer insulation layer **10127** may be continued to extend on toward the distal end of the lead **10108**, including filling the area where the electrode **10130** is located. Alternatively, prior to or contemporaneously with the addition of the outer insulation layer **10127**, an extension layer from the outer insulation layer **10120** may be created to extend further toward the distal end where electrodes such as

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electrode **10130** are located. The extension layer may be constructed of the same or similar materials as that of the outer insulation layers **10127**, **10120**.

While many embodiments have been particularly shown and described, it will be understood by those skilled in the art that various other changes in the form and details may be made therein without departing from the spirit and scope of the invention.

What is claimed is:

1. A method of terminating a shield within a jacket of an implantable medical lead, comprising:

providing the shield between an inner insulation layer of the jacket and an outer insulation layer of the jacket, the shield terminates so that a longitudinal end of the shield does not extend beyond a longitudinal end of the inner insulation layer, the outer insulation layer terminating prior to the longitudinal end of the shield and the longitudinal end of the inner insulation layer;

bonding an insulation extension layer onto a longitudinally facing surface of the longitudinal end of the inner insulation layer; and

positioning a metal ring over the shield and inner insulation layer where the outer insulation layer is not present.

2. The method of claim 1, wherein the metal ring abuts the insulation extension layer.

3. The method of claim 1, wherein the inner insulation layer and the shield terminate by an end of the metal ring.

4. The method of claim 1, wherein the inner insulation layer and the shield extend beyond the metal ring.

5. The method of claim 1, further comprising:

positioning a replacement outer insulation layer over the inner insulation layer and over the shield, the replacement outer insulation layer extending beyond the end of the inner insulation layer to also extend over the insulation extension layer, the replacement outer insulation layer abutting an end of the metal ring.

6. The method of claim 1, wherein positioning the metal ring over the shield and the inner insulation layer comprises crimping the metal ring against the shield.

7. The method of claim 1, wherein positioning the metal ring over the shield and the inner insulation layer comprises welding the metal ring against the shield.

8. An implantable medical lead, comprising:

a jacket comprising an inner insulation layer and an outer insulation layer;

a shield located between the inner insulation layer and the outer insulation layer, the shield terminates so that a longitudinal end of the shield does not extend beyond a longitudinal end of the inner insulation layer, the outer insulation layer terminating prior to the longitudinal end of the shield and the longitudinal end of the inner insulation layer;

an insulation extension layer bonded onto a longitudinally facing surface of the longitudinal end of the inner insulation layer; and

a metal ring positioned over the shield and inner insulation layer where the outer insulation layer is not present.

9. The implantable medical lead of claim 8, wherein the metal ring abuts the insulation extension layer.

10. The implantable medical lead of claim 8, wherein the inner insulation layer and the shield terminate by an end of the metal ring.

11. The implantable medical lead of claim 8, wherein the inner insulation layer and the shield extend beyond the metal ring.

12. The implantable medical lead of claim 8, further comprising:

a replacement outer insulation layer over the inner insulation layer and over the shield, the replacement outer insulation layer extending beyond the end of the inner insulation layer to also extend over the insulation extension layer, the replacement outer insulation layer being bonded to the inner insulation layer. 5

13. The implantable medical lead of claim 8, wherein the metal ring is crimped against the shield. 10

14. The implantable medical lead of claim 8, wherein the metal ring is welded against the shield.

15. An implantable medical lead, comprising:

a jacket comprising an inner insulation layer and an outer insulation layer; 15

a shield located between the inner insulation layer and the outer insulation layer, the shield terminates so that a longitudinal end of the shield does not extend beyond a longitudinal end of the inner insulation layer;

an insulation extension layer bonded onto a longitudinally facing surface of the longitudinal end of the inner insulation layer; and 20

a ring positioned over the shield and inner insulation layer, wherein the ring directly contacts the shield, and wherein the ring directly contacts the insulation extension layer. 25

16. The implantable medical lead of claim 15, further comprising a reflowed polymer around an outer surface of the ring.

17. The implantable medical lead of claim 15, wherein the reflowed polymer contacts the insulation extension layer and the outer insulation layer. 30

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